



De-Havilland DHC-8-400 B1 & B2 TRAINING MANUAL

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CHAPTER ATA-21 AIR CONDITIONING

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY.

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Contents

Contents	2
Air Conditioning - General	2
Airconditioning PACK.....	18
Airconditioning PACK-Normal System Operation	44
Flow Control System.....	49
Temperature Control and Indication	58
Distribution.....	78
Avionics Cooling	87
Pressurization Control	93

Air Conditioning - General

Introduction

The air conditioning pack supplies the flight compartment and cabin air distribution systems with conditioned air. The conditioned air is used for cabin and flight compartment heating, cooling, ventilation and pressurization.

General Description

[Refer Figure 21-1](#)

The air conditioning system consists of a bleed air system from each engine, one integrated pack containing two Air Cycle Machines (ACMs), distribution and recirculation air systems.

Air required to operate the system is supplied by each engine from high or low pressure bleed ports. On aircraft equipped with an Auxiliary Power Unit (APU), bleed air is available from the APU when the aircraft is on the ground. An optional ground air conditioning connection is available to supply conditioned air from a ground source.

Conditioned air is supplied to the cabin and flight compartment through upper and lower vents in the cabin ceiling and sidewalls and the flight compartment sidewalls. The gasper system provides air to the individually controlled air outlets in the cabin, flight compartment and lavatory. The recirculation system draws air from the cabin zone and routes it back to the mixer, where it is mixed with fresh air discharged from the Air Cycle Machines.

The conditioned air from the ACMs pressurizes the aircraft. Pressurization is controlled by regulating the amount of air leaving the aircraft. An avionics cooling system supplies cooling for the avionics equipment bay and the EFIS/ESID liquid crystal displays.

Air temperature is independently controlled for the cabin and flight compartment and can be controlled either automatically or manually. Cabin and flight compartment temperatures are selected using the switches on the AIR CONDITIONING control panel on the overhead console. Cabin temperature can also be selected using the switches on the flight attendant panel.

The air conditioning system has the following subsystems:

- Distribution
- Pressurization Control
- Cooling
- Temperature Control

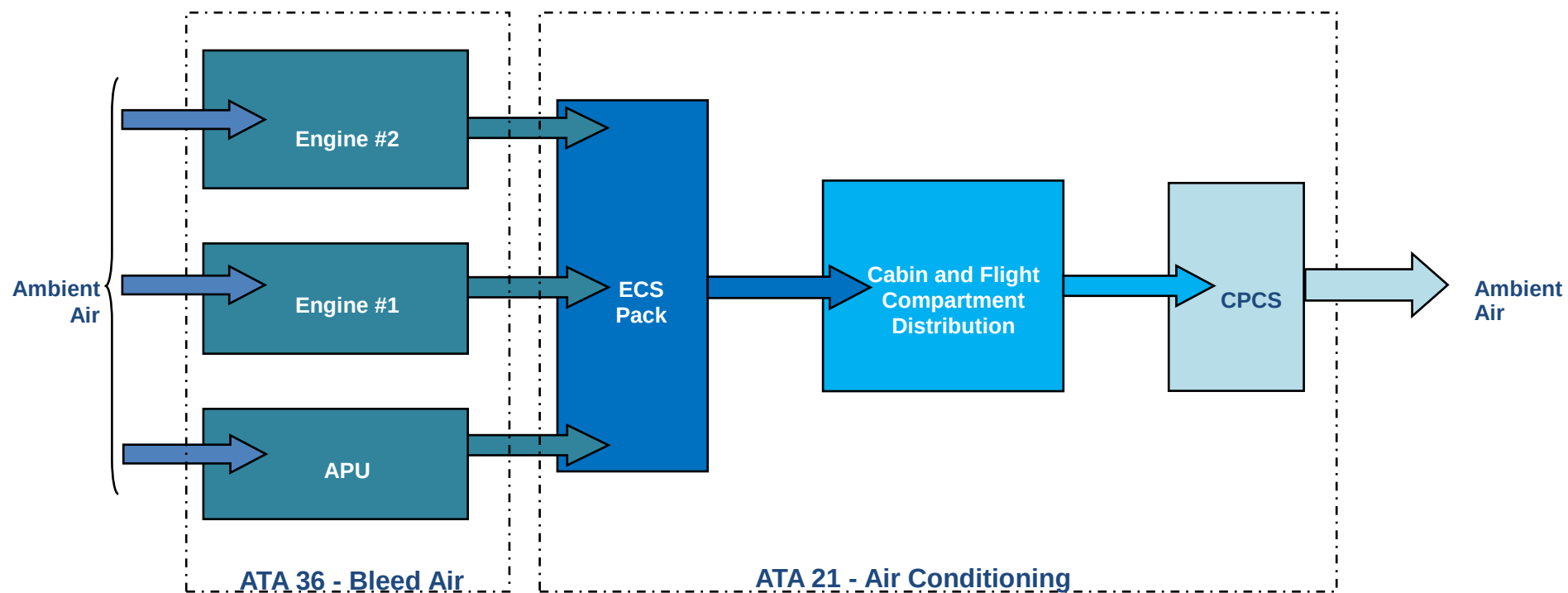


Figure 21- 1 Environmental Control System Block Diagram

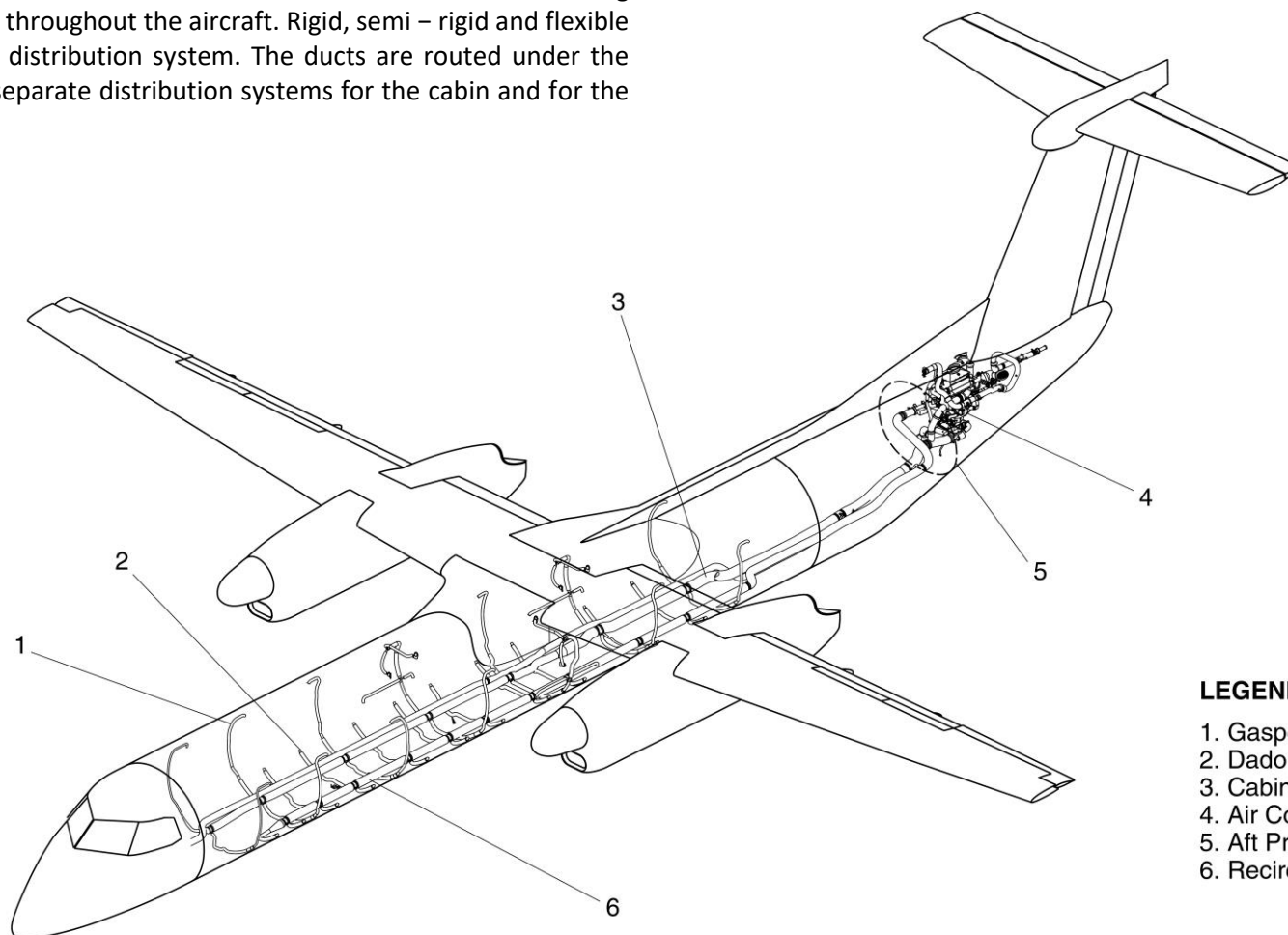
General Description

Distribution

[Refer Figures 21-2 & 21-3](#)

The air distribution system takes conditioned air from the air conditioning pack and distributes it throughout the aircraft. Rigid, semi – rigid and flexible ducts are used in the distribution system. The ducts are routed under the cabin floor and have separate distribution systems for the cabin and for the flight compartment.

The conditioned air is distributed to the cabin through lower dado panel vents, overhead vents and passenger gaspers. A recirculation fan helps to boost the supply of conditioned air. A gasper system distributes air to individual outlets in the cabin, flight compartment and the lavatory.



LEGEND

1. Gaspers.
2. Dado Panels.
3. Cabin Supply Duct.
4. Air Conditioning Pack.
5. Aft Pressure Bulkhead.
6. Recirculation Duct.

Figure 21- 2 Cabin Air Distribution

Conditioned air is also distributed to the flight compartment through lower vents near the rudder pedals, through side console gaspers and selectable vents for side window demisting.

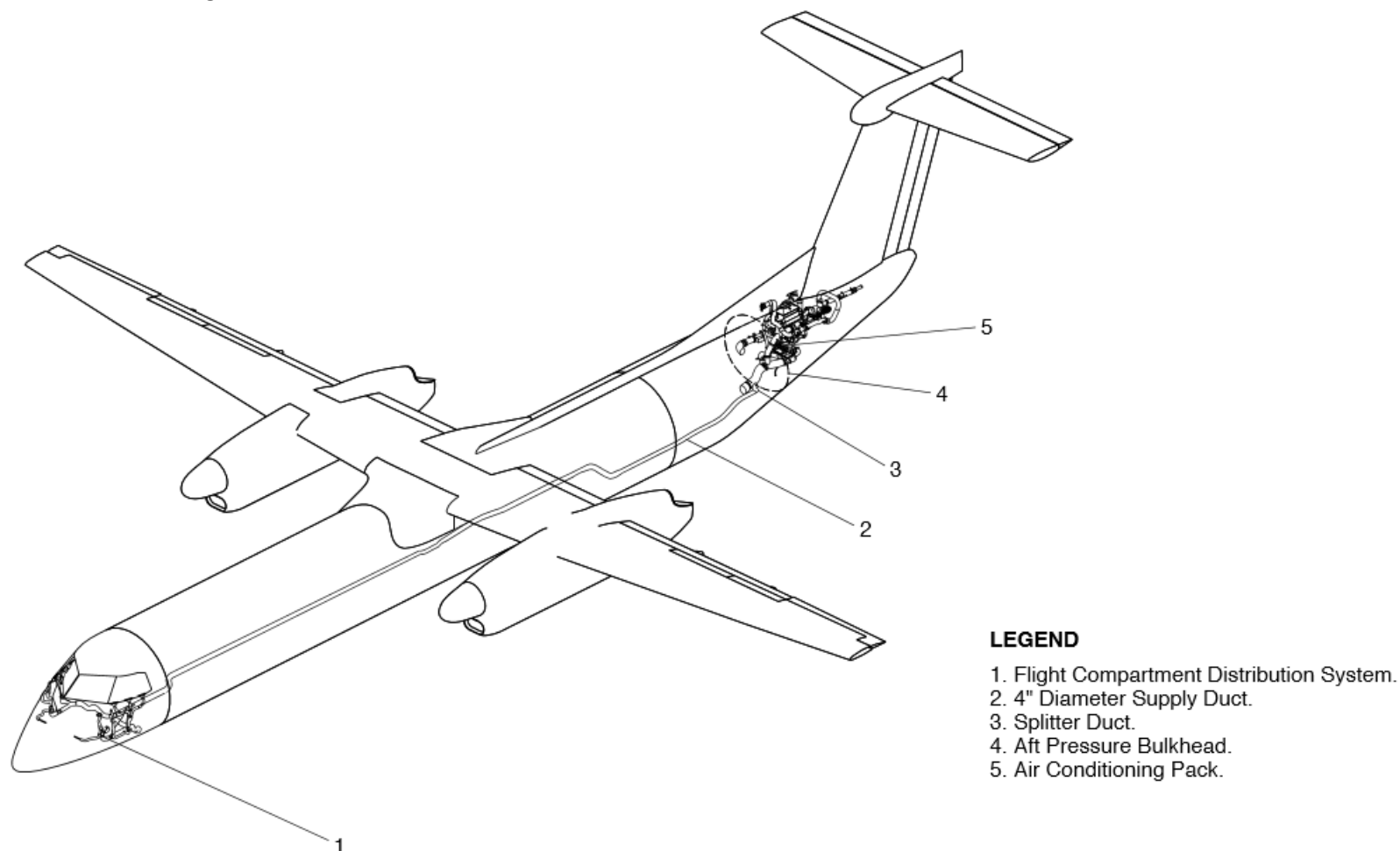


Figure 21- 3 Flight Compartment Air Distribution

[Refer Figure 21-4](#)

An external source of conditioned air is attached to the ground connection. Conditioned air flows through the ground connection and its ducting to the air conditioning supply ducts.

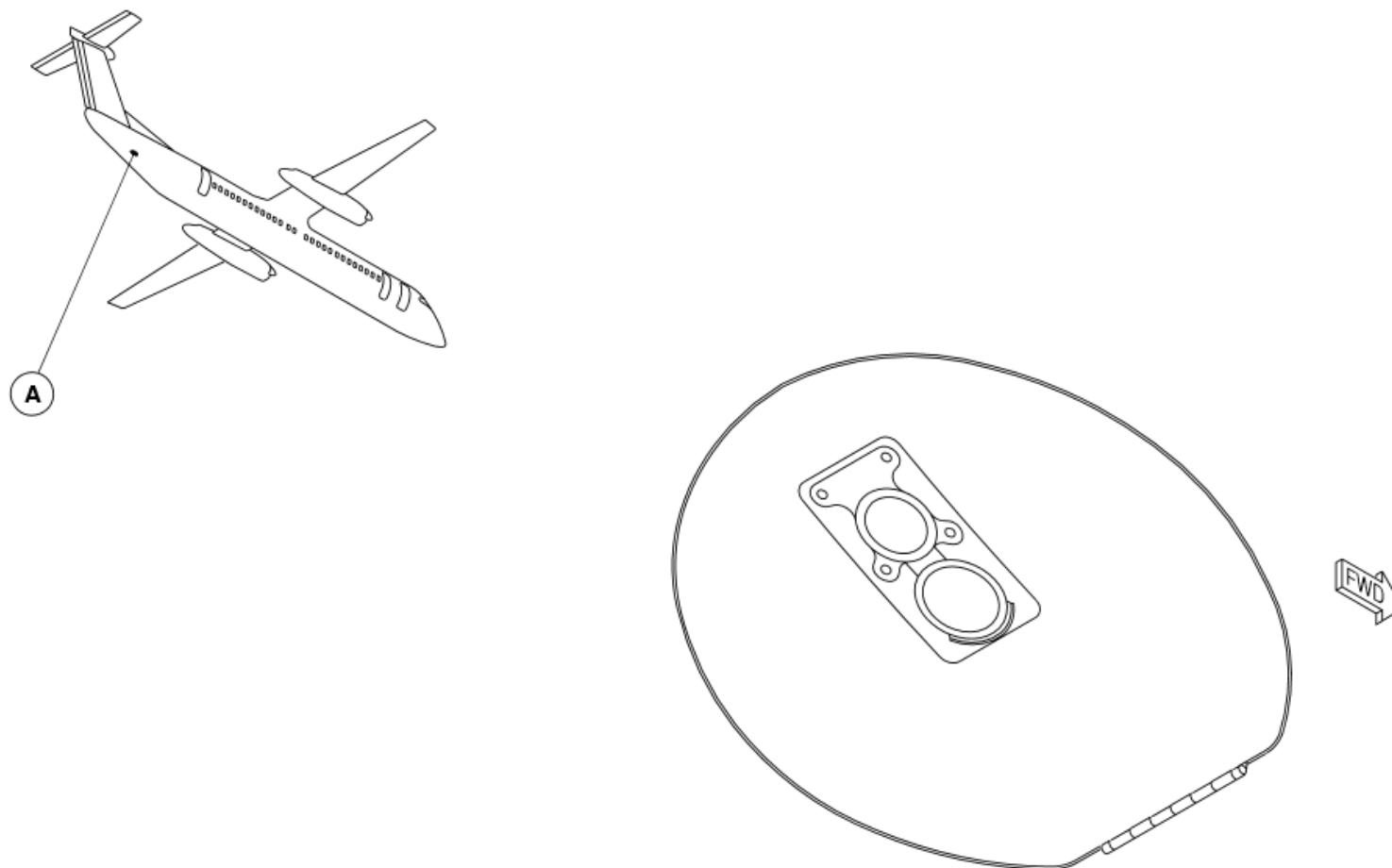


Figure 21- 4 External Conditioning Cart Connection

[Refer Figure 21-5](#)

The emergency ram air system supplies unconditioned air to the flight and cabin compartments in the event of a failure of the air conditioning pack.

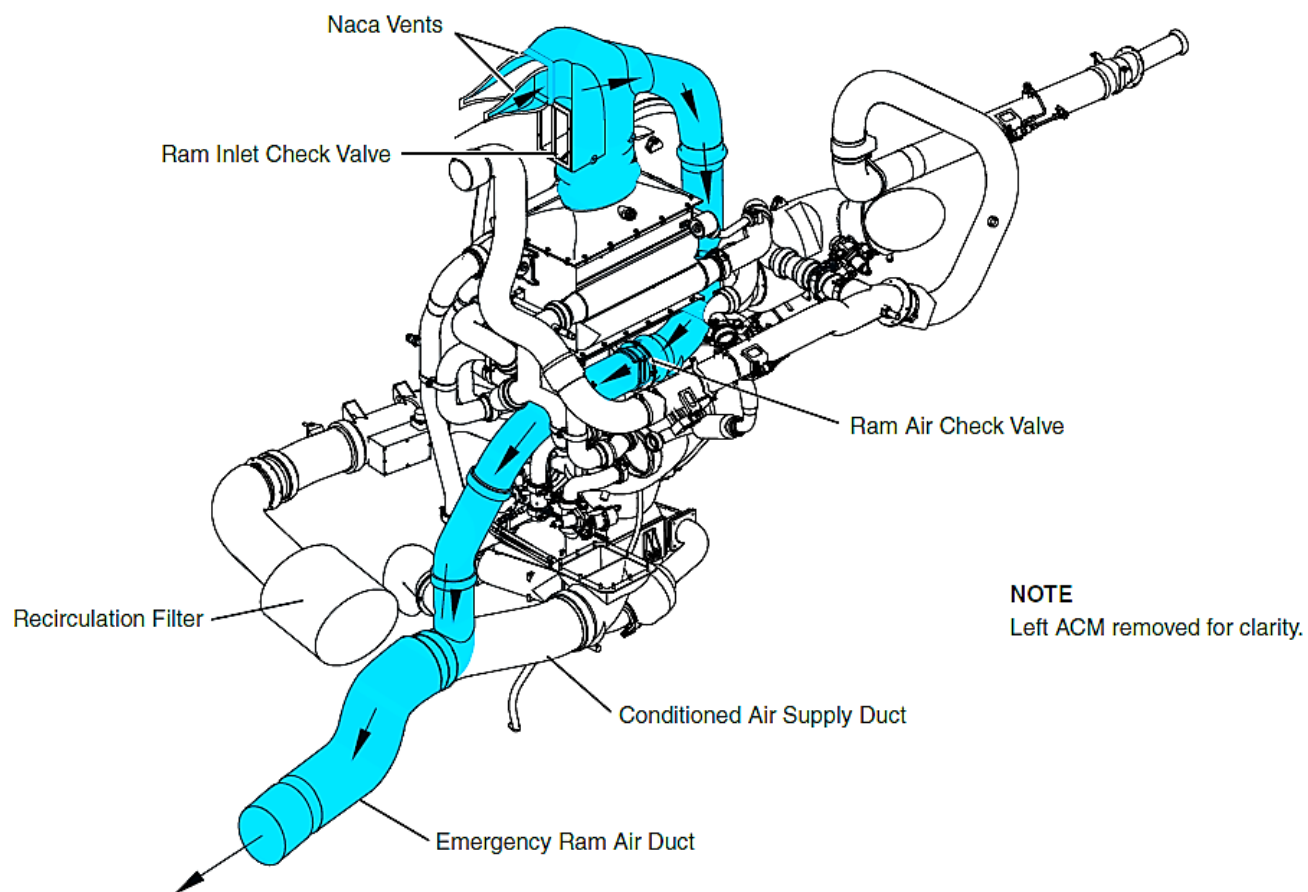


Figure 21- 5 Emergency Ram Air System

[Refer Figure 21-6](#)

The avionics cooling system has two independent duct networks and three fans. The system removes heat from the avionics equipment and the five liquid crystal displays. This improves the reliability and availability of the avionics systems.

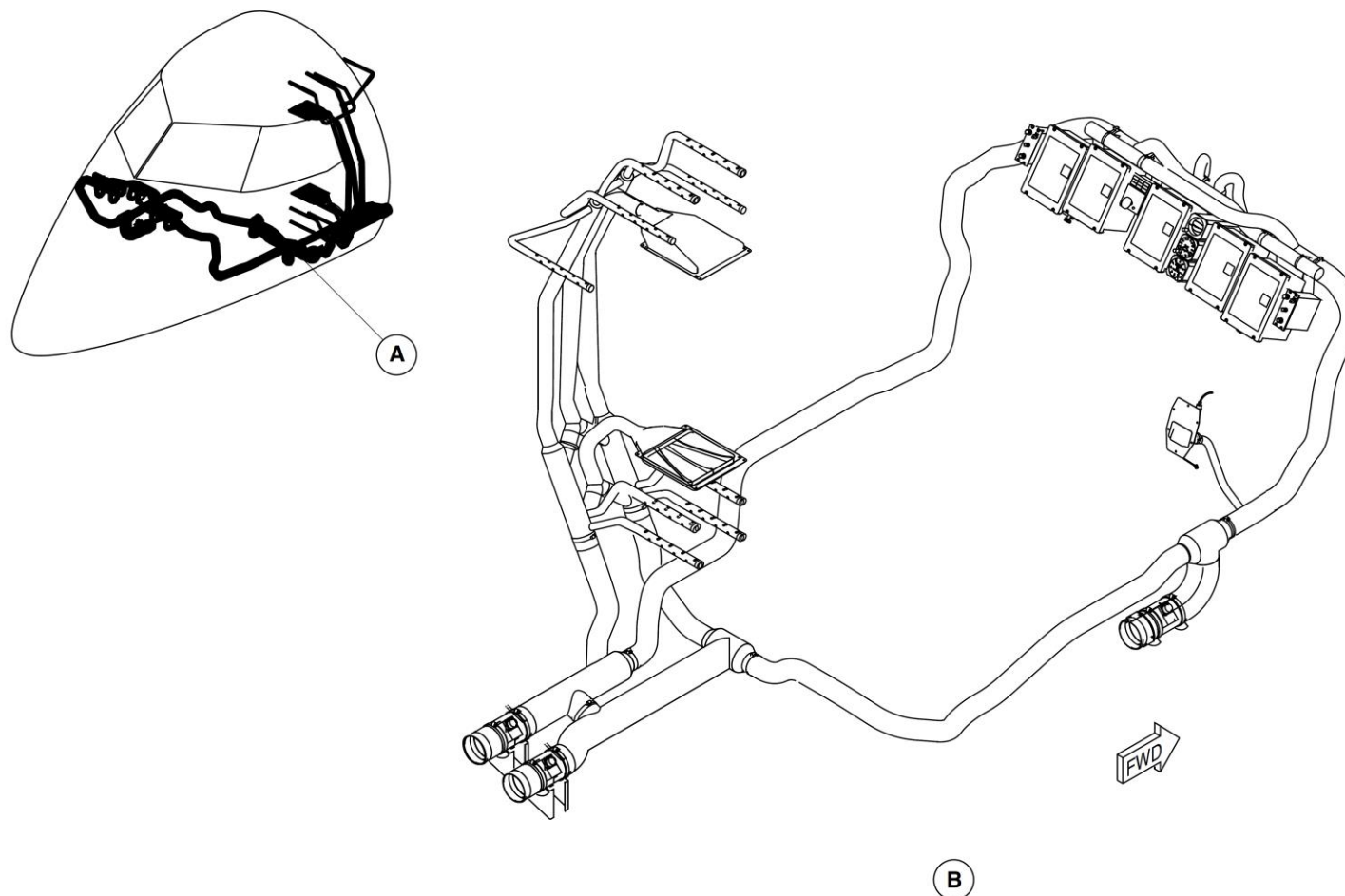


Figure 21- 6 Avionics Cooling System

Pressurization Control

[Refer Figure 21-7](#)

The pressurized area of the fuselage extends from the front pressure bulkhead immediately forward of the windshield at aircraft Sta X 111 to the rear pressure dome at Sta X 856. This area includes the flight compartment, passenger compartment, under floor space and baggage compartments.

LEGEND

1. Forward Pressure Bulkhead.
2. Forward Safety Valve.
3. Pressurization Control Panel.
4. Aft Pressure Dome.
5. Aft Outflow Valve.
6. Aft Safety Valve.

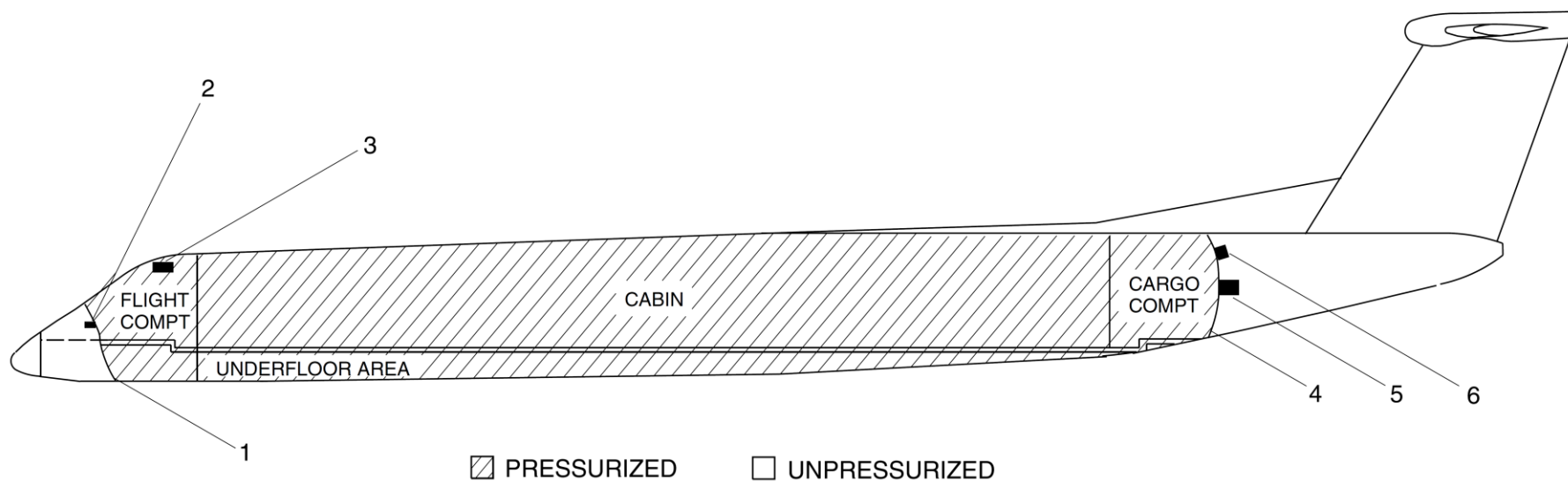


Figure 21- 7 Pressurized Zones

Refer Figures 21-8, 21-9 & 21-10

The aircraft is pressurized by engine bleed air supplied to and distributed by the air conditioning system. Pressure is maintained and controlled by the cabin pressure control system which governs the rate of air outflow from the pressurized areas of the aircraft.

An outflow valve is the primary control over the amount of air flowing out of the aircraft. It is assisted by the forward and aft safety valves. The outflow valve and the aft safety valve are located on the rear pressure dome. The forward safety valve is located on the forward pressure bulkhead.

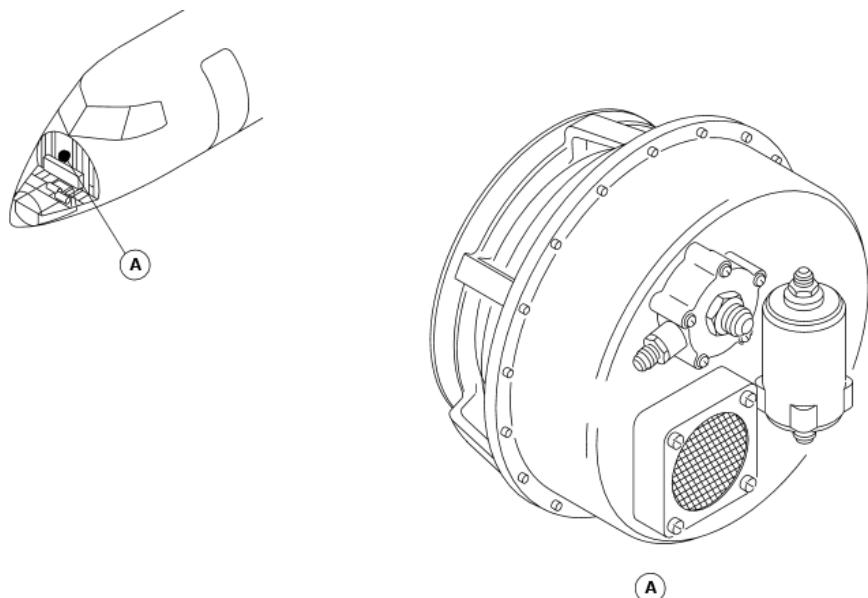


Figure 21- 8 Forward Safety Valve

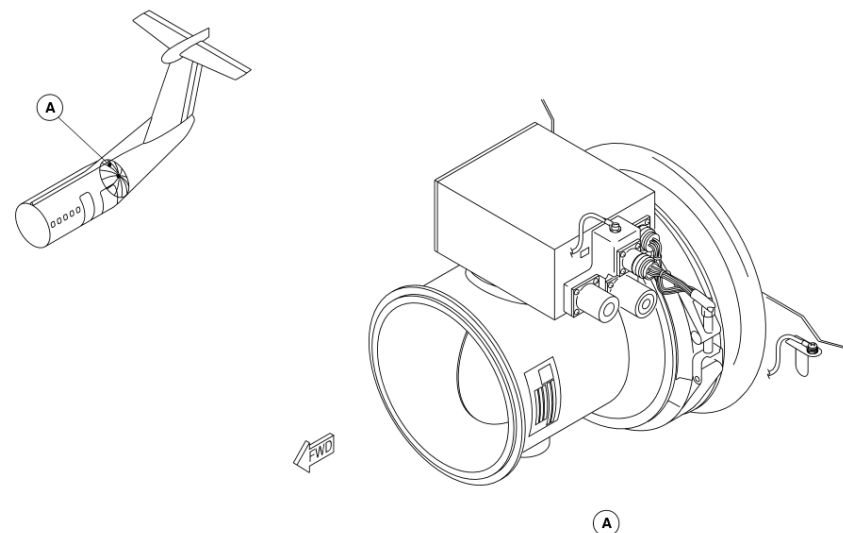


Figure 21- 9 Aft Outflow Valve

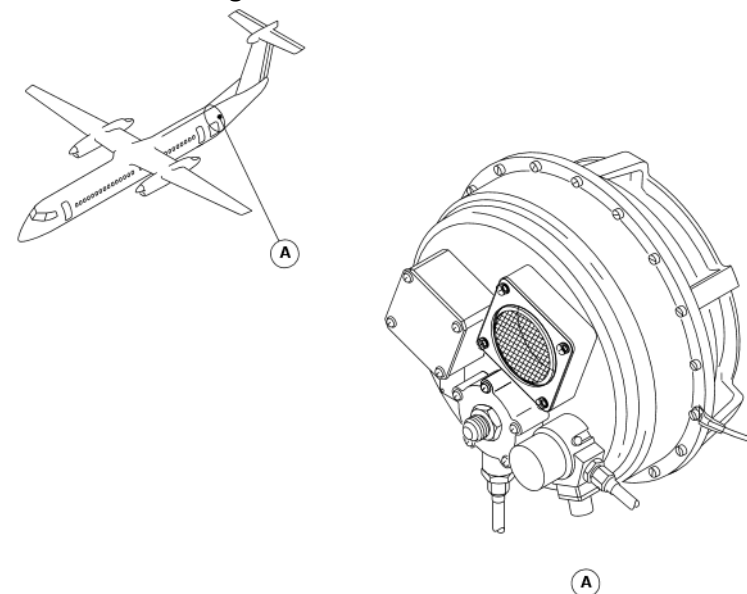


Figure 21- 10 Aft Safety Valve

Refer Figure 21-11

The outflow valve is controlled from the Cabin Pressure Control panel on the flight compartment overhead panel. The aft safety valve protects against cabin overpressure and negative pressure and supplies additional venting during ground operation.

A manually operated forward safety valve in the front pressure bulkhead can be used as a backup pressurization control or for smoke removal. This valve can be operated by the FWD OUTFLOW rotary knob on the Cabin Pressure Control Panel or by the forward safety valve selector on the copilot's side console.

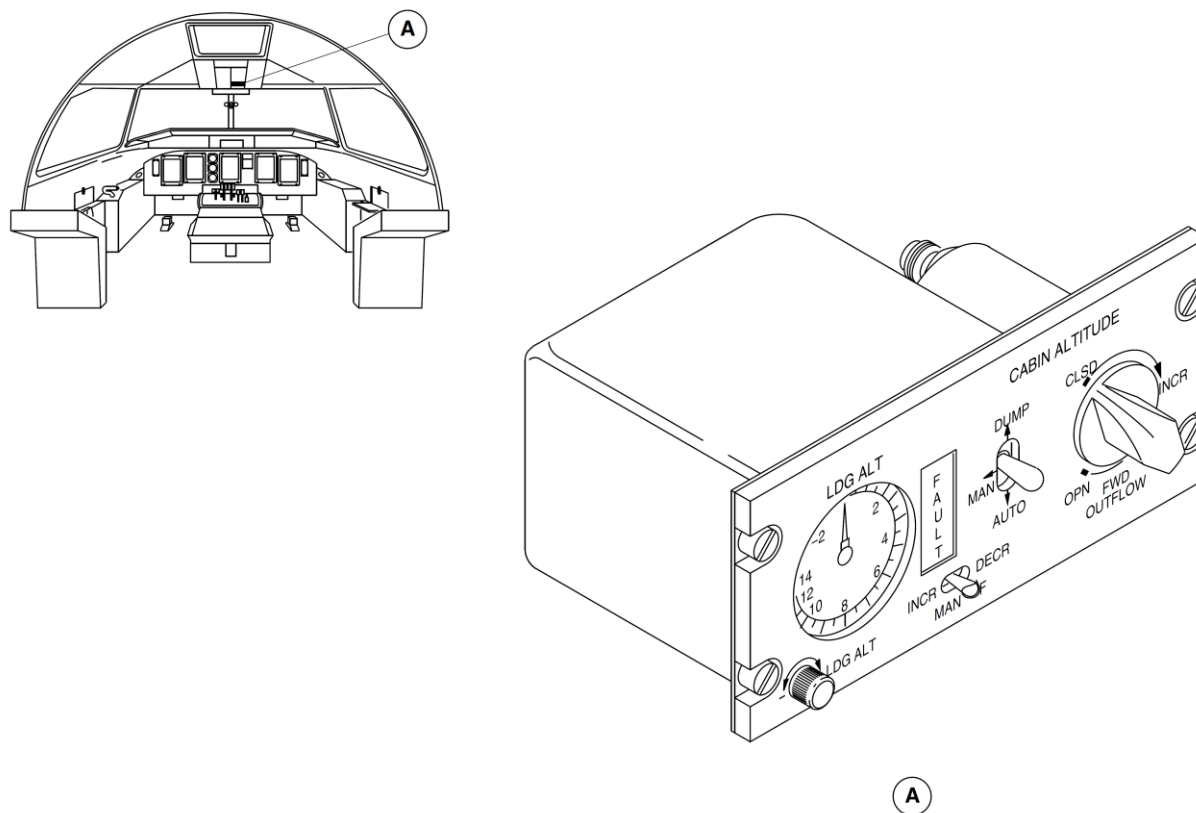


Figure 21- 11 Cabin Pressure Control Panel

Refer Figures 21-12 & 21-13

The Cabin Pressure Control Panel and the Cabin Pressure Controller control the system in the automatic mode. Exhaust air from the cabin and flight compartment is metered through the outflow valve in the rear pressure dome.

Refer Figure 21-14

Operation of the outflow valve is automatically controlled by the Cabin Pressure Controller according to the flight profile and landing elevation set by the aircrew. The valve can also be operated in the manual mode in the event of failure of the automatic mode of control.

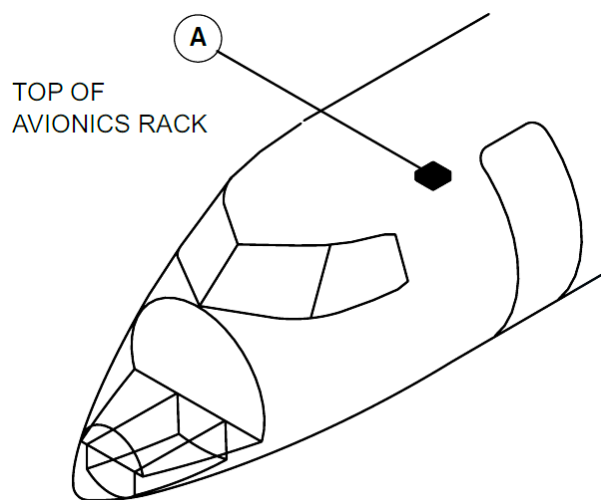


Figure 21- 12 Location of Cabin Pressure Controller

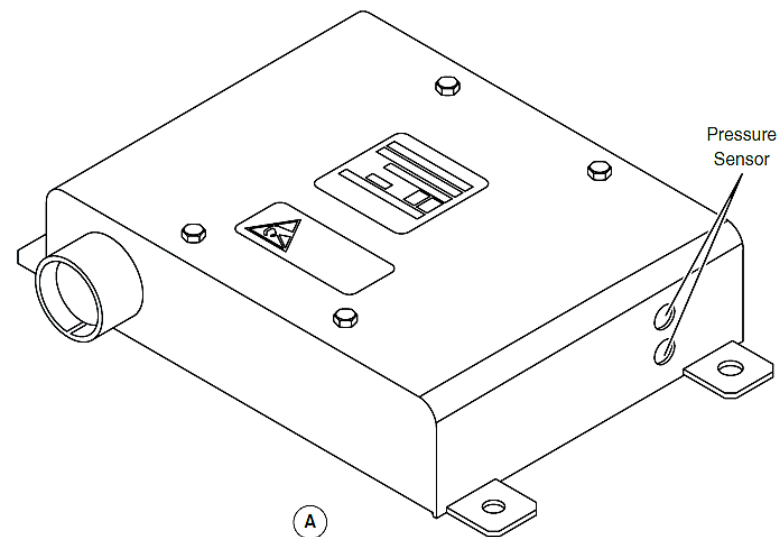


Figure 21- 13 Cabin Pressure Controller

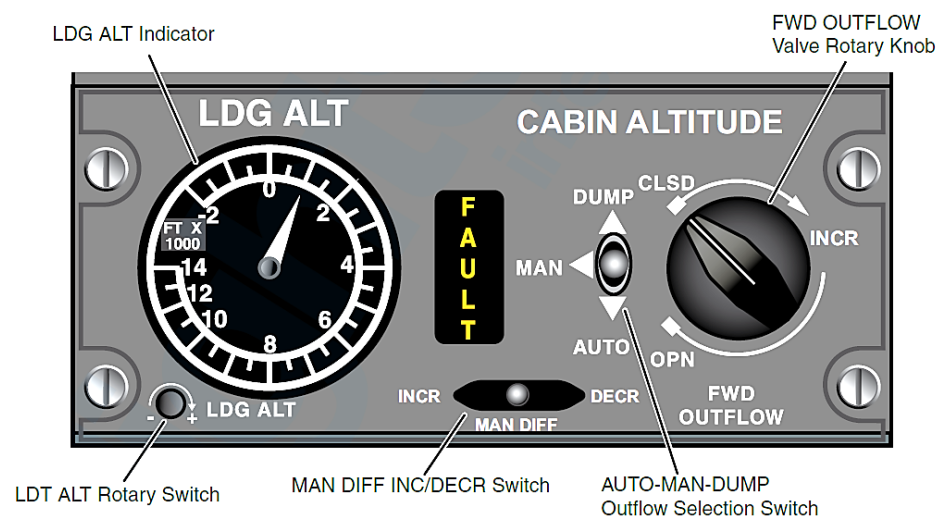


Figure 21- 14 Cabin Pressure Control Panel

Refer Figure 21-15

The cabin indication module shows the data to monitor the system. When cabin altitude is too high, the module sends a CABIN PRESS warning signal to the caution and warning panel.

Electrical power supply for pressurization control is supplied by the 28 Vdc left main bus through a CABIN PRESS AUTO CONT circuit breaker.

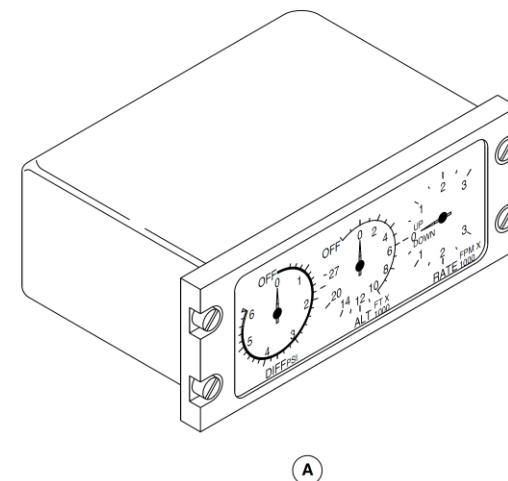
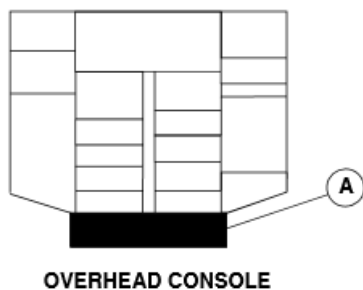
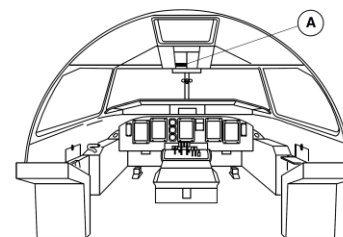


Figure 21- 15 Cabin Indication Module

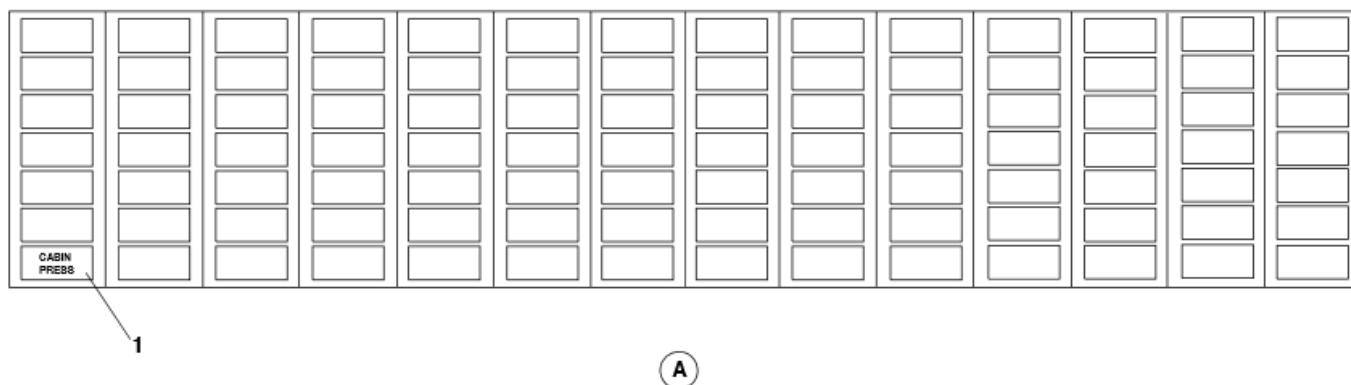


Figure 21- 16 Cabin Pressure Warning

Cooling

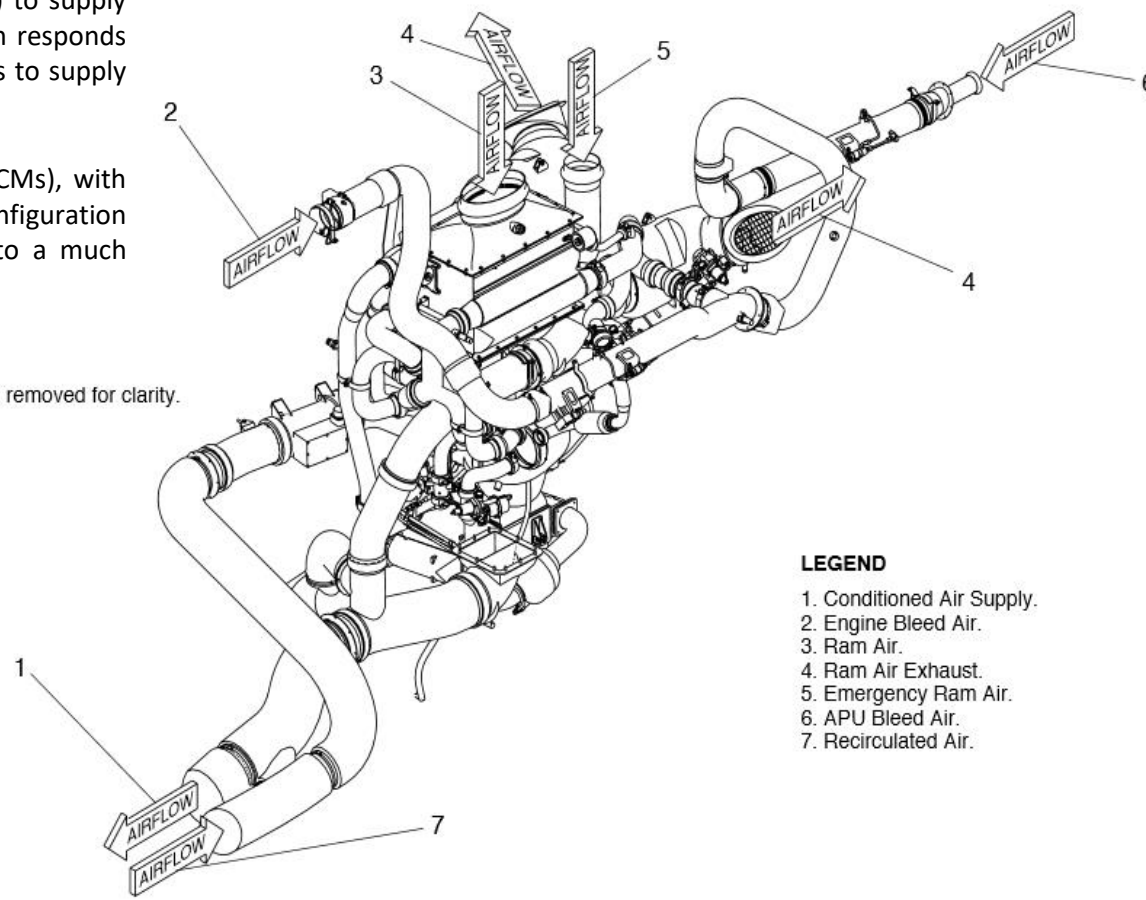
[Refer Figure 21-17](#)

The air conditioning pack is part of the Environmental Control System (ECS). It uses bleed air from the engines or Auxiliary Power Unit (APU) to supply conditioned air to the cabin and flight compartment. The system responds to flight crew commands and compartment temperature controls to supply a comfortable temperature environment for passengers and crew.

The air conditioning pack consists of two air cycle machines (ACMs), with one dual heat exchanger located in the aft fuselage. This configuration provides the redundancy of two ACMs while allowing access to a much larger dual heat exchanger during operation with a single ACM.

NOTE

Left ACM removed for clarity.



LEGEND

1. Conditioned Air Supply.
2. Engine Bleed Air.
3. Ram Air.
4. Ram Air Exhaust.
5. Emergency Ram Air.
6. APU Bleed Air.
7. Recirculated Air.

Figure 21- 17 Air Conditioning Pack

Refer Figure 21-18

The flow control system uses pressure and temperature data from the ECS Electronic Control Unit (ECU) to control the flow of the bleed air that enters the air conditioning pack. The Electronic Control Unit (ECU) opens and closes the pack flow control and shutoff valve to regulate the flow of bleed air.

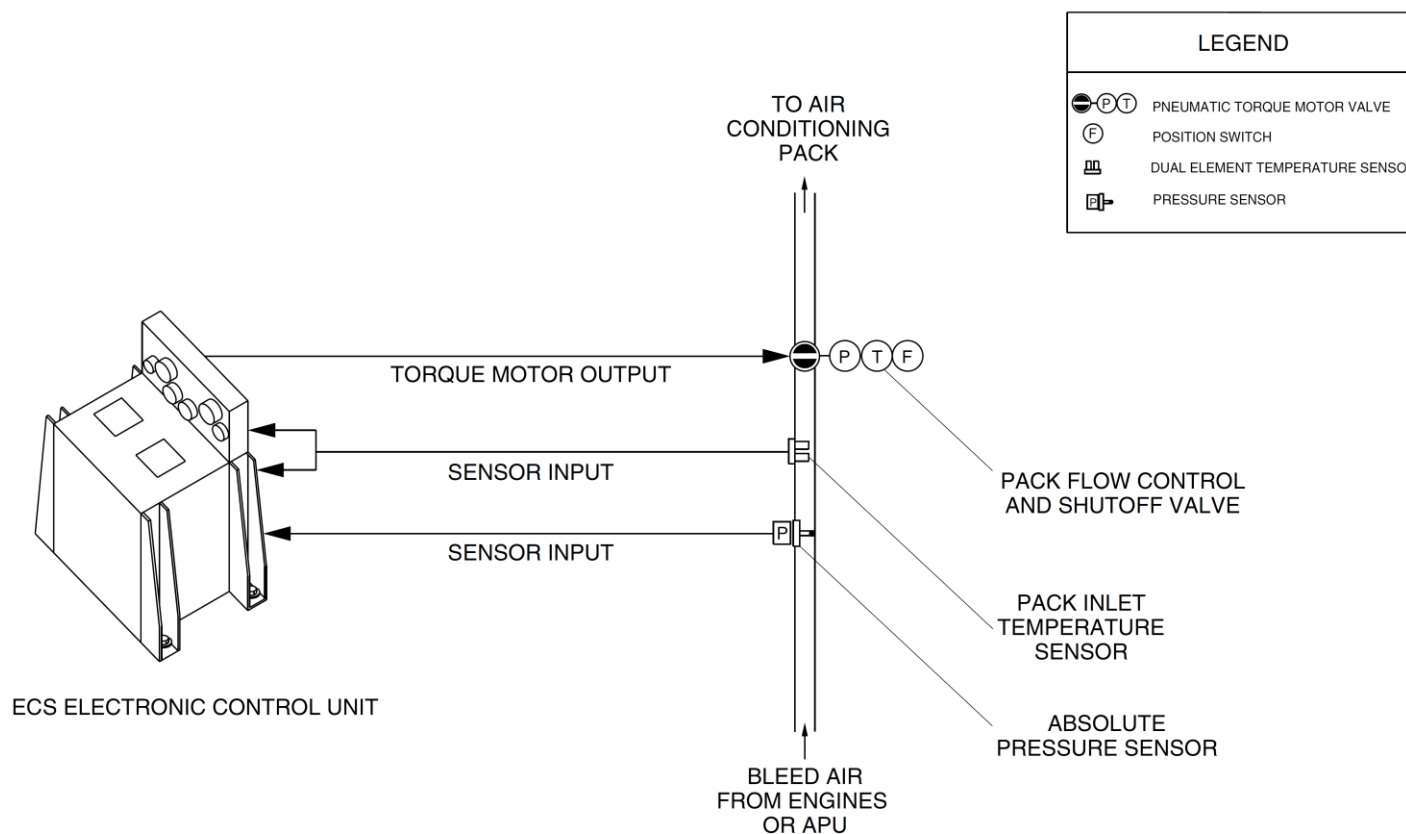
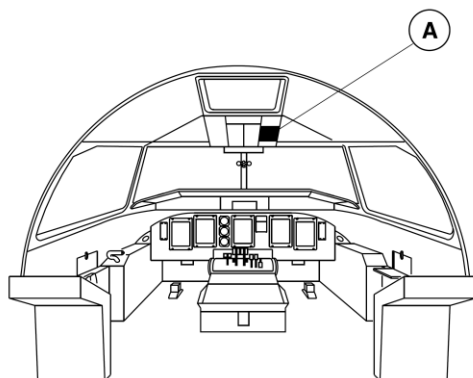


Figure 21- 18 Flow Control System

Temperature Control

The main functions of the temperature control and indication system are to:

- Supply temperature control of the Environmental Control System (ECS)
- Supply temperature indication to the flight crew.
- Detect malfunctions in the system and send fault codes to the fault indication system.



[Refer Figure 21-19](#)

Temperature control starts with selecting the desired temperature using the appropriate switches on the AIR CONDITIONING control panel in the overhead console in the flight compartment. The ECS Electronic Control Unit (ECU) monitors the temperature sensors and switches located throughout the aircraft. The ECU then operates the appropriate valves to maintain the selected temperature setting. The cabin temperature setting can also be controlled from the flight attendant panel.

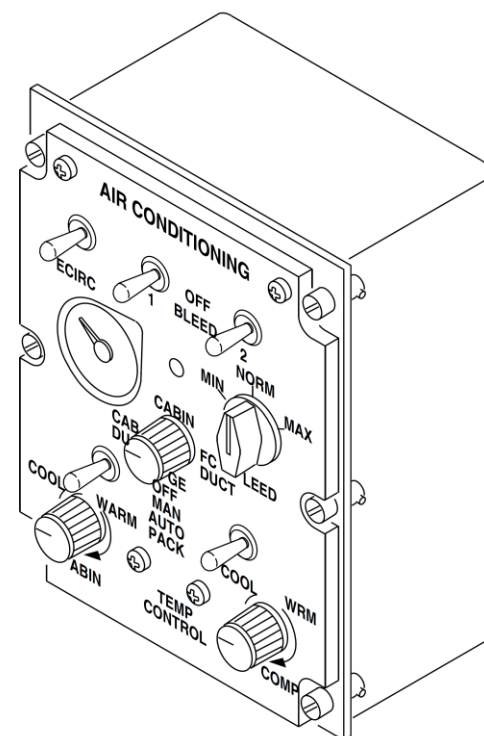
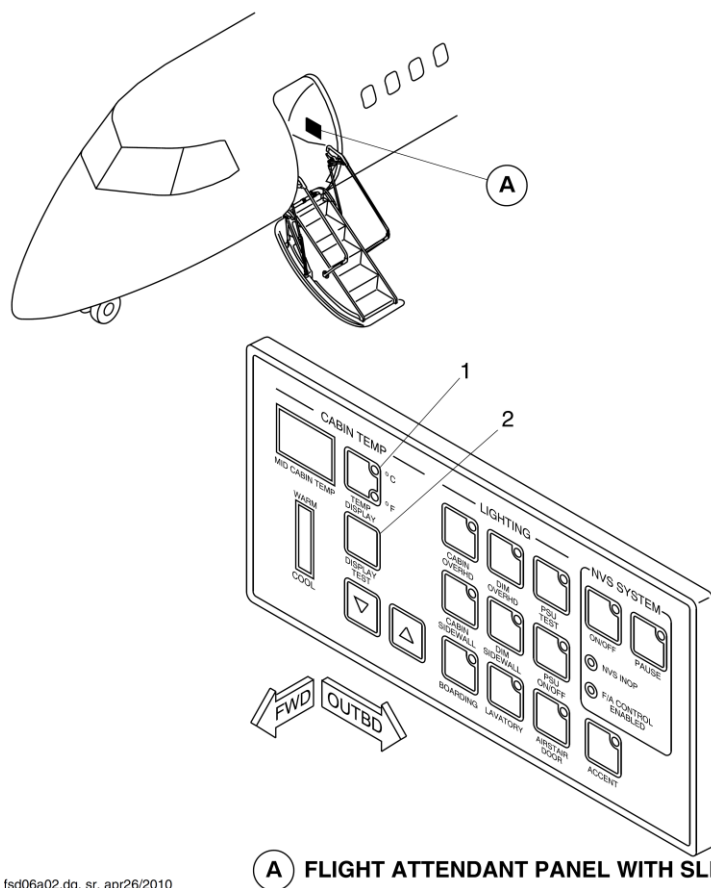


Figure 21- 19 Air Conditioning Control Panel

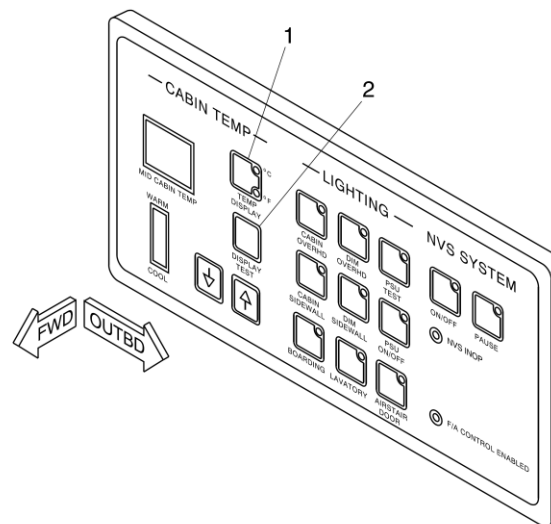
The ECU uses sensors and switches to monitor temperature, pressure, flow rates and valve positions. The ECU uses the pilot and crew commands to control the air conditioning system in manual or automatic mode, one or two ACM operation, bleed source selection and distribution of conditioned air to the flight and cabin compartments.

The ECU and the air conditioning system have redundant control systems for continued ECS operation with mechanical and electrical component malfunctions.

Caution lights indicate excessive temperatures in the Air Cycle Machine (ACM) compressor discharge, and in the cabin or flight compartment supply ducting.



A FLIGHT ATTENDANT PANEL WITH SLEEP MODE



A FLIGHT ATTENDANT PANEL WITHOUT SLEEP MODE

LEGEND

1. Temperature display pushbutton switch.
2. Display test pushbutton switch.

Figure 21- 20 Flight Attendant Control Panel

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Airconditioning PACK

Introduction

The air conditioning pack conditions the bleed air to the proper temperature and humidity and delivers it to the air distribution system for environmental control of the cabin and flight compartment.

General Description

The ECS Electronic Control Unit (ECSECU) monitors and controls the operation of the air conditioning packs. The bleed air is used from the engines or Auxiliary Power Unit (APU) to supply conditioned air to the cabin and flight compartment. The system responds to flight crew commands and compartment temperature controls to supply a comfortable temperature environment for passengers and crew.

Two air cycle machines (ACM), with one dual heat exchanger located in the aft fuselage, cool the bleed air coming from the two engines. This configuration provides the redundancy of two packs while allowing access to a much larger dual heat exchanger during operation with a single ACM.

The air conditioning pack has the components that follow:

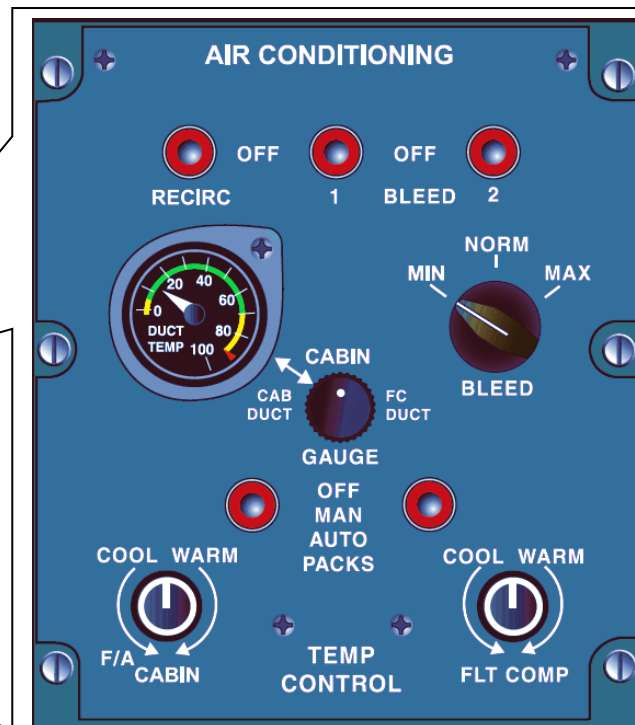
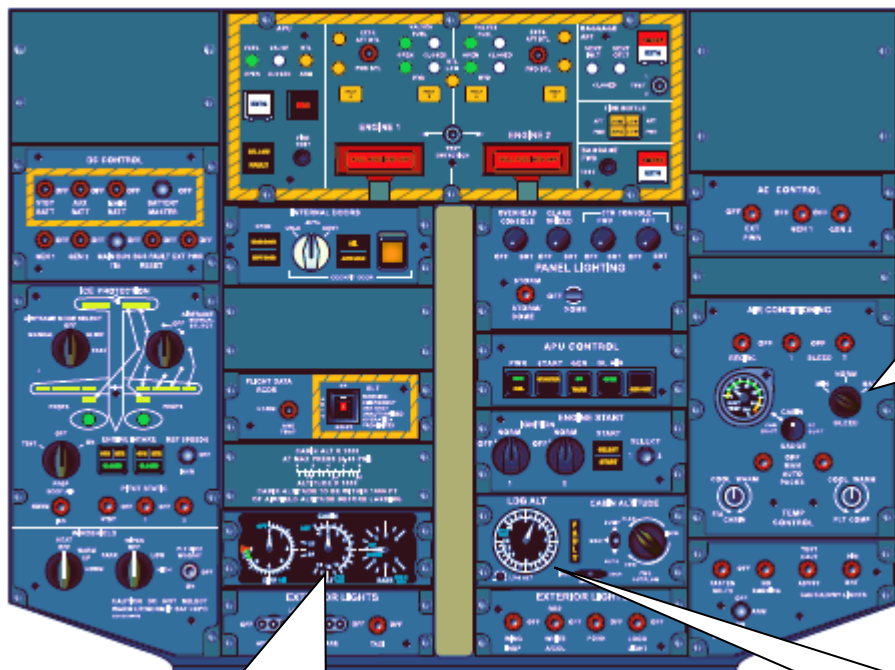
- Dual Heat Exchanger
- Condenser/Reheater/Mixer
- Spray Nozzles
- Collector, Water
- Machine, Air Cycle
- Housing, Fan Inlet Diffuser
- Valve, Pack Bypass
- Check Valve, Compressor Inlet
- Valve, Turbine Shutoff
- Check Valve, Fan Bypass
- Check Valve, Ram Outlet
- Check Valve, Ram Inlet
- Valve, Temperature Reduction Shutoff

- Ducts, Air Conditioning Pack
- Struts, Air Conditioning Pack
- Mounts, Air Conditioning Pack
- Switch, Valve – Temperature Reduction
- Switch, Compressor Overtemperature
- Sensor, Compressor Outlet Temperature
- Sensor, Outlet Temperature – Secondary Heat Exchanger

Air Conditioning System Control

Control panels for air conditioning and pressurization system are located on the overhead panel in the flight compartment. It includes:

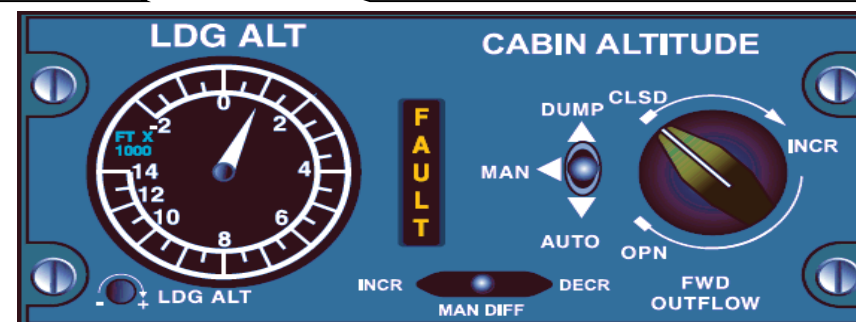
- Air Conditioning Control Panel
- Cabin Pressure Control Panel
- Cabin Indication Module



Air Conditioning Control Panel



Cabin Indication Module



Cabin Pressure Control Panel

Detailed Description

Operation

The air conditioning system is started by selecting BLEED 1 and/or BLEED 2 switches on the AIR CONDITIONING control panel, or the BL AIR switch light on the APU CONTROL panel. These switch settings determine the selection of the bleed air source, manual or automatic ECS operation and air flow rates for the flight and passenger compartments.

With the air conditioning PACKS switch set to AUTO, the bleed air comes from the left and right engines. With pilot and crew switch selections made, the ECS Electronic Control Unit (ECU) measures temperatures, pressures, and flow rates and uses this information to adjust valve positions in the ECS to get the selected conditioned air temperatures and flow rates.

The ECU controls the two nacelle shutoff valves to regulate the air flow to the air conditioning packs. The ECU sets these two valves to get an equal bleed air flow from each engine. Bleed air passes through the fully open pack flow control and shutoff valve to the air conditioning pack. The ECU receives the pressure and temperature data of the bleed air from the pack inlet absolute pressure sensor and the pack inlet temperature sensor. The ECU uses this data to control the flow of bleed air through the pack flow control and shutoff valve during single engine operation. The ECU also uses this data to control bleed air flow rate when APU bleed air is selected.

As soon as the bleed air flows into the air conditioning pack, it passes through the primary part of the dual heat exchanger. The primary heat exchanger uses ram air to remove some of the heat from the bleed air. The primary – cooled bleed air then divides into two air flow paths which pass through two compressor inlet check valves (one check valve for each air flow path) to the compressors of both Air Cycle Machines (ACMs). Whenever the air conditioning system is on, both air cycle machines are operating.

The ACM compressors increase the temperature and pressure of the bleed air. The two compressor outlet flows combine into one flow to pass through

the secondary part of the dual heat exchanger. Two compressor outlet temperature sensors (one for each compressor outlet) measure the compressor outlet temperatures. The secondary heat exchanger also uses cooler ram air to remove heat from the compressed bleed air. A temperature sensor measures the cooled bleed air as it leaves the secondary heat exchanger.

The cooled bleed air from the secondary heat exchanger passes through the reheater and then to the condenser. The condenser, mixer, and reheater are all part of the Condenser/Mixer/Reheater unit.

The condenser is a heat exchanger which uses the cooler air from the ACM turbines to remove enough heat from the bleed air. This decrease in temperature causes water vapor in the bleed air to condense. The bleed air with condensed water vapor then passes to the water collector. The water collector uses centrifugal action (no moving parts) to remove the condensed water from the bleed air flow. The water in the collector then passes through a water line to a spray nozzle in the secondary heat exchanger. The high pressure of the bleed air forces the water out of the nozzle as a spray. The spray nozzle has two functions:

- Evaporates the water sprayed into the air flow to increase the efficiency of the secondary heat exchanger
- Discards the water into the ram air flow which gets exhausted out of the aircraft.

Dried bleed air from the water collector then passes through the reheater (another heat exchanger). The reheating of the bleed air vaporizes any condensation that gets past the water collector.

Bleed air from the reheater divides into two streams. Each air stream passes through a fully open turbine shutoff valve to an ACM expansion turbine. The bleed air expands as it passes through the turbine.

The expanding air spins the turbine to supply the mechanical power to rotate the ACM compressor and fan. The temperature of the expanding bleed air greatly decreases to become the conditioned (cooled) air for the flight and cabin compartments. The volume of the expanding air increases and the pressure decreases to approximately 1 atmosphere (14.7 psi (101.35 kPa)).

The expanded air from both turbines enters the mixer which combines it with recirculated air from the cabin. This mixed air enters the condenser to cool the bleed air from the secondary heat exchanger and condense water vapor. The mixed air then passes to the cabin and flight compartment distribution system.

The ECU controls two pack bypass valves to regulate the temperature of the conditioned air sent to the cabin and flight compartment. These valves pass a portion of the warmer and higher – pressure bleed air (taken before the dual heat exchanger) to the outputs of each ACM expansion turbine. Opening the pack bypass valve increases the flow of warm bleed air into the turbine outlet. This raises the temperature of the air supplied to the flight and cabin compartment. Closing the pack bypass valve decreases the flow of warm bleed air into the turbine outlet. This decreases the temperature of the air supplied to the flight and cabin compartments. Bleed air, bypassed by the pack bypass valves, raises the humidity of the very dry conditioned air sent to the flight and cabin compartments.

Each ACM turbine drives a fan which keeps the ram air flow through the dual heat exchanger at acceptable levels. This becomes necessary when the aircraft's speed cannot generate enough ram air flow through the dual heat exchanger (such as when the aircraft is on the ground). When the aircraft is airborne, the ACM fans can limit the flow of ram air. As the flow of ram air increases, the ram air pressure on the fans increases. This increasing pressure opens two bypass valves in the ram air outlet ducts to permit the maximum flow of ram air.

The emergency ram air system supplies unconditioned air to the flight and cabin compartments in the event of a failure of the air conditioning packs. If there is no air flow from the air conditioning packs, the pressure from the ram air is enough to overcome the spring in the emergency ram check valve. The check valve will open allowing cooling ram air to flow directly into the cabin and flight compartment supply ducts.

A series of ducts carry the emergency ram air to the flight and cabin compartments. The emergency ram air ducting starts at the aft side of the ram air inlet duct. The air is ducted down and then forward between the two ACM's to the front of the air conditioning pack. The ram air ducting then joins the conditioned air supply duct at a tee connection just upstream of the splitter duct. Thus the emergency ram air is distributed to the cabin and the flight compartments.

During ground operations when there is no ram effect, the airflow through the NACA is insufficient for cooling. The Air Cycle Machine (ACM) fans continue to draw in air through the ram air inlet creating a negative pressure in the ram inlet. This negative pressure opens a rectangular-shaped ram inlet check valve to let in additional cooling air.

A recirculation fan takes air from the cabin to mix with the conditioned air from the ACM turbines. A RECIRC switch turns the recirculation fan on or off. Operating conditions determine the automatic control of the recirculation fan speed.

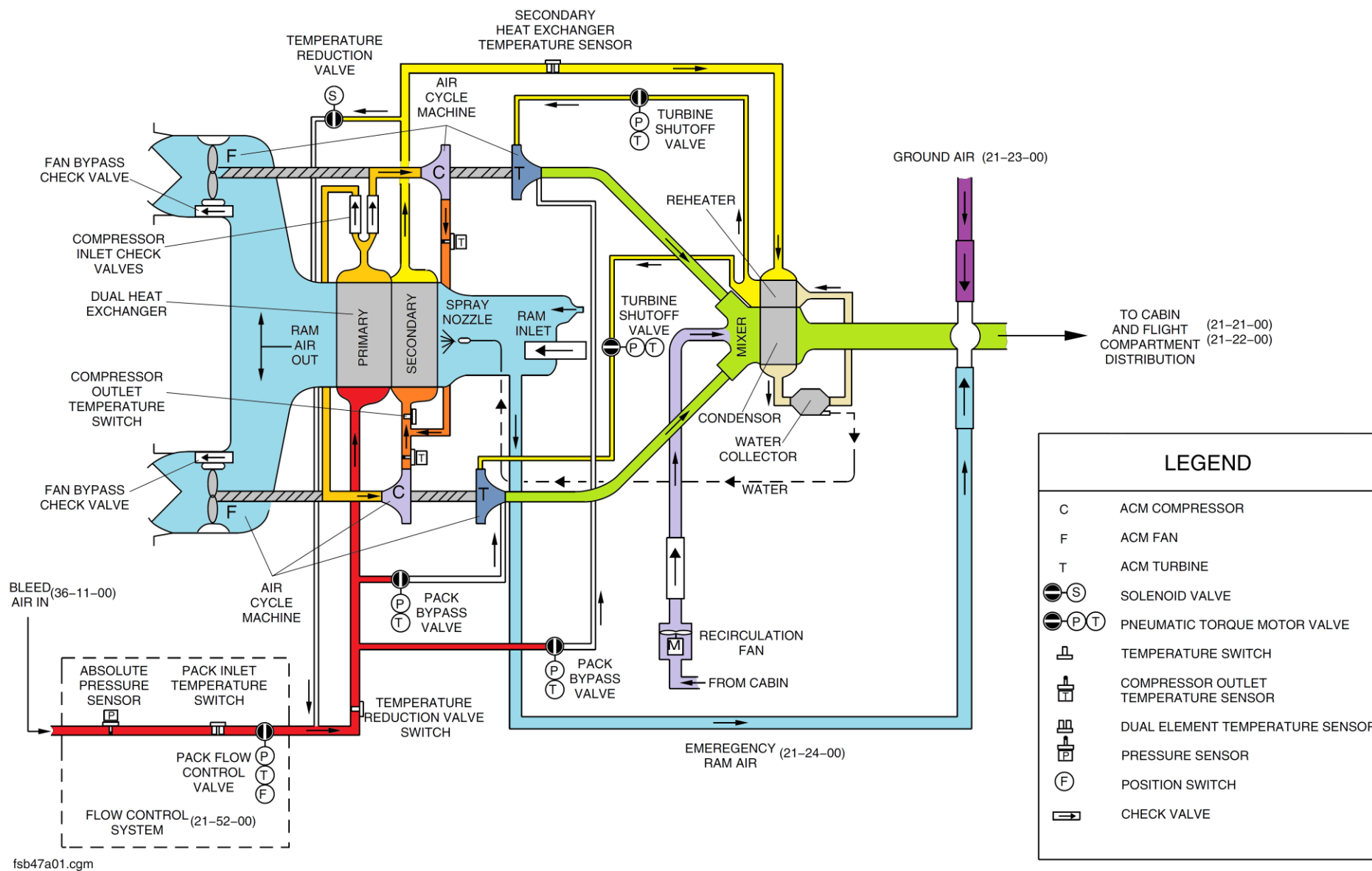


Figure 21- 21 Air Conditioning Schematic

Emergency Ram Air Operation

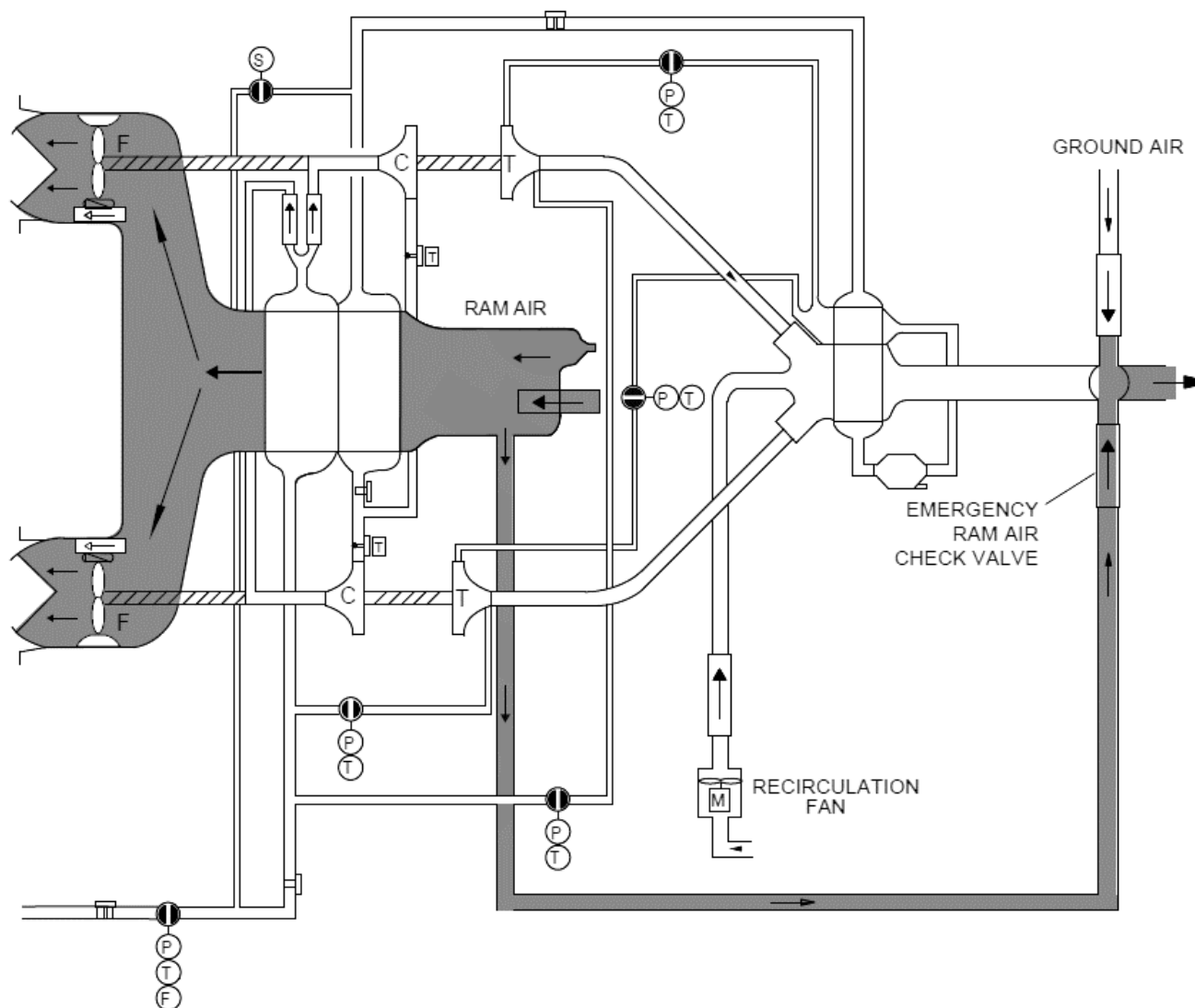


Figure 21- 22 Emergency Ram Air System

Component Description

Heat Exchanger

[Refer Figure 21-23](#)

The function of the dual heat exchanger is to use ram air to cool the engine bleed air to levels compatible with the ACM and system performance requirements

The dual heat exchanger is an aluminum plate and fin, primary crossflow/secondary counterflow heat exchanger. It provides primary cooling of the pack inlet air and secondary cooling of the air from the ACM compressors.

The dual heat exchanger is mounted at the top of the air conditioning pack between the ram air inlet duct and fan inlet diffuser housing.

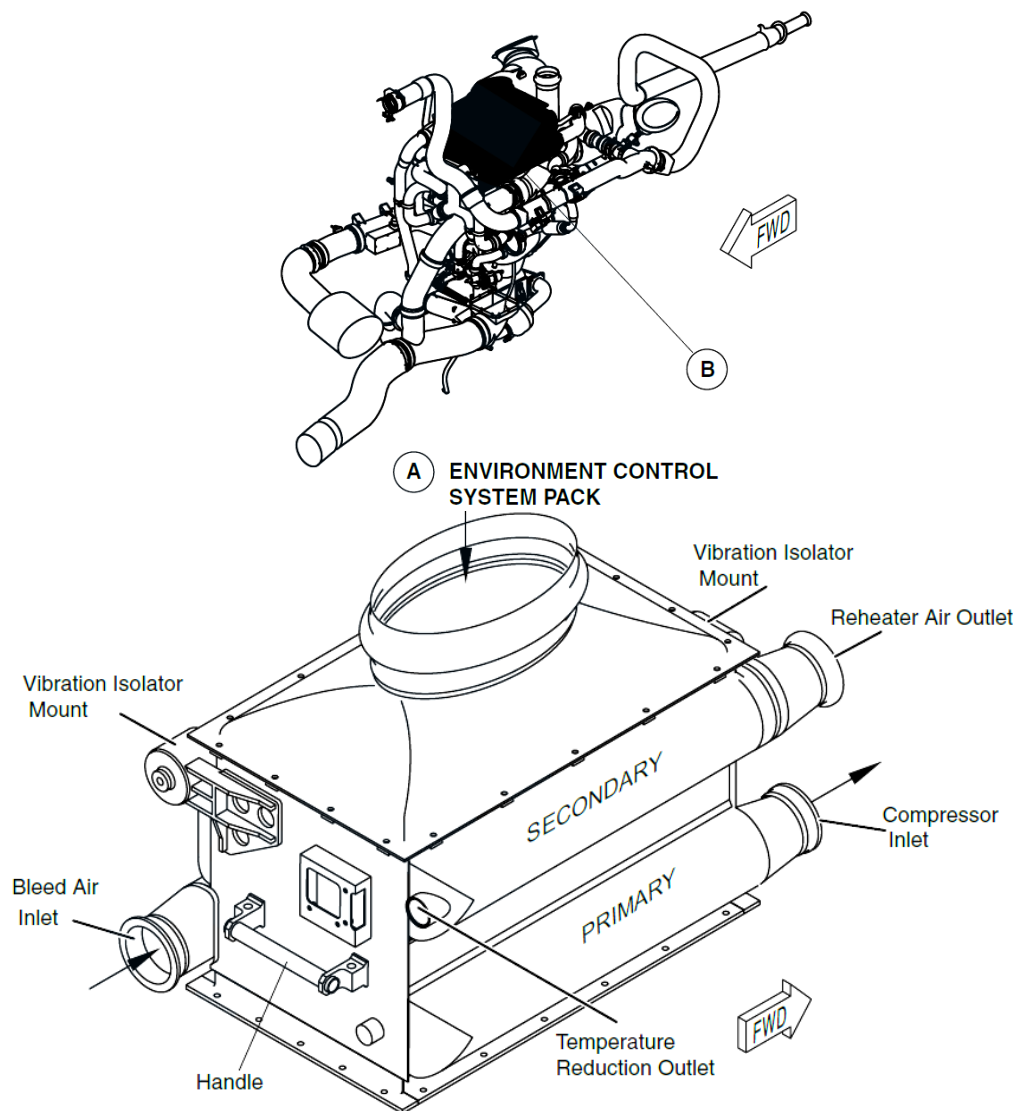
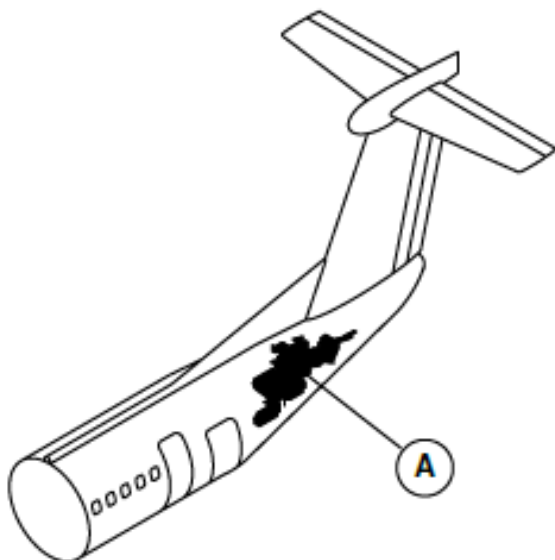


Figure 21- 23 Dual Heat Exchanger

Condenser/Reheater/ Mixer

[Refer Figure 21-24](#)

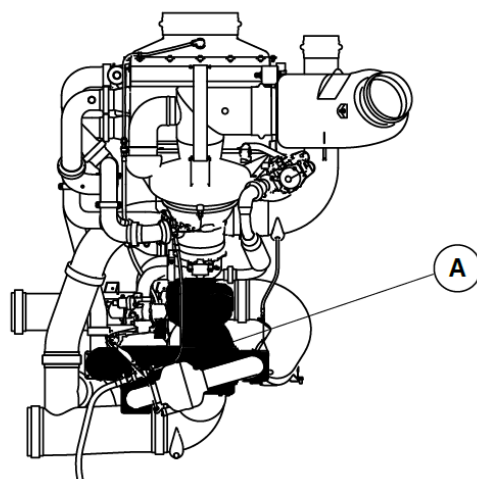
The condenser/reheater/mixer is a crossflow, plate and fin, aluminum heat exchanger. It has three air circuits:

The condenser circuit receives warm, moist (ACM compressor) air from the secondary heat exchanger and the cool air from the ACM turbines. The air in this circuit is cool enough to condense the water vapour from the surrounding air. The condensed water then goes to the water collector.

The reheater circuit receives the air after it has passed through the water collector. This air is reheated before it is delivered to the ACM turbine inlet.

The mixer circuit combines the air from the two turbine exit ducts and recirculated air from the passenger compartment.

The condenser/reheater/mixer is located at the bottom of the air conditioning pack.



VIEW LOOKING STARBOARD

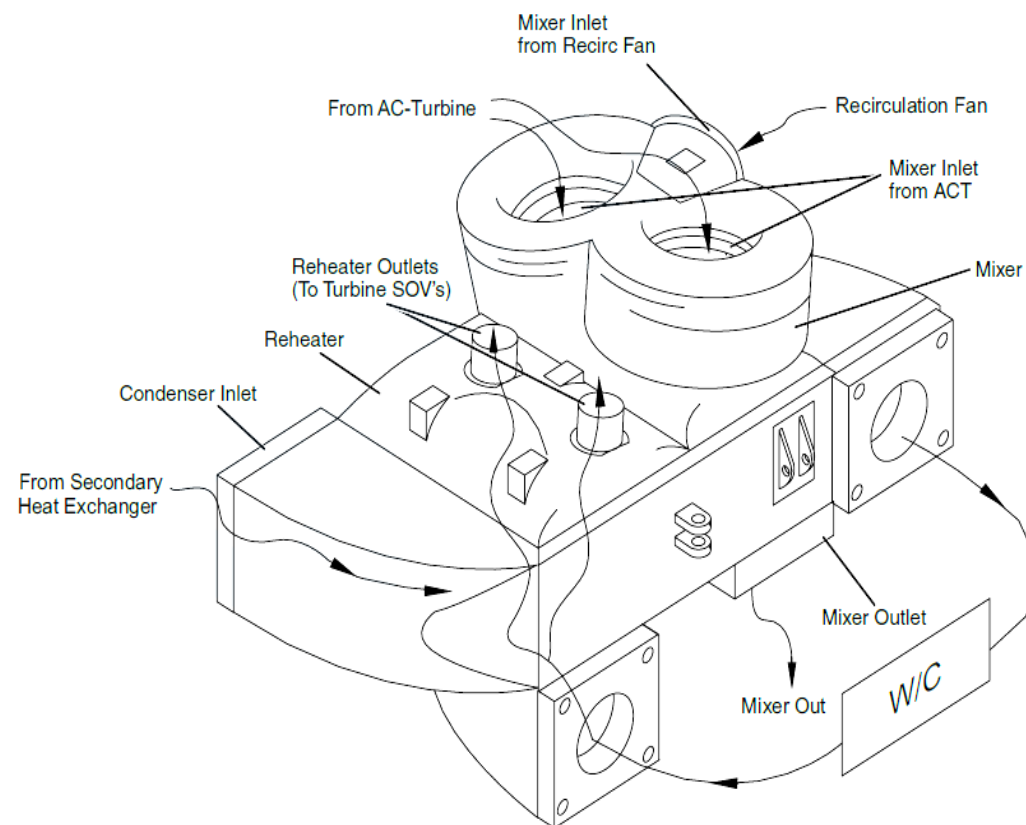
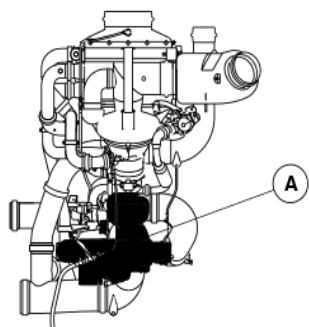


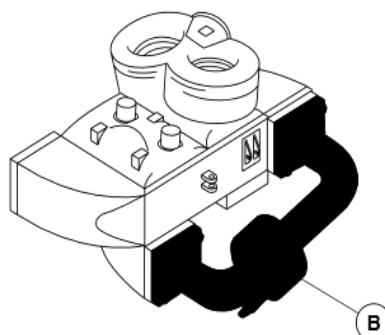
Figure 21- 24 Condenser/Reheater/Mixer

Water Collector

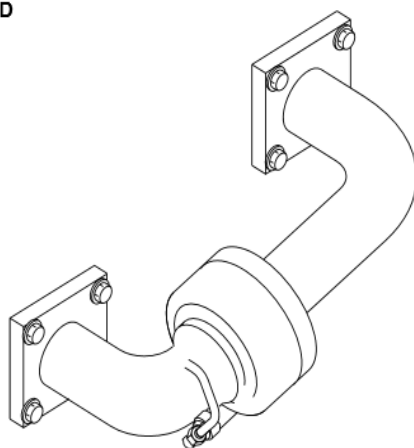
The water collector is installed on the left side of the condenser and removes water from the bleed air. It has swirl vanes which use centrifugal action to push the condensed water to its inside walls. The water goes out through a tube to the spray nozzle, for spray against the ram inlet face of the dual heat exchanger. A small overflow provides an outlet for excess water.



VIEW LOOKING STARBOARD

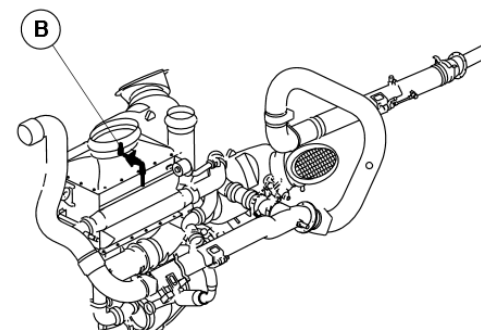


A CONDENSER/REHEATER/MIXER

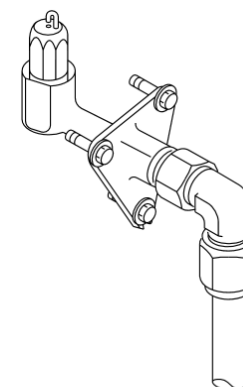


Spray Nozzle

The spray nozzle is installed on the top left side of the ram air header duct. The nozzles spray water against the ram air inlet face of the dual heat exchanger. This water evaporates and improves the efficiency of the heat exchanger. Removal of the water from the system is important for extending the life of the air cycle machine. If the water cannot be removed from the air before it enters the ACM turbines, it causes turbine nozzle erosion, which degrades ACM performance and decreases its reliability.



A AIR CONDITIONING PACK



B

Air Cycle Machine

[Refer Figure 21-25](#)

The air cycle machine (ACM) is a 3 wheeled, air bearing device that provides cooling of the warm bleed air. There are three rotors in each ACM, namely:

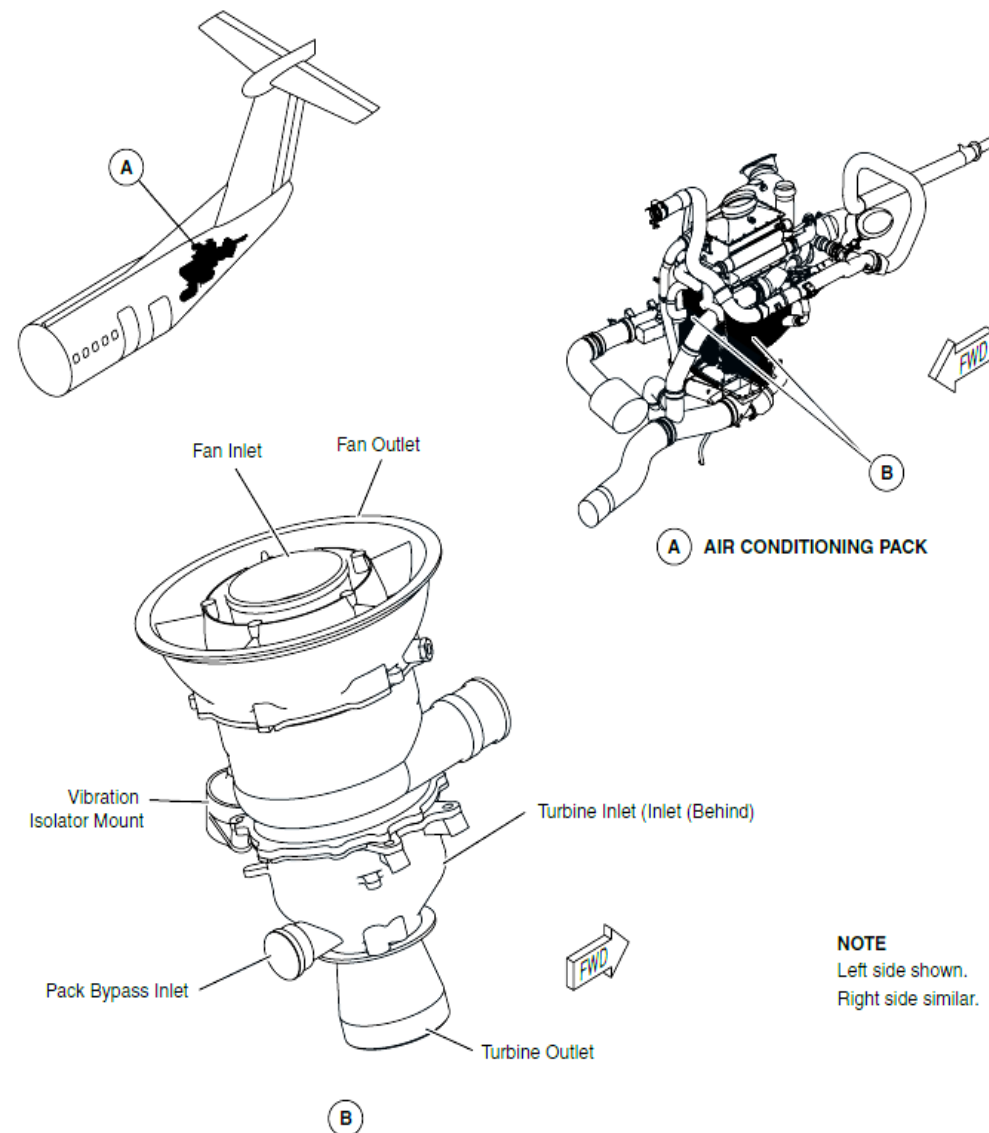
- Compressor
- Turbine
- Fan

The rotors are held together with interference fits on shaft segments so the three rotors turn together. The ACM compresses the air (compressor section) and then expands it (turbine section). The expansion of the air causes a decrease in temperature. The fan section draws cooling air through the dual heat exchanger during ground operations.

There are two air cycle machines which are located below the fan inlet diffuser housing.



Figure 21- 25 Air Cycle Machines



Fan Inlet Diffuser Housing

[Refer Figure 21-26](#)

The fan inlet diffuser housing is located in the air conditioning pack between the dual heat exchanger outlet and the ACM fans. The housing has a common duct for cooling airflow from the dual heat exchanger to the ACM fans and ram air exhaust outlets. It collects cooling air from the dual heat exchanger outlet and delivers it to the fans. It also collects the air from the fan outlets and delivers it to the ram air exhaust outlets which pass the air overboard.

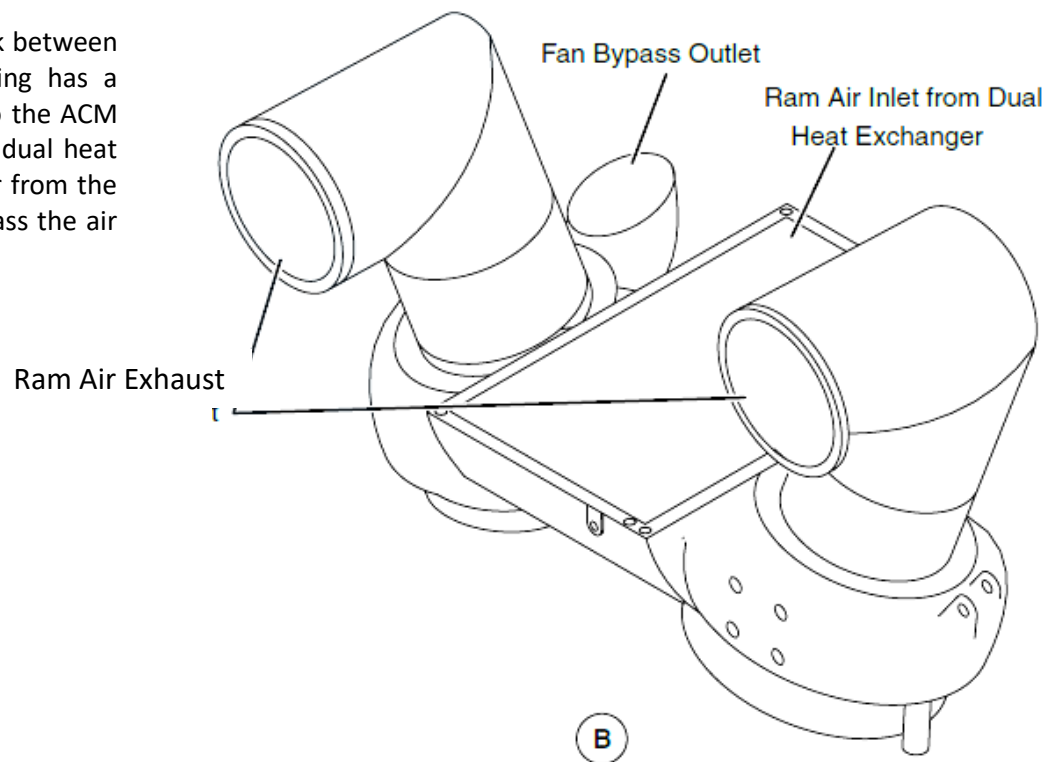
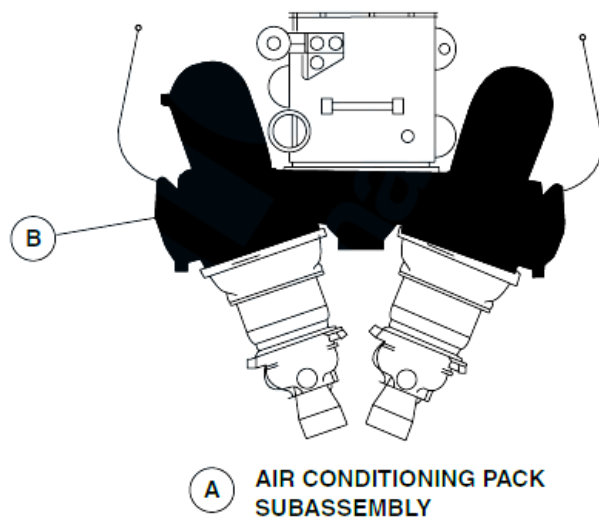


Figure 21- 26 Fan Inlet Diffuser

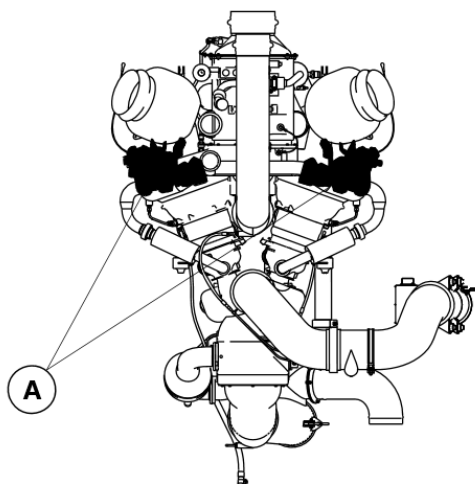
Pack Bypass Valves

[Refer Figure 21-27](#)

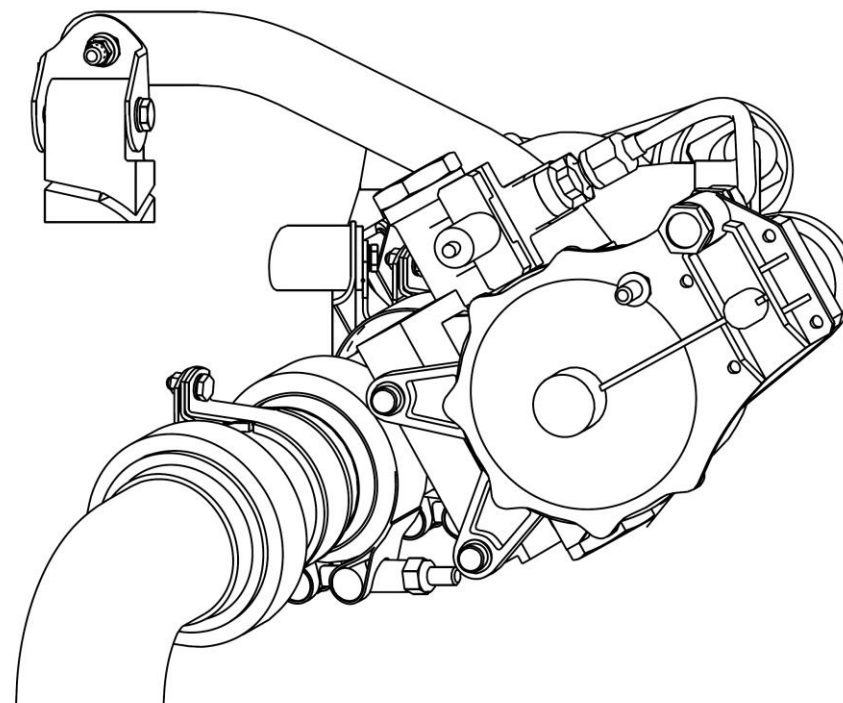
The pack bypass valve is an electrically controlled, pneumatic valve, which controls the pack discharge temperature for all operating conditions. The valve opens to let warm bleed air go to the ACM turbine outlet when warmer air is needed. The valve consists of:

- Torque motor
- Pneumatic actuator
- Butterfly valve

The valve is fitted with a metal wire mesh filter in the servo air line, to prevent contamination from entering the torque motor and pneumatic actuator. The valve also has a locking pin that, when in the locked position, bleeds servo air to ambient and mechanically locks the valve in the closed position. The pack (ACM) bypass valves are located between the bleed air inlet and the ACM turbine outlets. The valve is used to regulate the temperature of conditioned air sent to the cabin and flight compartments.



VIEW LOOKING FORWARD



A

Figure 21- 27 Pack Bypass Valve

Turbine Shut-off Valve

[Refer Figure 21-28](#)

The turbine shutoff valve is an electrically controlled, pneumatic valve that opens to allow air into the ACM turbine. The valve consists of:

- Torque motor
- Pneumatic actuator
- Position switch
- Butterfly valve

The valve has a metal wire mesh filter in the servo air-line to prevent contamination from entering the torque motor and pneumatic actuator. The valve also has a locking pin that, when in the locked position, bleeds servo air to ambient and mechanically locks the valve in the closed position. The turbine shutoff valves are located between the condenser/reheater/mixer and the ACM turbine inlets.

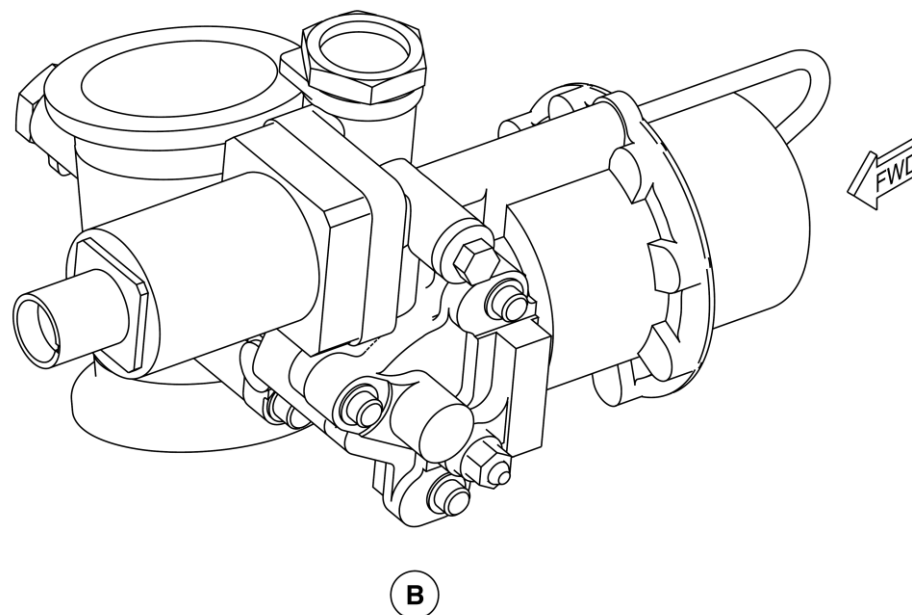
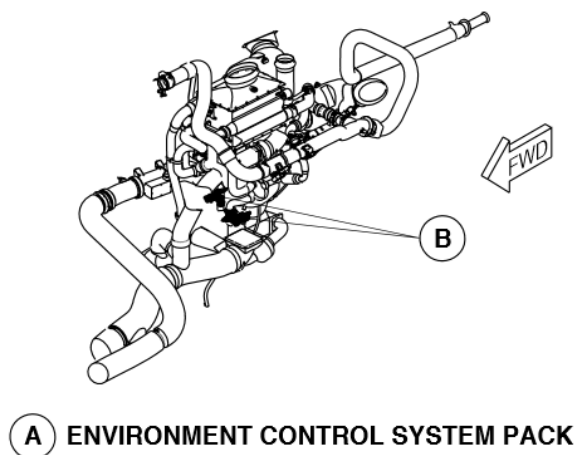


Figure 21- 28 Turbine Shut-off Valve



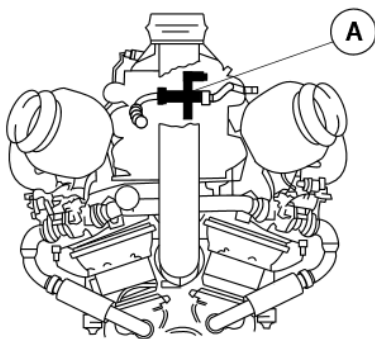
A ENVIRONMENT CONTROL SYSTEM PACK

Temperature Reduction Shutoff Valve

[Refer Figure 21-29](#)

The temperature reduction shutoff valve is a pneumatically actuated, solenoid controlled gate type valve that opens to recirculate secondary heat exchanger air, back into the primary heat exchanger inlet. This is done when the temperature at the primary heat exchanger inlet is too hot.

The temperature reduction shutoff valve is installed on the aft side of the heat exchanger.



A VIEW LOOKING FORWARD

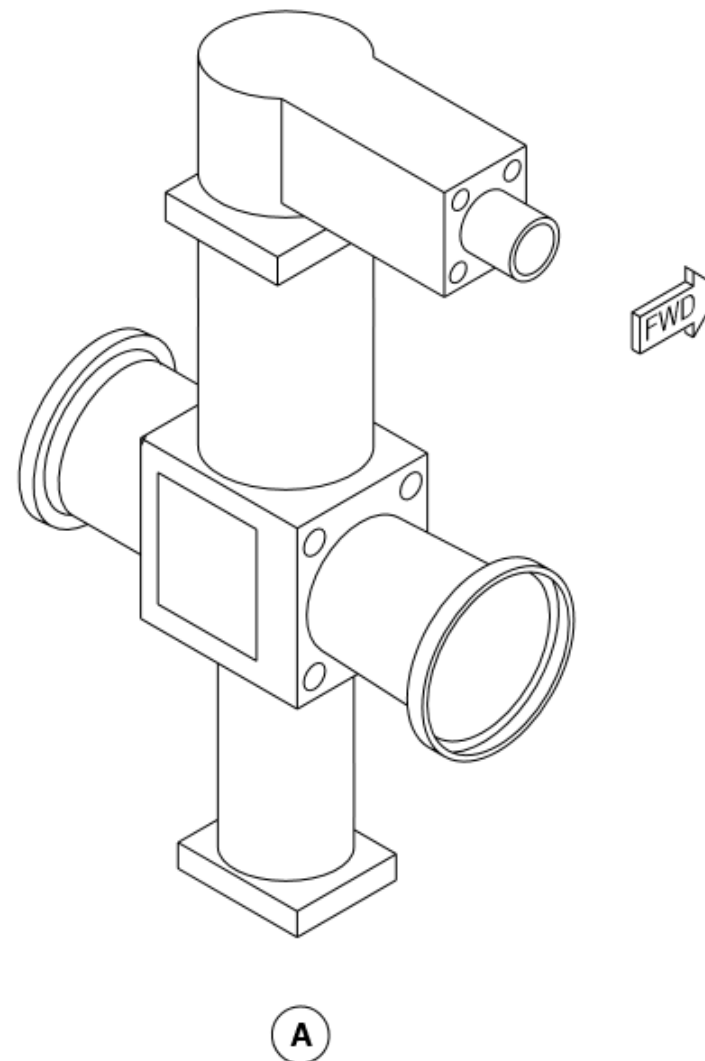


Figure 21- 29 Temperature Reduction Shutoff Valve

Temperature Reduction Valve Switch

[Refer Figure 21-30](#)

The temperature reduction valve switch, installed on the bleed air inlet ducting on the aft side of the air conditioning pack, determines if the temperature reduction valve has failed to open.

The temperature reduction valve switch is a thermal switch which trips if the pack inlet temperature exceeds 560 °F (293 °C). This action signals the ECU to generate a fault code to the CDS for a temperature reduction valve failure.

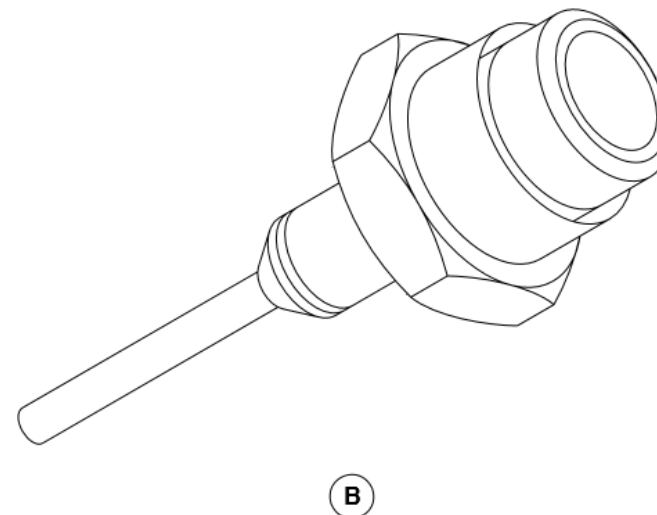
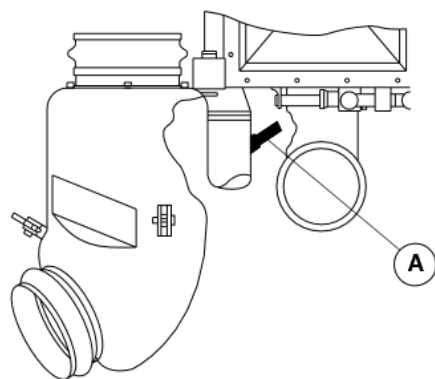
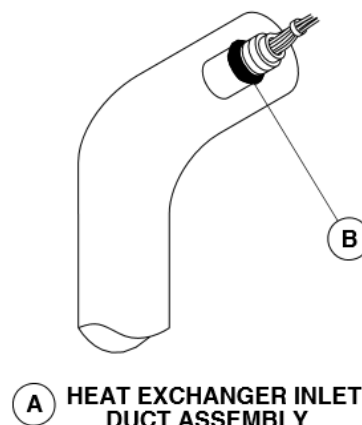


Figure 21- 30 Temperature Reduction Valve Switch



VIEW LOOKING DOWN



**(A) HEAT EXCHANGER INLET
DUCT ASSEMBLY**

Compressor Outlet Temperature Sensors

[Refer Figure 21-31](#)

The compressor outlet temperature sensors are located in ACM compressor outlet ducting. They monitor the temperature of the air leaving the ACM compressors. The temperature signal is used to reduce air flow through the air conditioning pack at 380 °F (193.33 °C) and to close the pack flow control and shutoff valve at 415 °F (212.78 °C).

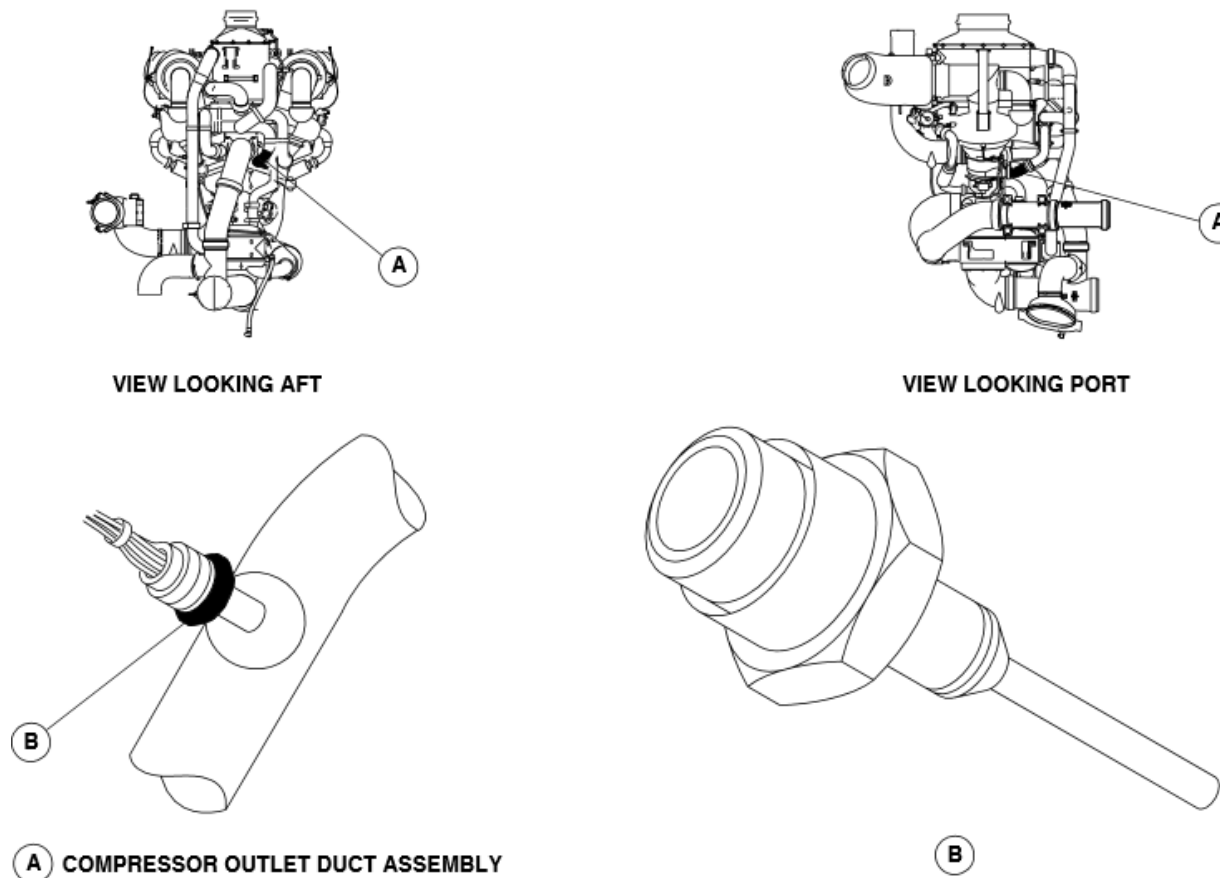
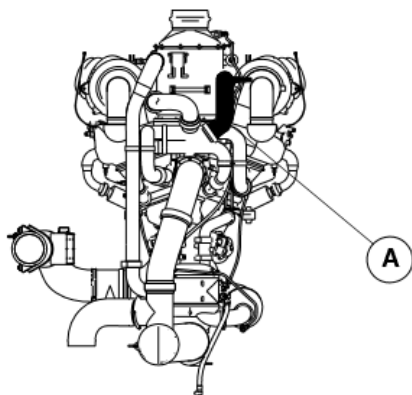


Figure 21- 31 Compressor Outlet Temperature Switch

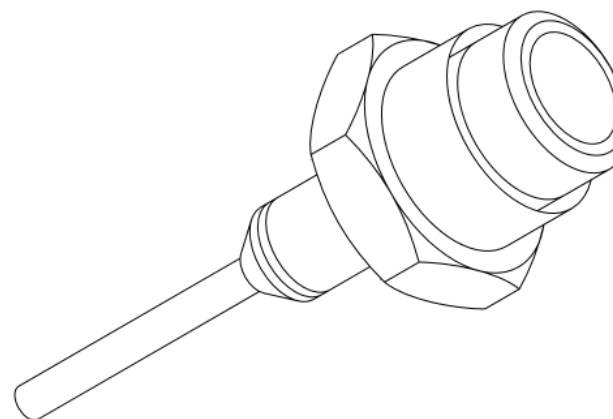
Compressor Over Temperature Switch

[Refer Figure 21-32](#)

The compressor over temperature switch is mounted in the secondary heat exchanger inlet duct and monitors air flow temperature entering the secondary heat exchanger. It is a thermal switch which trips if the compressor outlet temperature exceeds 415 °F (212 °C).

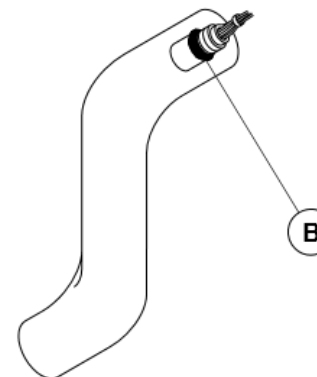


VIEW LOOKING AFT



B

Figure 21- 32 Compressor Over-Temperature Switch

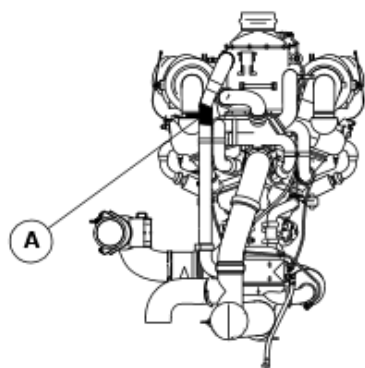


A COMPRESSOR OUTLET DUCT ASSEMBLY

Secondary Heat Exchanger Outlet Temperature Sensor

[Refer Figure 21-33](#)

The secondary heat exchanger outlet temperature sensor is installed in the secondary heat exchanger outlet duct and senses the temperature of the air leaving the heat exchanger. This signal is used only for Built-in-Test (BIT). The ECU compares the signal from this switch, to the signal from the pack inlet temperature sensor, to determine whether the dual heat exchanger is operating efficiently or not.



VIEW LOOKING AFT

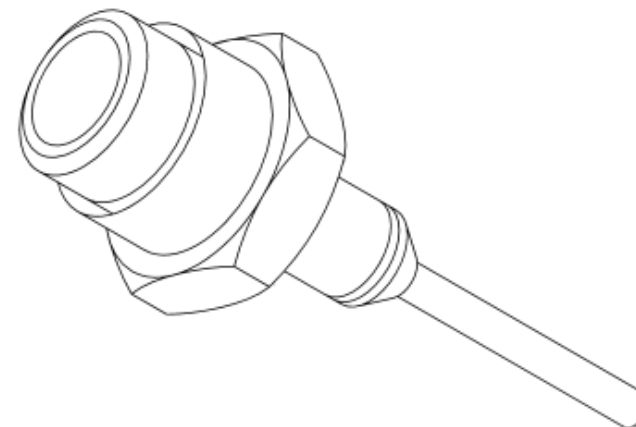
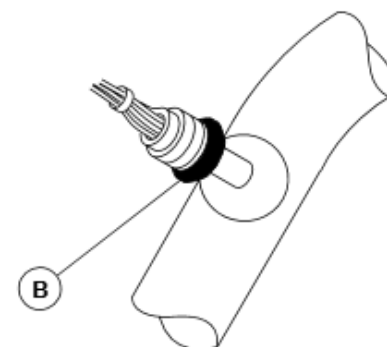


Figure 21- 33 Secondary Heat Exchanger Outlet Temperature Switch

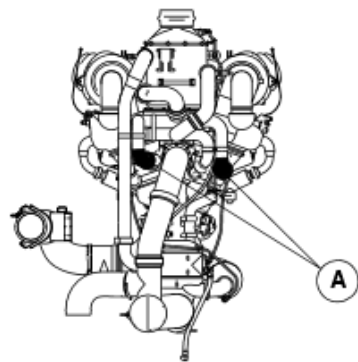


(A) SECONDARY HEAT EXCHANGER
OUTLET DUCT ASSEMBLY

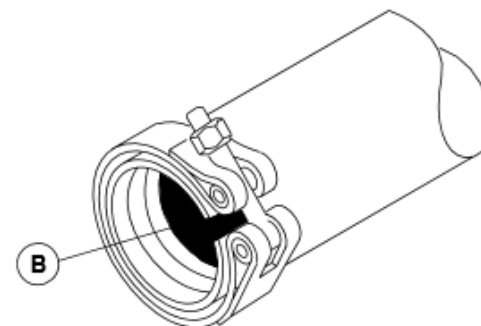
Compressor Inlet Check Valve

[Refer Figure 21-34](#)

The compressor inlet check valve is a spring loaded, dual flapper valve that prevents back flow of air into the primary heat exchanger.



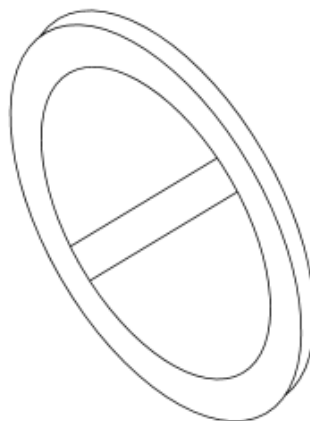
VIEW LOOKING AFT



A COMPRESSOR INLET DUCT ASSEMBLY

NOTE

Left component shown.
Right component similar.



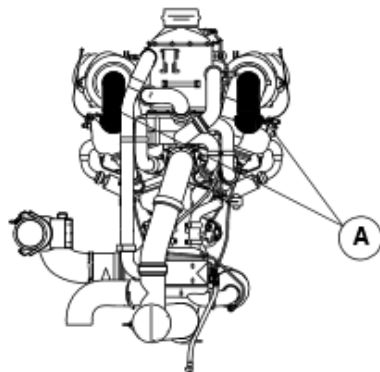
B

Figure 21- 34 Compressor Inlet Check Valve

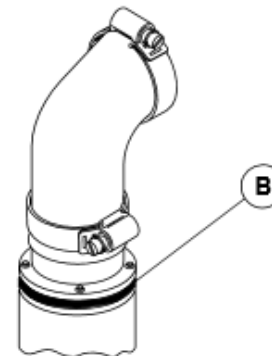
Fan Bypass Check Valve

[Refer Figure 21-35](#)

The fan bypass check valve is a spring – loaded, dual – flapper device that allows ram air to flow overboard in parallel with the ACM fan to supply the necessary ram flow during flight.



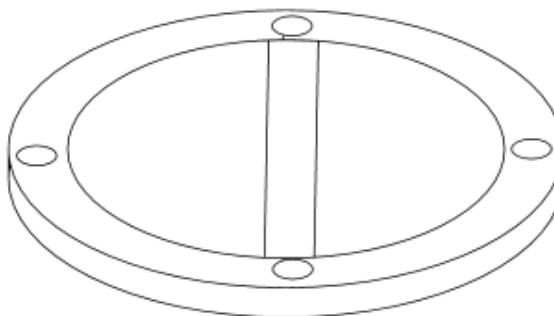
VIEW LOOKING AFT



A FAN BYPASS DUCT ASSEMBLY

NOTE

Left component shown.
Right component similar.



B

Figure 21- 35 Fan Bypass Check Valve

Ram Inlet Check Valve

[Refer Figure 21-36](#)

The ram inlet check valve is a rectangular, aluminum, bolted flange, spring loaded, dual flapper valve. It is visible and accessible from inside the ECS compartment in the tail of the aircraft. The valve allows additional ram air to flow through the dual heat exchanger during ground operation.

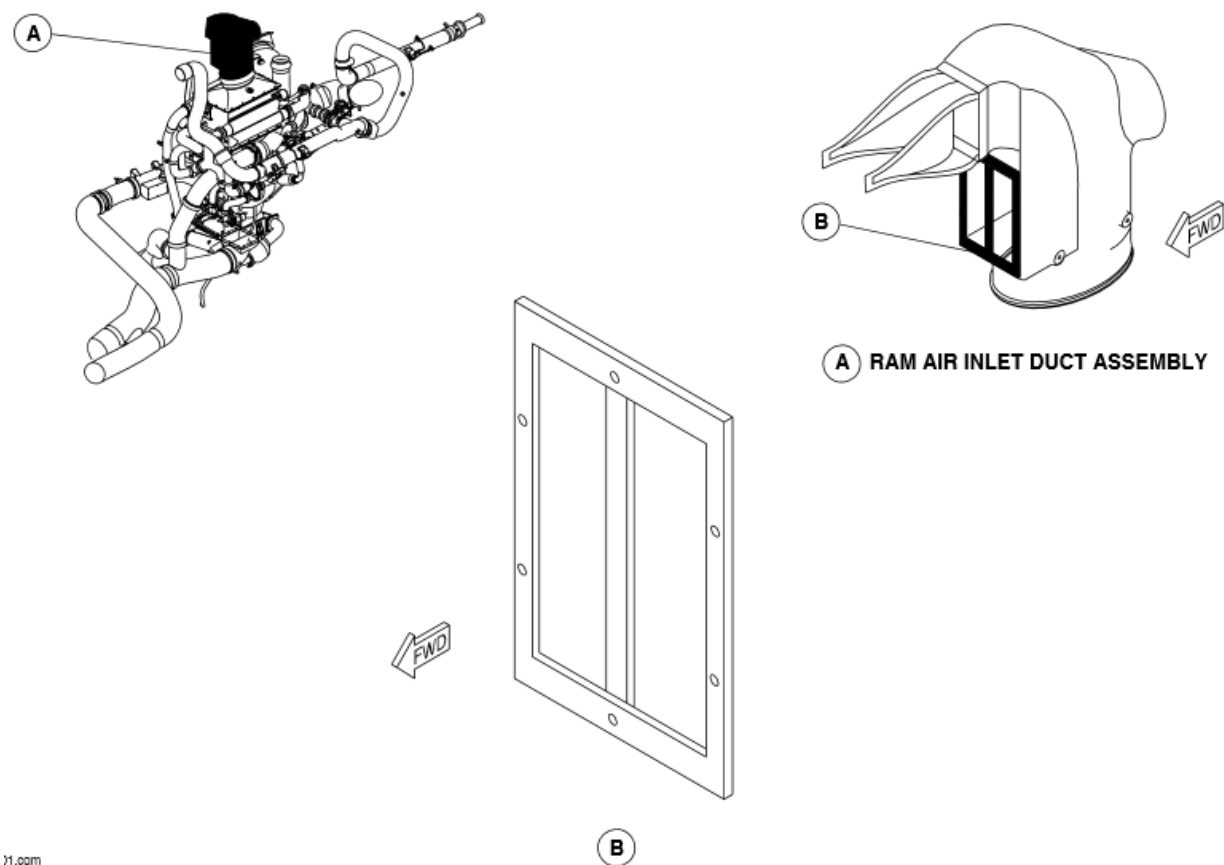


Figure 21- 36 Ram Inlet Check Valve

Ram Outlet Check Valve

[Refer Figure 21-37](#)

The ram outlet check valve is a spring loaded, dual flapper device that keeps debris from entering fan exhaust ducts during pack shutdown operation.

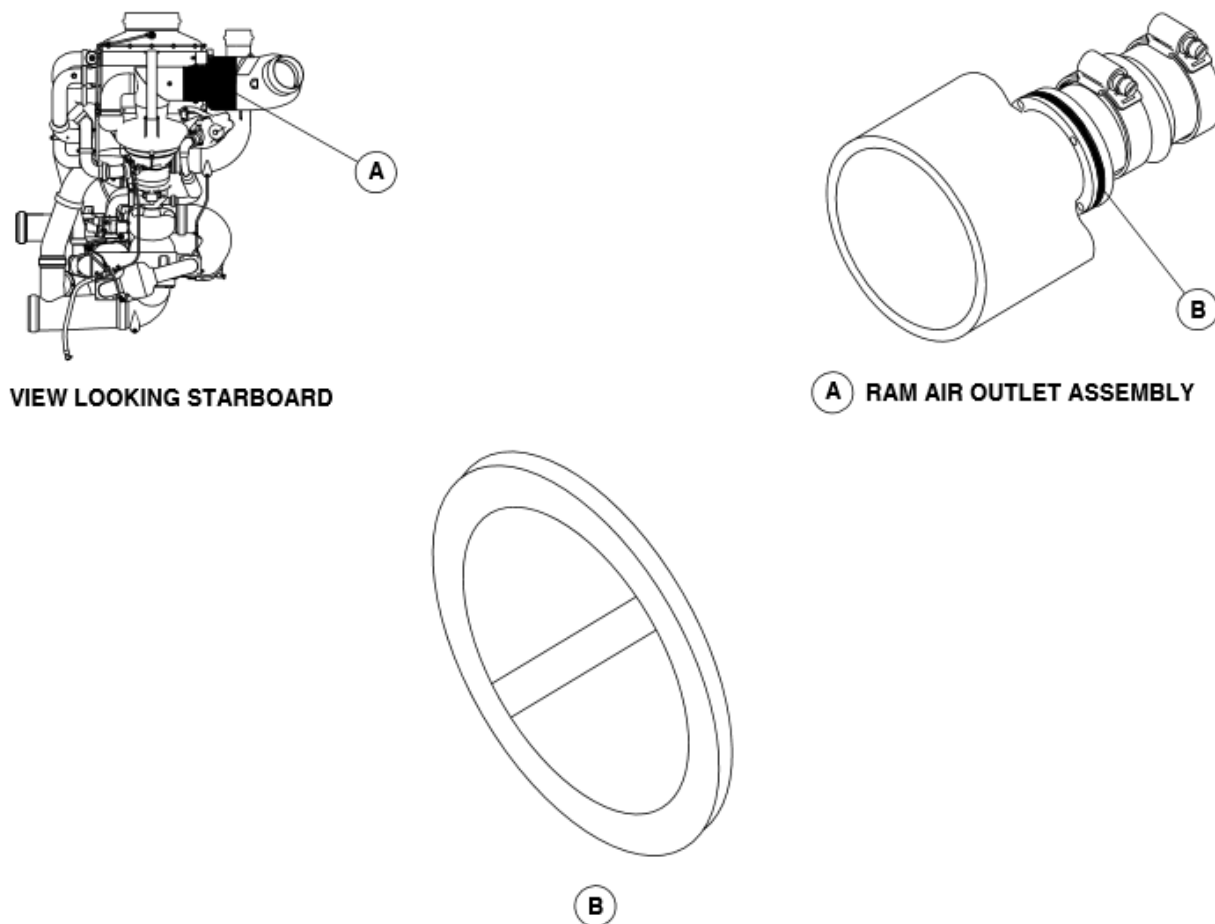
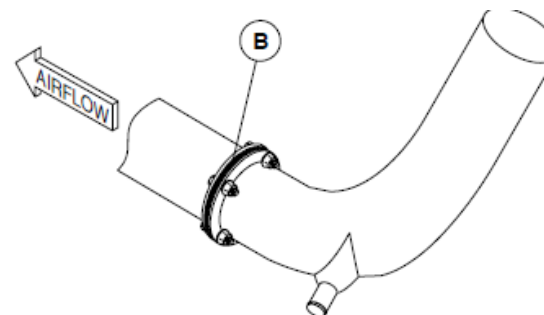
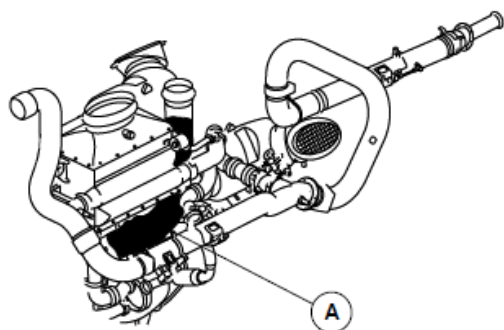


Figure 21- 37 Ram Outlet Check Valve

Emergency Ram Air Check Valve

Refer Figure 21-38

The emergency ram air check valve is a circular, 5 in. (127 mm) diameter, dual flapper valve. It is spring-loaded to the closed position. The valve is installed between the upper and mid emergency ram air ducts.



A EMERGENCY RAM AIR DUCT

NOTE

Parts removed for clarity.

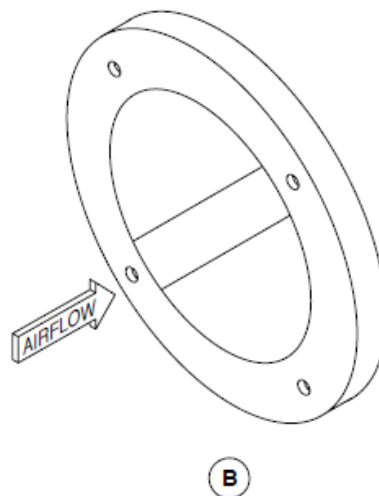
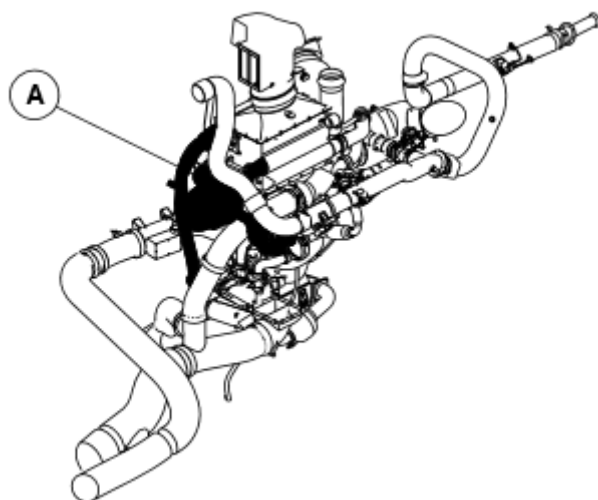


Figure 21- 38 Emergency Ram Air Check Valve

Air Conditioning Packs Ducts

[Refer Figure 21-39](#)

After the bleed air flows through the primary heat exchanger of the air conditioning pack, it divides into two separate airflows. One airflow is ducted through a check valve to the right ACM compressor inlet. Similarly, the other airflow is ducted through a check valve to the left ACM compressor inlet. The two compressor outlet flows combine into one flow to pass through the secondary part of the dual heat exchanger. Two compressor outlet temperature sensors (one for each compressor outlet) measure the compressor outlet temperatures. After the cooled air leaves the secondary heat exchanger, it flows down a long vertical duct to the reheater part of the condenser/reheater/mixer unit. A sensor monitors the temperature of the air leaving the secondary heat exchanger.



Compressor
Outlet Temp
Sensors

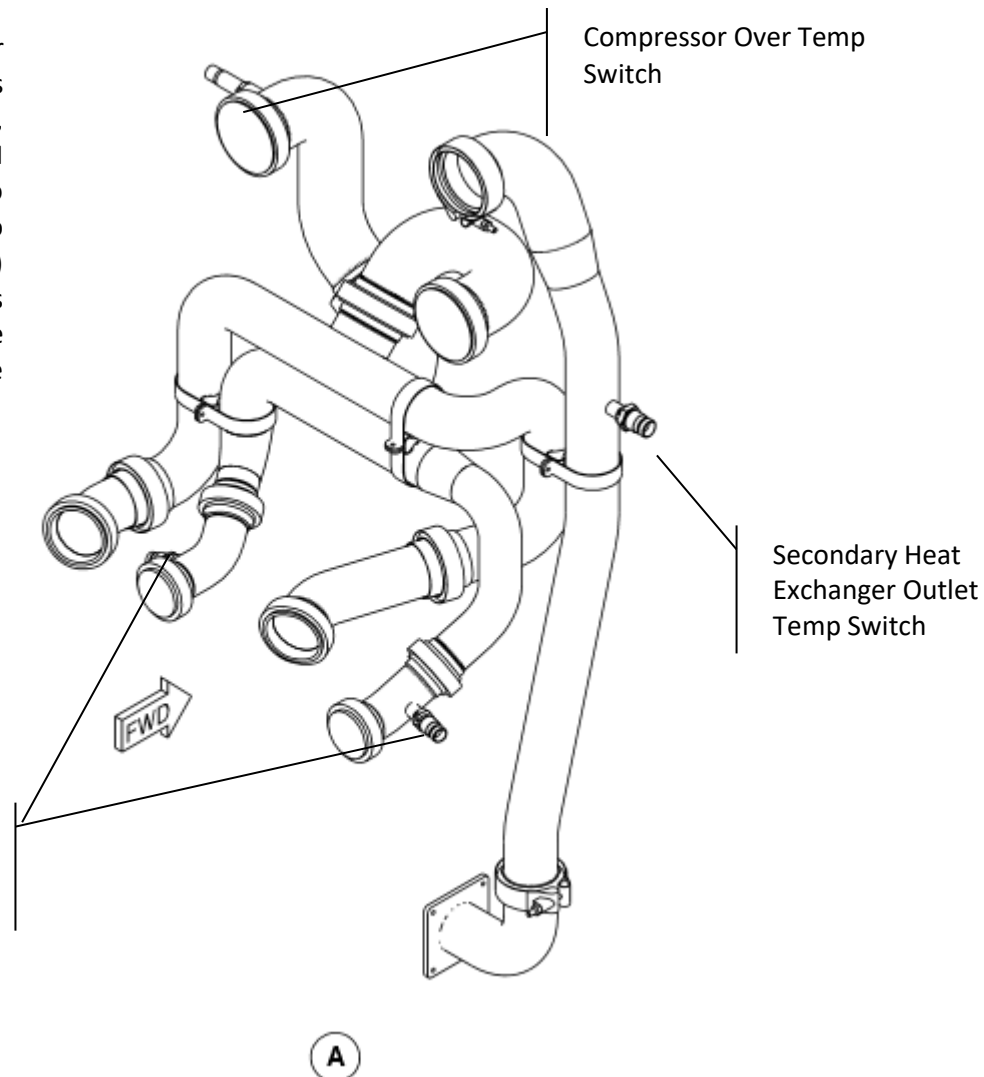


Figure 21- 39 Air Conditioning Pack Ducts

Air Conditioning Pack Struts and Mounts

[Refer Figure 21-40](#)

There is a mounting strap attached to either side of the fan inlet diffuser housing. The straps are also attached to the airframe and support the weight of the air conditioning pack.

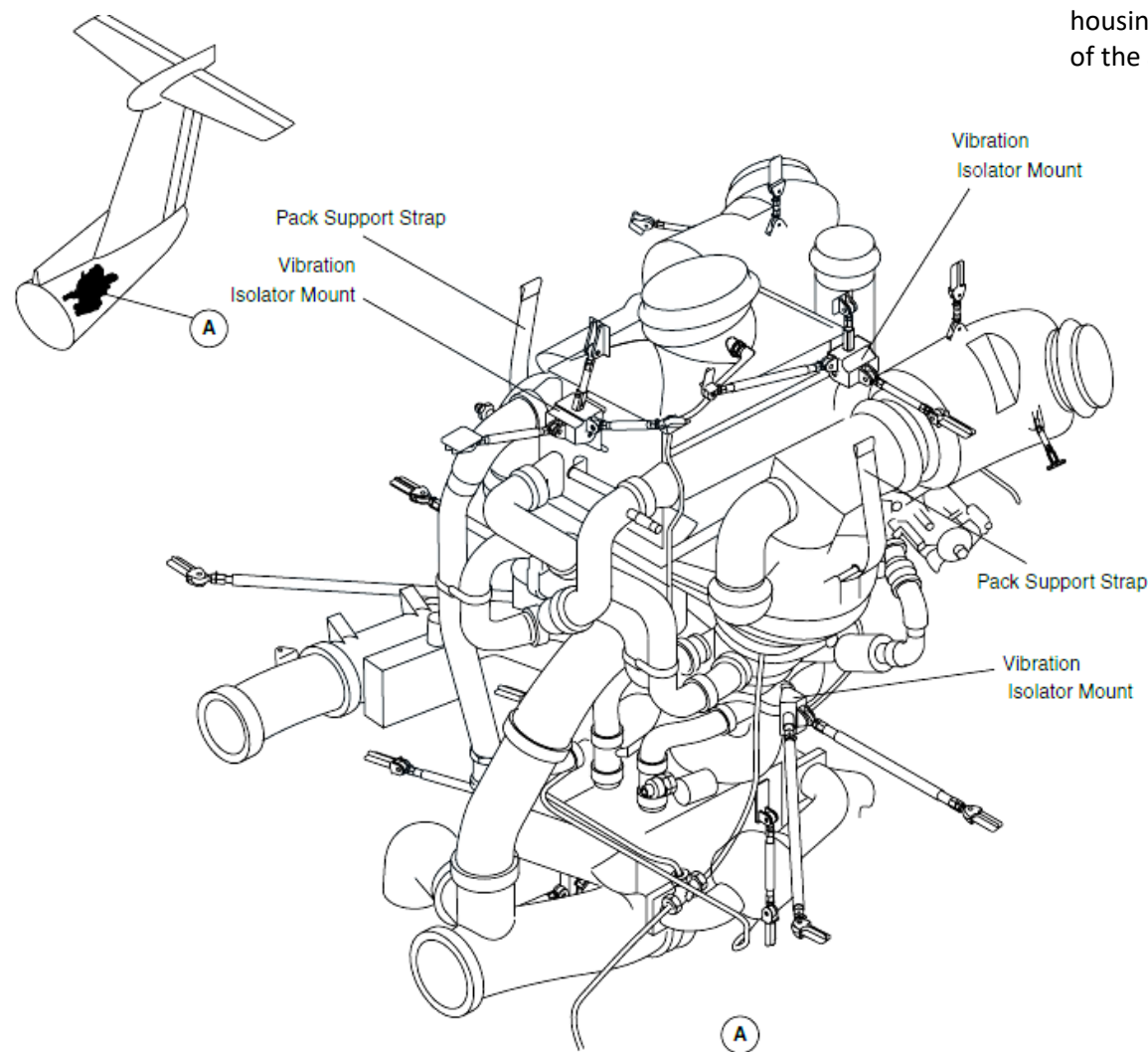


Figure 21- 40 Air conditioning Pack Struts and Mounts

Pack Control and Indications

Pack Switches

[Refer Figure 21-41](#)

The three-position packs control switches are located on the AIR CONDITIONING control panel.

- OFF - Closes related Turbine Shut-off Valve (TSOV).
- MAN - ECU temperature control based on related flight compartment or cabin rotary switch temperature selection.
- AUTO - ECU temperature control based on actual cabin temperature modulates the pack bypass valve.

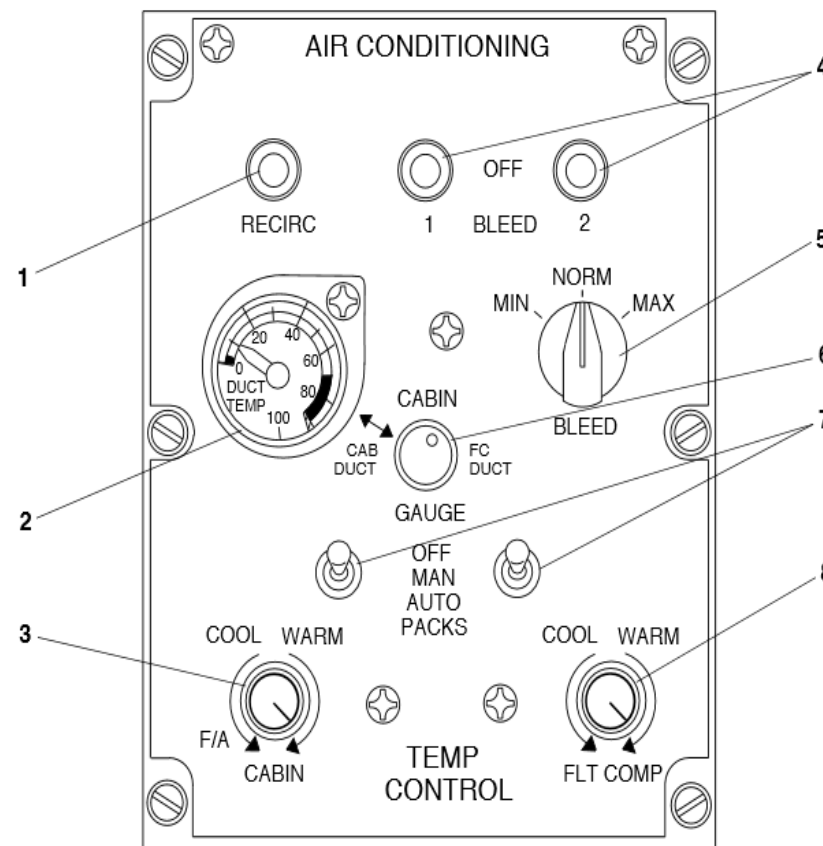


Figure 21- 41 Air Conditioning Control Panel

LEGEND

1. Recirculation Fan Switch.
2. Ducts Temperature Gauge.
3. Cabin Temperature Selector.
4. Bleed Control Switches.
5. Bleed Control Selector.
6. Ducts Temperature Gauge Switch.
7. Packs Control Switches.
8. Flight Compartment Temperature Selector.

Airconditioning PACK-Normal System Operation

Cooling

[Refer Figure 21-42](#)

The air conditioning system supplies the desired flow of conditioned air to the aircraft air distribution system. The air is conditioned to the proper temperature and humidity required to maintain a comfortable environment for the crew and passengers.

The main functions of the air conditioning packs are:

- Condition the bleed air from the engines (or APU) and supply it to the air distribution system.
- Protect the Air Cycle Machines (ACMs) from excessively high temperatures.
- Recirculate cool air to the pack inlet to counteract high bleed air temperatures
- Remove condensed water from the conditioned air.

Selection of the BLEED 1 and/or 2 switches on the AIR CONDITIONING panel provides power to the ECS ECU. The ECS ECU automatically controls the packs airflow through the modulation of the PFCSOV and the selected NFCSOV. In normal operation the PFCSOV modulates at approximately 10% > NFCSOVs' output.

Selection of the PACKS switches to AUTO, causes:

- ECU monitoring of temperatures, pressures, and flow rates and
- Adjusting of the pack valves to achieve the selected conditioned air temperatures and flow rates.

The modulated bleed air flows through the primary heat exchanger. The heat exchanger uses ram air to remove heat from the bleed air. The primary cooled bleed air is then divided into two air flows.

The air flows through one check valve for each of the two compressors of the ACM.

These compressors increase the temperature and pressure of the bleed air.

The two high pressure air flows are combined into one flow to pass through the secondary heat exchanger. The secondary heat exchanger uses cool ram air to remove heat from the ACM compressed bleed air.

The secondary heat exchanger temperature sensor measures bleed air as it leaves the secondary heat exchanger. This data is used by the ECU to determine the effective cooling effects of the secondary heat exchanger.

Each compressor outlet temperature sensor sends data to the ECS ECU in order to maintain the outlet temperature below 380°F (193°C). If the temperature reaches 380°F, the ECU will modulate the PFCSOV towards the closed position. If the temperature reaches 210°C (415°F), the ECU closes the TSOV and the bypass valve. After a time delay of 10 seconds, if the temperature is still high, the PFCSOV will be commanded closed.

The cooler air from the ACM turbine enters the condenser causing the water moisture in the hot wet air to accumulate and be removed by the water collector. The removed water is sent to the secondary heat exchanger spray nozzle.

After the air passes through the condenser, it is ducted back through the reheater section. The reheater warms the air and vaporizes any remaining water before it is used to power the ACM turbines.

The turbine SOV controls the bleed air flow into the ACM turbine. The energy to power the turbines and the air expansion absorb heat from the air.

After passing through the turbine, the cold discharge air is ducted into the mixer to combine with the recirculation fan air. The recirculation fan takes air from the cabin. After the mixer, the air is ducted into the cabin and flight compartments.

The ECU controls two pack bypass valves to regulate the temperature of the conditioned air sent to the cabin and flight compartment. These valves can send warm, high pressure bleed air from the pack inlet to the outputs of each ACM turbine. Such action raises the temperature of the air supplied to the cabin and flight compartments.

Closing the pack bypass valve decreases the air flow, decreasing the air temperature flowing to the cabin and flight compartments. The turbines also drive a fan which sends the ram air flow through the dual heat exchanger. This is required when there is no ram air flow through the dual heat exchanger (During ground runs).

When the aircraft is in-flight, the ACM fans can limit the flow of ram air. During flight the ram air increases, causing the ram air pressure to increase. The increase in pressure opens two bypass valves in the ram air outlet ducts, allowing maximum flow of ram air.

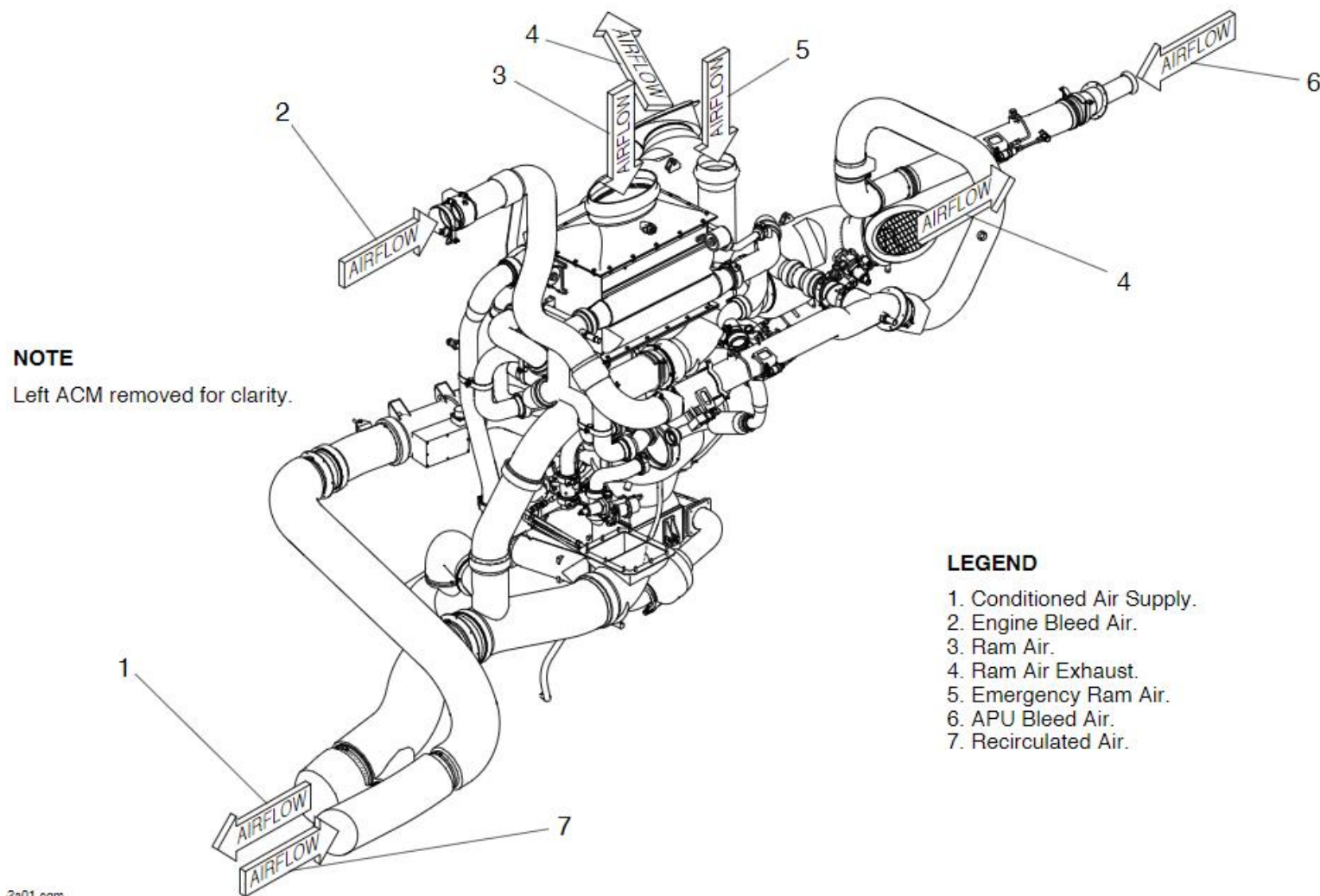


Figure 21- 42 Air Conditioning Pack

Recirculating Fan

[Refer Figure 21-43](#)

The recirculation fan, draws cabin air through the recirculation filter mounted behind the AFT class "C" baggage compartment. The air is routed aft, where it is mixed with pack conditioned air. The recirculation fan switch on the AIR CONDITIONING control panel controls the on/off operation of the recirculation fan independent of the ECU. However, the ECU will use the operating conditions and the length of time the fan was on to determine the fan speed.

When the switch is selected to the RECIRC position, the fan starts at low speed (to reduce initial current draw), then switches to high speed. Operating conditions determine the automatic control of the recirculation fan speed. The fan operates at low speed if one pack is turned off.

The recirculation fan is installed in the recirculation duct located on the right side of the air conditioning pack in the aft equipment bay.

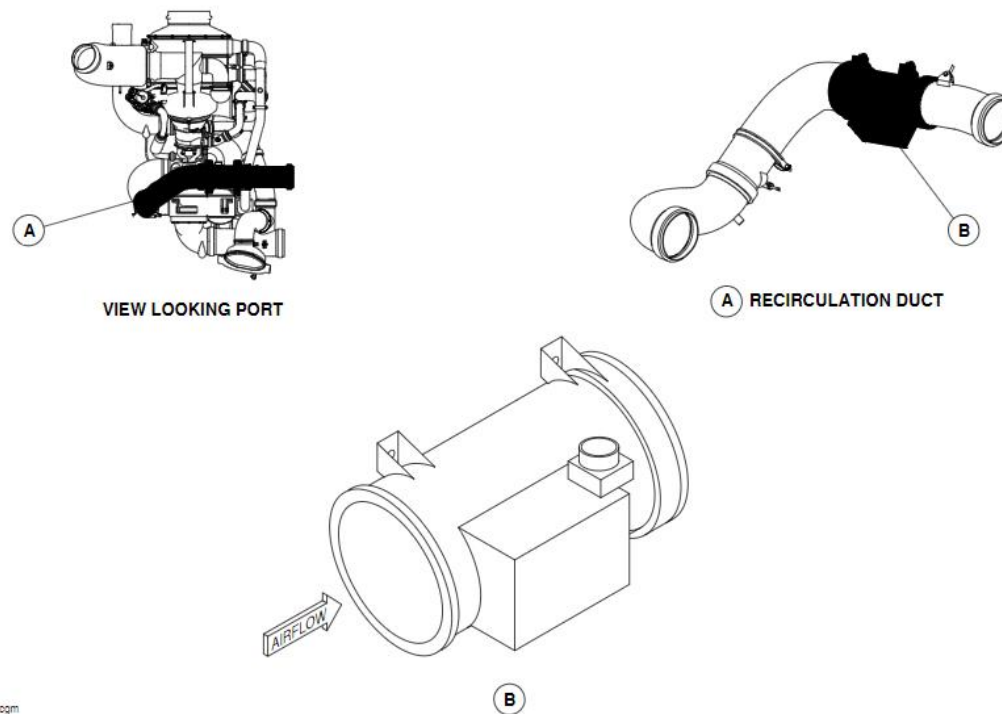


Figure 21- 43 Recirculation Fan

Recirculation Filter

[Refer Figures 21-44 & 21-45](#)

The recirculation HEPA filter is installed on the aft face of the baggage canted bulkhead and forward of the aft pressure bulkhead in the aft baggage compartment. The recirculation duct coming from the recirculation fan inlet is connected to the recirculation filter. The fan inlet is located in the aft equipment bay.

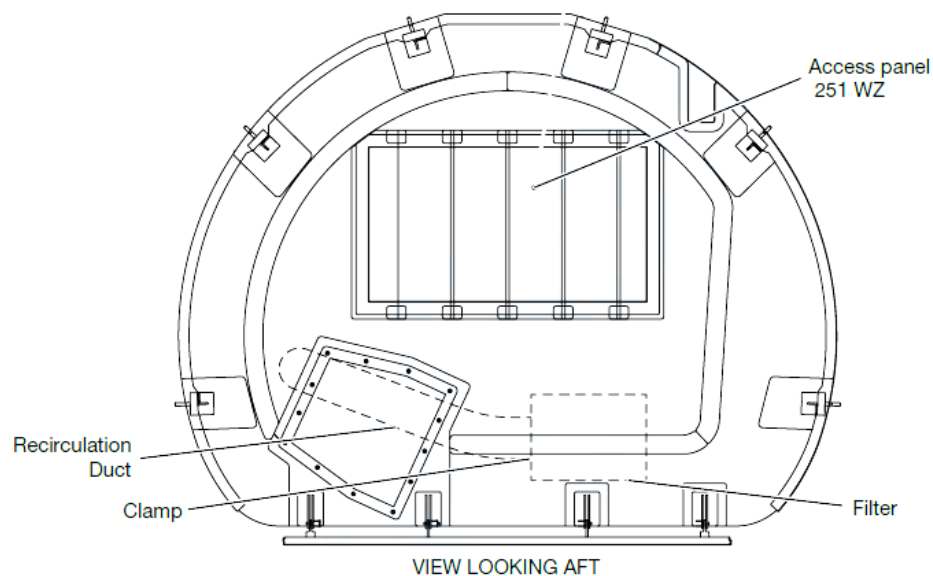


Figure 21- 44 Aft Cargo Compartment Panels

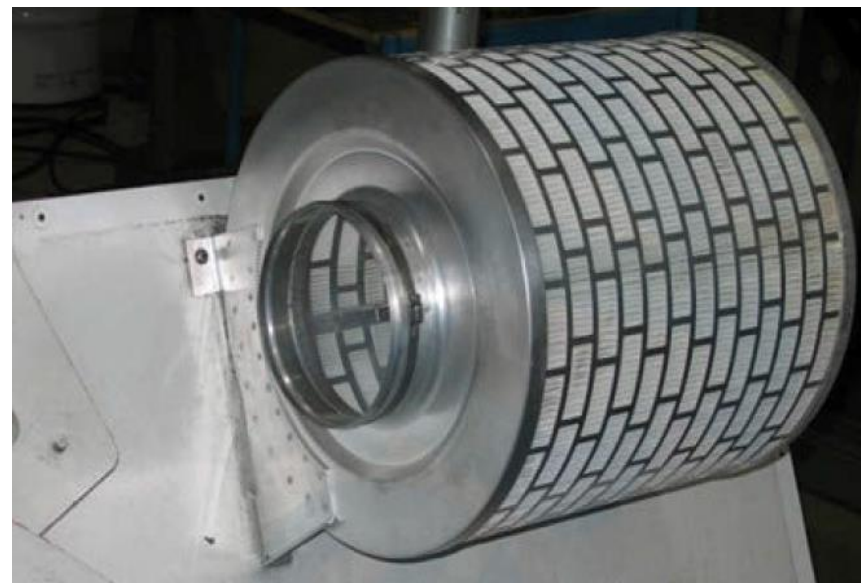


Figure 21- 45 Recirculation Filter

NOTE: The recirculation filter can only be replaced. Used filters should be discarded.

Flow Control System

Introduction

[Refer Figure 21-46](#)

The flow control system supplies temperature and pressure signals to the Electronic Control Unit (ECU) which regulates the airflow from the bleed source to the air conditioning pack.

General Description

The flow control system uses pressure and temperature data to control the flow rate of the bleed air that enters the air conditioning pack. The flow control system has three components:

- Valve, Pack Flow Control and Shutoff
- Sensor, Absolute Pressure
- Sensor, Pack Inlet Temperature

These components are installed in the ducting that supplies bleed air to the air conditioning pack. The two sensors supply the pressure and temperature data of the bleed air to the Electronic Control Unit (ECU). The ECU uses this data to open and close the pack flow control and shutoff valve to regulate the flow of bleed air to the air conditioning pack.

Detailed Description

The functions of the flow control system are:

- Control the flow of bleed air to the air conditioning pack.
- Supply pack inlet pressure and temperature signals to the ECU.
- Supply a valve closed signal to the ECU.

The flow of bleed air to the air conditioning pack is determined by three flow venturi differential pressure sensors. There is one differential pressure sensor located in each wing bleed air duct and one sensor located in the APU duct.

Pack inlet pressure and temperature sensors, located upstream of the pack flow control and shutoff valve, are also used in regulating airflow to the air conditioning pack.

When both engines are operating and supplying bleed air, the ECU keeps the pack flow control and shutoff valve approximately 10% more than Nacelle Flow Control Shut-off Valve (NFCSOV) flow output.

The ECU regulates the nacelle shutoff valves in the bleed air system to control the flow of bleed air. If the left or right side of the bleed air system is turned off (one nacelle shutoff valve is closed) the ECU regulates bleed airflow with the Pack Flow Control and Shutoff Valve.

The ECU also regulates bleed airflow with the Pack Flow Control and Shutoff Valve if the APU is supplying the bleed air.

The pack inlet pressure and temperature sensor signals are supplied to the ECU. Both digital channels of the ECU use these signals to share control of the Pack Flow Control and Shutoff Valve.

During a flight, one digital channel gets full control of the Pack Flow Control and Shutoff Valve (the other channel gets full control during the next flight).

If the digital channel in control loses power or fails, the other digital channel takes control of the pack flow control and shutoff valve.

The analog backup channels do not control the Pack Flow Control and Shutoff Valve.

The digital channel modulates the pack flow control and shutoff valve to provide either minimum, normal, or maximum pack flow. The MIN/NORM/MAX switch on the AIR CONDITIONING control panel sets the desired flow rate.

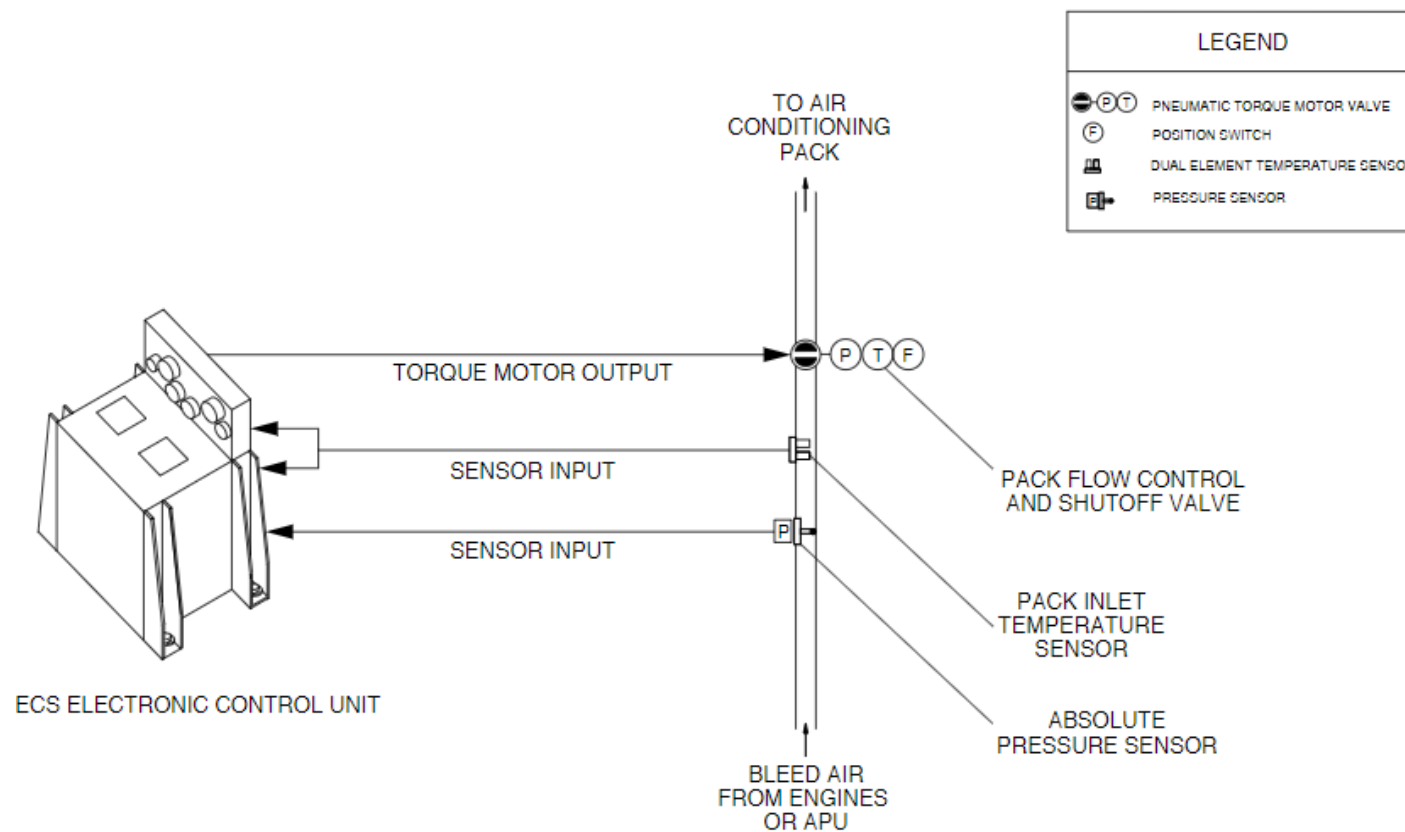


Figure 21- 46 Flow Control System

Component Description

Pack Flow Control and Shutoff (PFCSOV)

[Refer Figure 21-47](#)

The pack flow control and shutoff valve is a pneumatically operated, torque motor actuated butterfly valve. It has a position switch that indicates to the ECU when the valve is closed.

The pack flow control and shutoff valve is installed in line with the air conditioning ducting in the aft fuselage. It controls bleed air flow to the air conditioning pack according to the demands of the Environmental Control System (ECS).

The PFCSOV defaults pneumatically to the open position for mechanical malfunction and defaults to the closed position for electrical malfunction. This electrical default can occur only if both digital channels of the ECU lose power or fail.

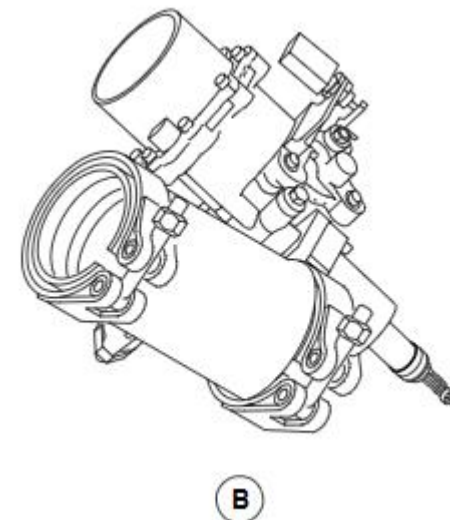
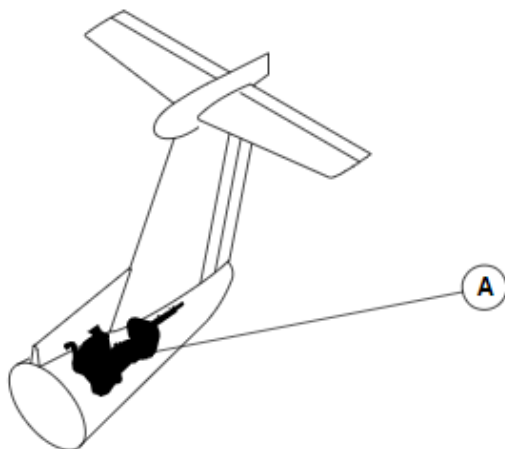
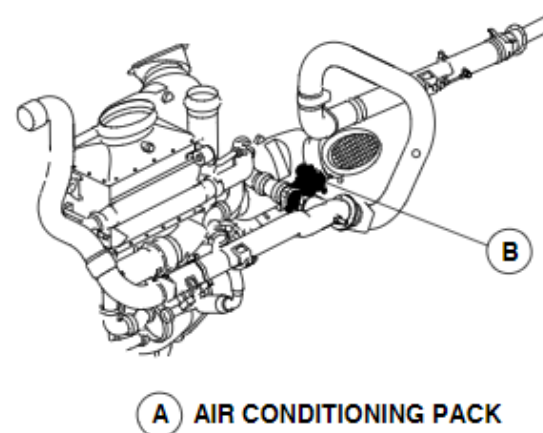


Figure 21- 47 Pack Flow Control and Shutoff valve



Absolute Pressure Sensor

[Refer Figure 21-48](#)

The absolute pressure sensor is installed in the air conditioning ducting in the aft fuselage. It senses the pressure of the bleed air as it flows into the air conditioning pack. The sensor supplies a linear pressure signal to the ECU. The absolute pressure data is used by the ECU to determine the bleed air flow rate.

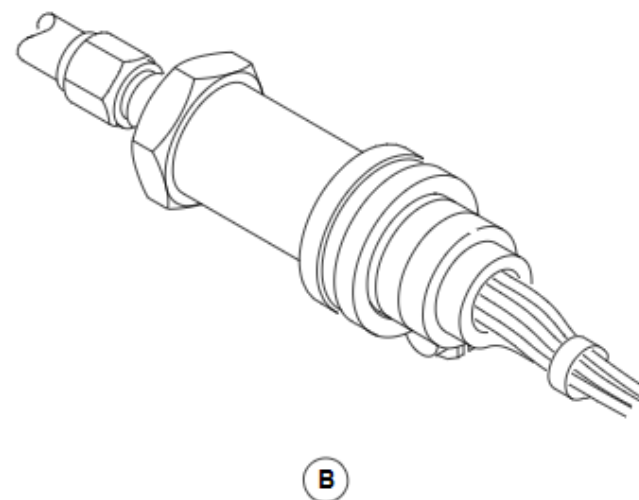
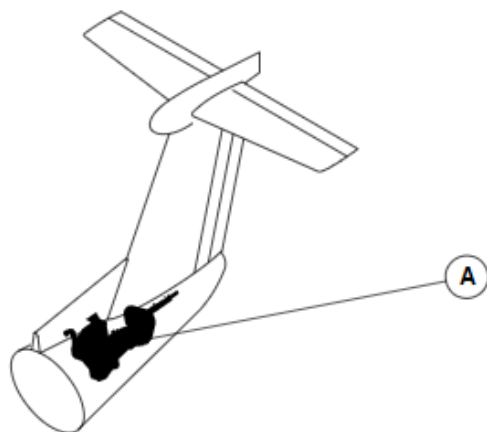
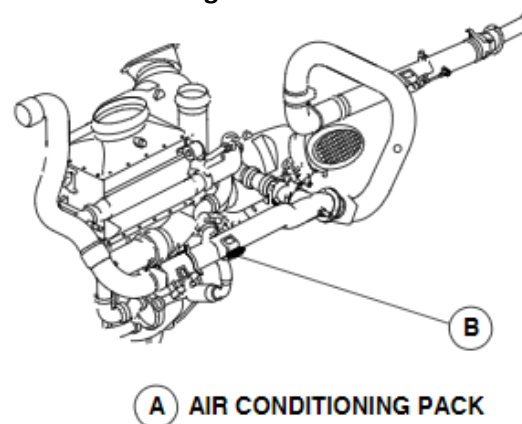


Figure 21- 48 Absolute Pressure Sensor



A AIR CONDITIONING PACK

Pack Inlet Temperature Sensor

[Refer Figure 21-49](#)

The pack inlet temperature sensor is a dual resistive temperature device (RTD) that supplies two independent temperature signals to the ECU (one to each digital channel). The pack inlet temperature sensor is installed in the air conditioning ducting in the aft fuselage. It senses the temperature of the bleed air as it flows into the air conditioning pack. The ECU uses this temperature data to determine the bleed air flow rate.

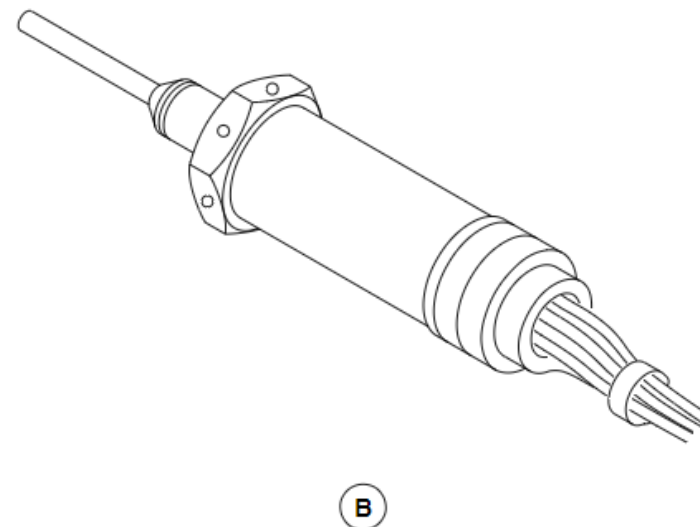
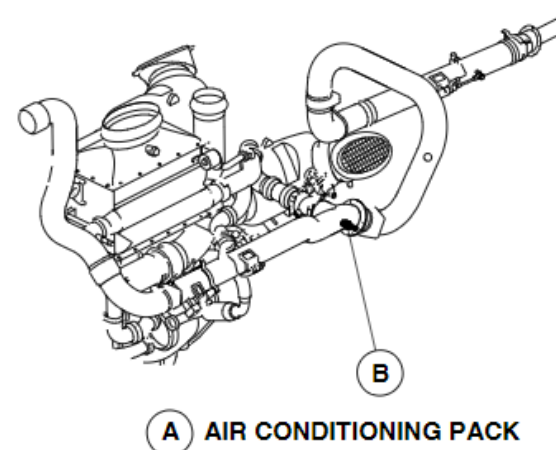
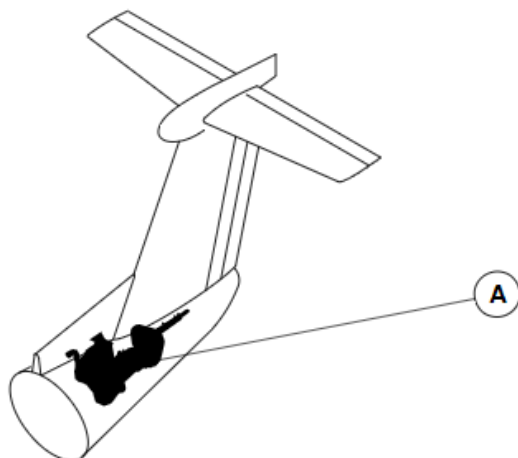


Figure 21- 49 Pack Inlet Temperature Sensor



A AIR CONDITIONING PACK

Air Conditioning Panel Switches

1. RECIRC SWITCH (Two Position)

OFF-selection turns recirculation fan off

RECIRC - turns fan on high speed.

ECU automatically selects slow speed when required. The fan operates at slow speed only:

- during single pack operation and
- at start-up

2. DUCTS TEMP GAUGE

Displays temperatures when selected by GAUGE selector:

- CABIN DUCT (supply) temperature
- CABIN temperature
- FLIGHT DECK DUCT (supply) temperature

3. GAUGE SELECTOR (Rotary Action)

Selects temperatures for display on the DUCT TEMP gauge:

- CAB DUCT (cabin supply) temperature
- CABIN temperature
- FC DUCT (flight compartment supply) temperature

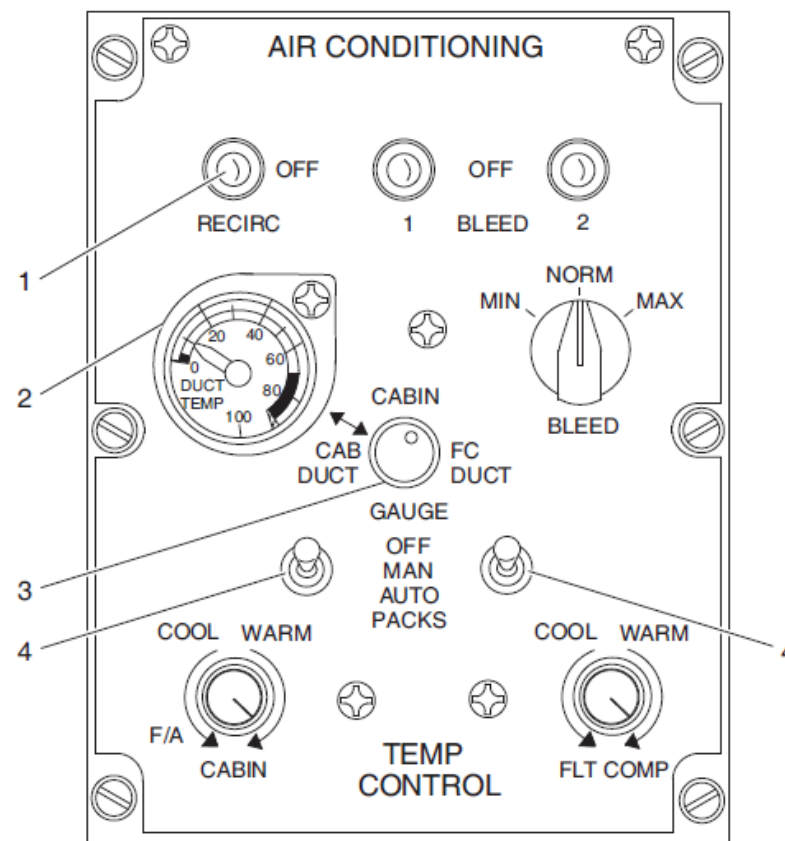
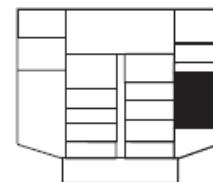
4. PACKS CONTROL SWITCHES (Three Position)

OFF - shuts related ACM off

MAN, AUTO

For both MAN and AUTO positions: -

- * modulates the pack Flow Control and Shut Off Valve
- packs start operating and supply conditioned air to the cabin and flight deck
- * modulates the Pack Bypass Valve in accordance with the temp selections from the cockpit.



5. BLEED SWITCHES (two position)

1 AND/OR 2 - starts bleed air flow from the No.1 and/or No.2 engines to the air conditioning packs.

OFF - stops bleed air flow from selected engine

6. BLEED CONTROL SELECTOR (rotary action) - signals ECU to modulate the nacelle shut off valves - provides air flow for one of three selected settings (for engine bleed operation)

MIN - for minimum bleed air flow

NORM - for moderate bleed air flow

MAX - for maximum bleed air flow

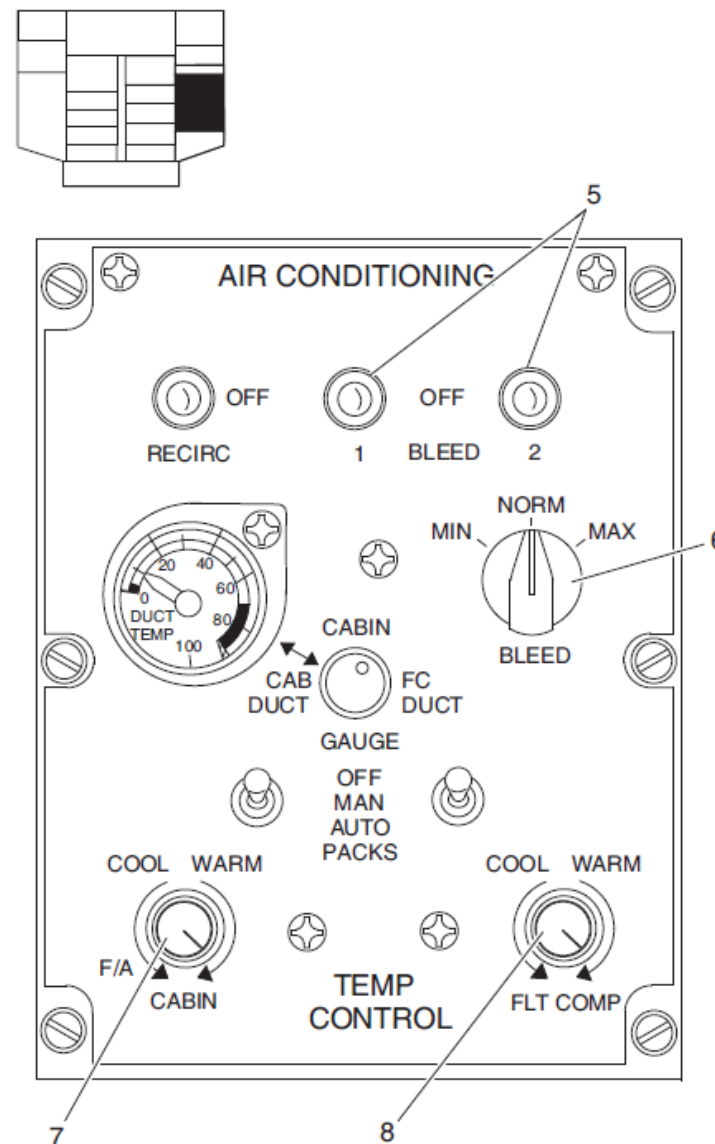
7. CABIN TEMPERATURE SELECTOR (rotary action)

ROTATE - adjusts cabin temperature when PACKS control switch is selected to AUTO or MAN

F/A - rotating fully counterclockwise to the cabin attendant position, allows the cabin crew member to control cabin temperature from the FLIGHT ATTENDANT'S panel - indicated by a light on the flight attendant's panel.

8. FLT COMP TEMPERATURE SELECTOR (rotary action)

ROTATE - adjusts flight deck temperature when the associated PACK control switch is selected to AUTO or MAN



Normal System Operation

ECS Flow Control

[Refer Figure 21-50](#)

The aircraft flow control system is automatically controlled. Selection of the BLEED 1 and/or 2 switches on the AIR CONDITIONING control panel provides a signal to the ECS ECU. The ECU will open the selected engine NFCSOV.

The ECS ECU automatically controls airflow to the ECS Packs through modulation of the Pack Flow Control and Shut-Off Valve (PFCSOV) or the selected NFCSOV.

During a single failure of a digital channel, the other digital channel will take over control of the PFCSOV and the analog control will operate for that side. If both digital channels fail, the analog channel is unable to control the PFCSOV and it will close.

Operation with one channel on analog control will cause the related NFCSOV to open and flow to be controlled through the PFCSOV. This is also the case with both pack switches selected to MAN. An ECU failure or power loss causes the PFCSOV to close and stop the ECS operation.

Refer Figure 21-50: The ECU uses four input sensors to calculate the airflow into the packs:

- The left wing differential pressure sensor signal comes from connector P8 pins 3 and 4 to the ECU connector 2100-P3 pins 50 and 54.
- The right wing differential pressure sensor signal comes from connector P25 pins 3 and 4 to the ECU connector 2100-P4 pins 50 and 54.
- The APU differential pressure sensor signal comes from connector 2121- P9 pins 3 and 4 to the ECU connector 2100-P4 pins 49 and 53.
- The Pack inlet pressure sensor signal comes from connector P26 pin 3 to the ECU connector P3 pin 1.

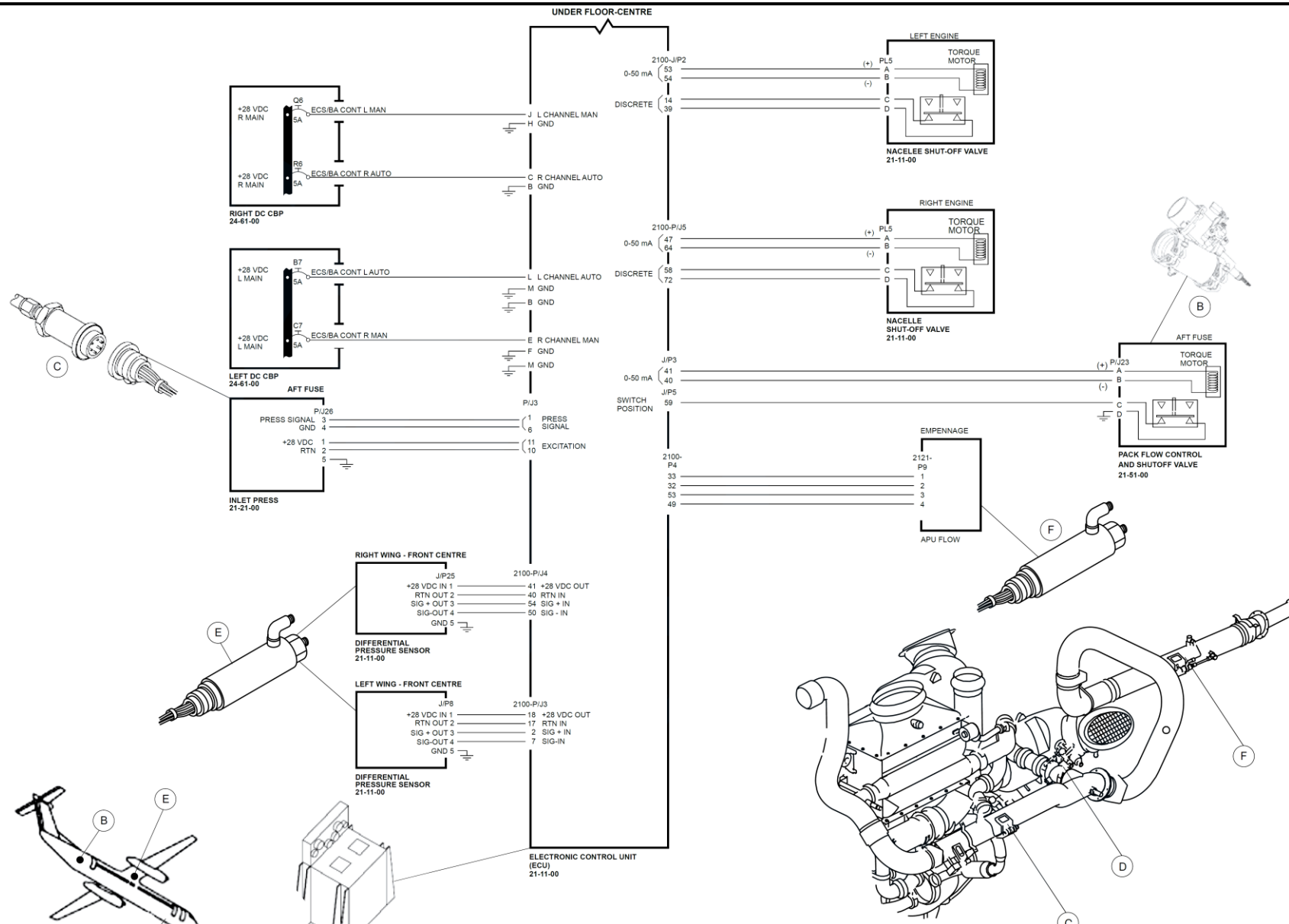


Figure 21- 50 Flow Control Schematic

Temperature Control and Indication

General

Refer Figure 21-51

The temperature control and indication system is part of the Environmental Control System (ECS), which supplies conditioned air to the flight and cabin compartments of the aircraft.

The main functions of the temperature control and indication system are to:

- supply temperature control of the Environmental Control System
- supply temperature indication to the flight crew
- sense malfunctions in the system and send fault codes to the fault indication system

Components of the temperature control and indication system are listed below:

- Electronic control unit -environmental control system zone supply indication sensors
- Temperature sensors
- Zone supply temperature sensors
- Zone supply overtemperature switches
- AIR CONDITIONING control panel
- Temperature selector - Cabin
- Temperature selector - Flight compartment
- Control switches - Packs
- Ducts temperature gauge
- Ducts temperature switch

Temperature control starts with setting the desired temperature using the appropriate switches on the AIR CONDITIONING control panel in the overhead console in the flight compartment.

The Electronic Control Unit (ECU) monitors the temperature sensors and switches located throughout the aircraft.

The ECU then operates the appropriate valves to maintain the selected temperature setting. The ECU uses sensors and switches to monitor temperature, pressure, flow rates, and valve positions. The ECU uses the pilot and crew commands to control the air conditioning system in manual or automatic operation, one or two ACM operation, bleed source selection, and distribution of conditioned air to the flight and cabin compartments. The ECU and the air conditioning system have redundant control systems for continued ECS operation with mechanical and electrical component malfunctions.

Detailed System Description

The two zone supply temperature sensors measure the temperature of the air in the cabin and flight compartment supply ducts. The ECU uses these signals to control the temperature of the air leaving the air conditioning pack.

The ECU keeps the temperature in the supply ducts between 37 and 160 °F (2.8 and 71 °C). The actual temperature in the supply ducts depends on the settings of the CABIN and FLT COMP temperature selectors on the AIR CONDITIONING control panel. The zone supply temperature sensors are located in the cabin and flight compartment supply ducts below the aft baggage compartment floor.

The three zone temperature sensors measure the temperature of the air in the cabin and in the flight compartment zones. The ECU uses two of the three zone temperature sensor signals to control the conditioned air temperature in AUTO mode. The cabin zone sensor also feeds the data to the temperature gauge on the flight attendant's panel. The ECU keeps the temperature in the cabin and flight compartments between 59 and 80.6 °F (15 and 27.0 °C).

The third zone temperature sensor (in the cabin compartment) is an indication sensor. It supplies the cabin temperature signal to the DUCT TEMP gauge on the AIR CONDITIONING control panel.

The two zone supply indication sensors are located in the flight compartment and cabin supply ducts near the zone supply temperature sensors. These indication sensors send the supply duct temperature signals to the DUCT TEMP gauge on the AIR CONDITIONING control panel. These temperature signals are supplied to the gauge independent of the Environmental Control Unit (ECU). The flight crew can select the two temperatures for display on the DUCT TEMP gauge:

- CABIN (supply) DUCT
- FLT COMP (supply) DUCT

The two zone supply overtemperature switches are protective devices that turn off the air conditioning system if it overheats. They send discrete signals to the ECU if the temperature of conditioned air in the supply ducts is greater than 190 °F (87.8 °C). These signals cause the ECU to turn off the applicable (left or right) air cycle machine in the air conditioning system. The ECU also turns on the applicable FLT COMPT DUCT HOT or CABIN DUCT HOT caution lights. These switches are located in the cabin and flight compartment supply ducts near the zone supply indication and temperature sensors.

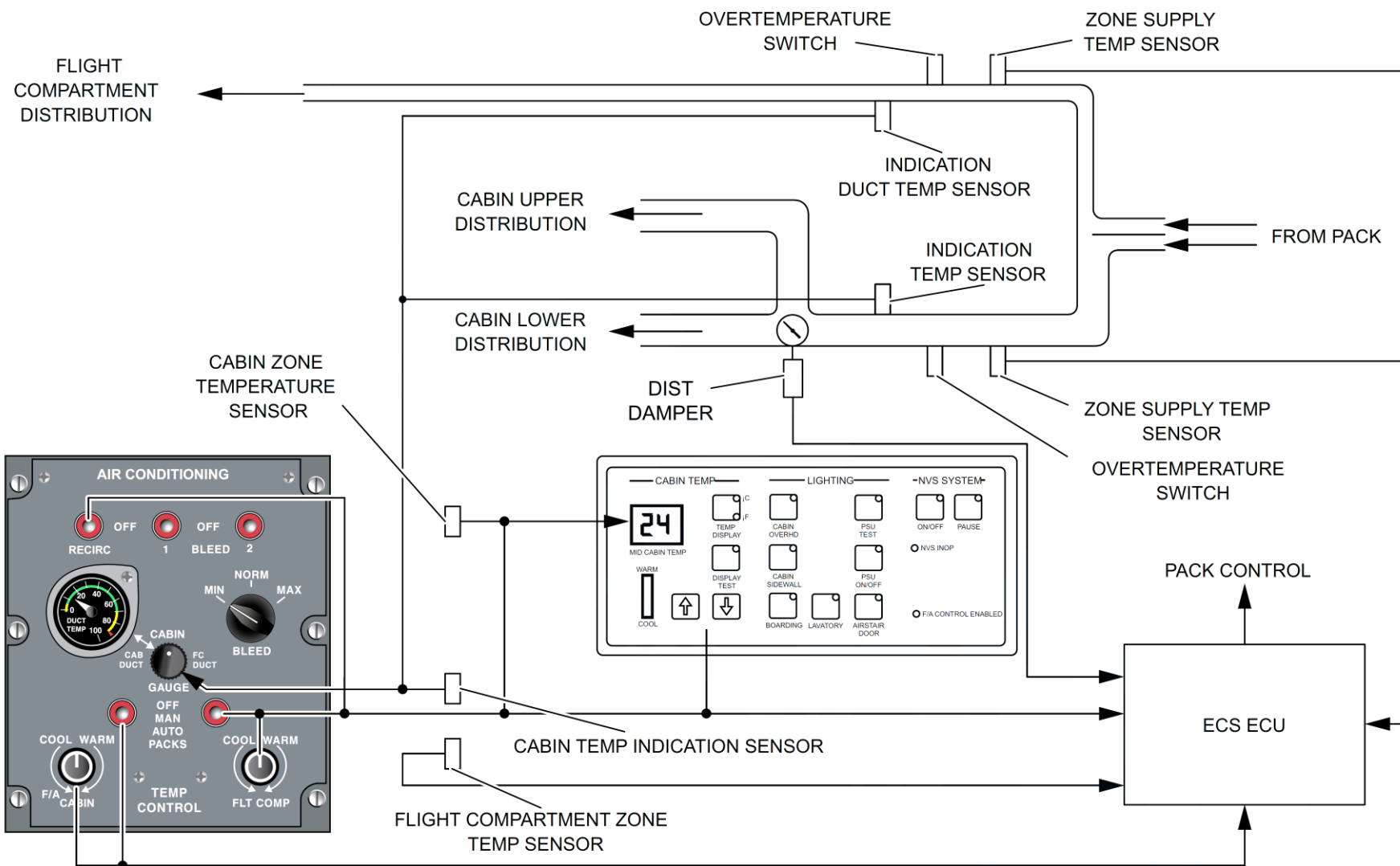


Figure 21- 51 Temperature Control and Indication Block Diagram

Electronic Control Unit

[Refer Figure 21-52](#)

The ECU is the interface between the AIR CONDITIONING control panel and the mechanical and electrical components of the air conditioning system. It is divided into a left digital channel with a left backup analog channel and a right digital channel with a right backup analog channel.

The left digital channel and right analog channel receive power from the left 28 Vdc bus. The right digital channel and left analog channel receive power from the right 28 Vdc bus. This permits continued left and right channel control if one 28 Vdc bus loses power. For example, if the left digital channel loses power, the left analog channel continues operation.

Inside the ECU, the digital channels use an RS 422 digital data bus to communicate with each other. The ECU uses the ARINC 429 digital data bus to communicate with the other aircraft systems. The ARINC 429 receivers of both digital channels are connected to the aircraft wiring. Only the ARINC 429 transmitter of the left digital channel is connected to the aircraft wiring.

The left and right digital channels share control of the pack flow control and shutoff valve, the temperature reduction valve, and the recirculation fan.

One channel is selected to be the digital channel in control. The digital channel in control is responsible for all functions applicable to the shared components. Loss of the digital channel in control causes the other digital channel to take control of the shared components. The digital channel in control automatically changes (between the left and right channels) after every landing. The digital channel in control of shared functions has air conditioning system ON/OFF control. For normal operation (both packs operating) the digital channel in control of shared functions opens the pack flow control and shutoff valve to enable bleed air flow.

The digital channel in control will shut off the pack flow control and shutoff valve for these conditions:

- Both PACKS switches on the AIR CONDITIONING control panel are set to OFF
- One or more of the three pack overtemperature protection switches (compressor outlet temperature switch, two zone supply overtemperature switches) are set and turning off the applicable ACM does not remove the overtemperature condition.
- The BIT function of the ECU detects an unacceptable condition.

The two digital channels also share control of the four ECS caution lights:

- FLT COMP DUCT HOT
- CABIN DUCT HOT
- CABIN PACK HOT
- FLT COMPT PACK HOT

Both digital channels generate discrete outputs to turn on the four ECS caution lamps according to the pack over-temperature protection switches or the BIT logic. Either digital channel turns on the applicable caution light if the channel detects a condition which causes pack shutdown. The caution light stays on until the over-temperature condition or the BIT fault that caused the pack to turn off no longer exists.

The digital channel in control sets pack air flow. Differential pressure sensors in the nacelle bleed ducts and the APU bleed duct measure pack air flow. The digital channel in control uses the differential pressure sensors in the nacelle bleed ducts when bleed air is supplied by the engines. The sensors in the APU bleed duct are used when bleed air is supplied by the APU. The pack inlet (absolute) pressure and temperature sensors are used to determine pack air flow. All of these sensor inputs go to both digital channels.

For operation with both engines and both packs operating, the digital channels regulate the nacelle shutoff valves for minimum, normal, or maximum pack air flow. The pack air flow is selected on the BLEED MIN/NORM/MAX switch on the AIR CONDITIONING control panel. The flight condition determines the amount of flow for each of these levels.

With digital channel failure or APU bleed selection, the pack flow control and shutoff valve regulates the pack air flow. To limit compressor outlet temperature, pack outlet temperature or to prevent ACM overspeed, the digital channel decreases the pack air flow below the setting of the BLEED switch. This lets system operation continue but with reduced heat exchanger performance.

The left digital channel uses approximately 50% of the air from the left ACM to control the flight compartment temperature. The right digital channel uses the other 50% of the air from the left ACM and all the air from the right ACM to control the cabin compartment temperature.

Each digital channel has its own compressor outlet temperature sensor and pack outlet sensor, and receives the other channel's compressor outlet temperature sensor and pack outlet sensor data by the internal RS 422 digital data bus. The digital channel in control uses all four temperature sensors for temperature limiting control.

The right digital channel receives two cabin compartment temperature reference inputs. One comes from the AIR CONDITIONING control panel and the other comes from the flight attendant's control panel. The right digital channel also receives a discrete input that enables or disables the flight attendant's control panel. If the flight attendant's control panel is enabled, it supplies the cabin compartment temperature reference. If the flight attendant's control panel is disabled, the AIR CONDITIONING control panel gives the cabin compartment temperature reference.

The CABIN temperature selector on the AIR CONDITIONING control panel has a switch at the full counter clockwise position. Turning the selector knob to full COOL activates the switch which sends the discrete signal to the right digital channel to enable the flight attendant's control panel. This switch also turns on the ENABLED light on the flight attendant's control panel. This indicates the flight attendant has control of the cabin temperature selection. The temperature gauge on the panel receives its signal from the cabin zone

temperature indication sensor. Both the ENABLED light and the cabin temperature gauge on the flight attendant's control panel operate independently from the ECU.

AUTO Temperature Control

When the PACKS switch is set to AUTO, the digital channel in control opens the pack flow control and shutoff valve and the turbine shutoff valves. This starts the ACMs which supply cold air. The digital channel in control (ECU) opens and closes the ACM bypass valves to add warm air to the cool air coming out of the turbines.

The ECU controls the pack outlet temperature based on the settings of the CABIN and FLT COMP temperature selectors on the AIR CONDITIONING control panel. The digital channel uses the two zone supply temperature sensors (one in the cabin supply duct and one in the flight compartment duct) for temperature feedback.

The ECU can vary the cabin and flight compartment temperatures from 59 to 80.6 °F (15 to 27.0 °C). To prevent the possible formation of ice in the condenser, the ECU also keeps the supply duct temperature at or above 37 °F (2.8 °C). This removes the need for ice detection devices and deice controls.

MANUAL Temperature Control

The ECU reverts to manual mode by selection to MAN. Automatic reversion to manual occurs without indication to the flight compartment when the ECU detects digital control has been lost.

An ECU amber fault light on the Maintenance panel illuminates whenever a major fault is detected by BIT. This indication is only displayed with WOW.

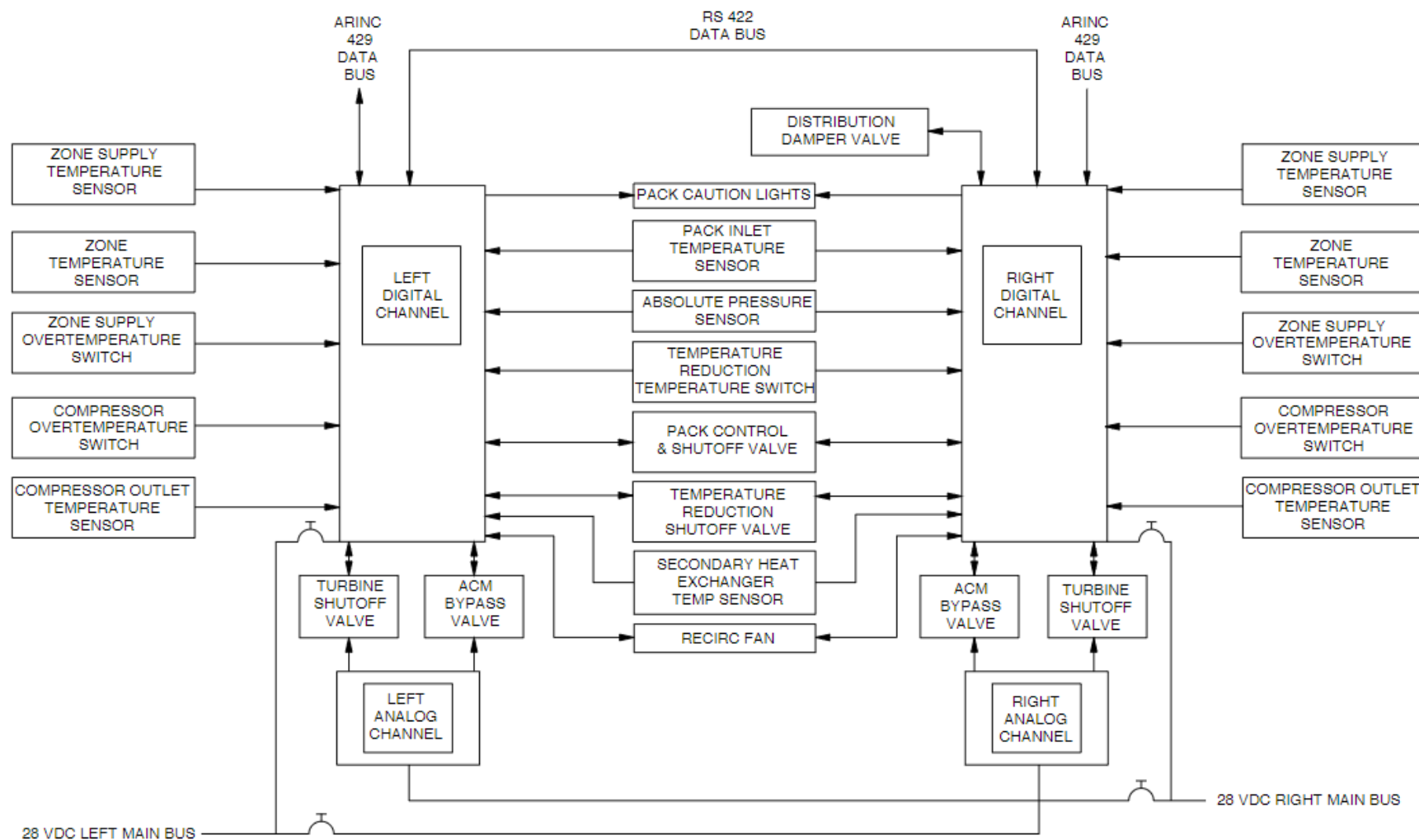
The analog channel controls the temperature of the supply ducts by opening and closing the pack bypass valves. The analog channel uses the settings of the CABIN and FLT COMP temperature selectors as a reference for controlling the temperature.

Temperature Reduction Valve Control

If the temperature reduction valve switch identifies that the bleed air temperature to the pack is too hot, the digital channel in control opens the temperature reduction shutoff valve. This lets the cooler air from the secondary heat exchanger outlet mix with the hot bleed air. The temperature reduction shutoff valve stays open until the pack inlet temperature sensor identifies that the bleed air temperature has decreased to normal values. Each digital channel has its own discrete input for the temperature reduction valve switch and its own temperature sensor input for the pack inlet temperature. Both digital channels share control of the temperature reduction shutoff valve.

Recirculation Fan Control

The digital channels share control of the recirculation fan speed. The digital channel in control generates the discrete signal that sets the recirculation fan speed to high or low. The RECIRC switch on the AIR CONDITIONING control panel commands the recirculation fan on or off independently of the ECU. The ARINC 429 digital data bus gives hot/cold day operation condition information to the digital channels. Control logic uses the operation conditions and the length of time the fan is on to determine fan speed. The recirculation fan starts operation at low speed and runs at low speed when the air conditioning system operates with one ACM turned off (limited flow capacity). The recirculation fan runs at high speed for all other conditions.



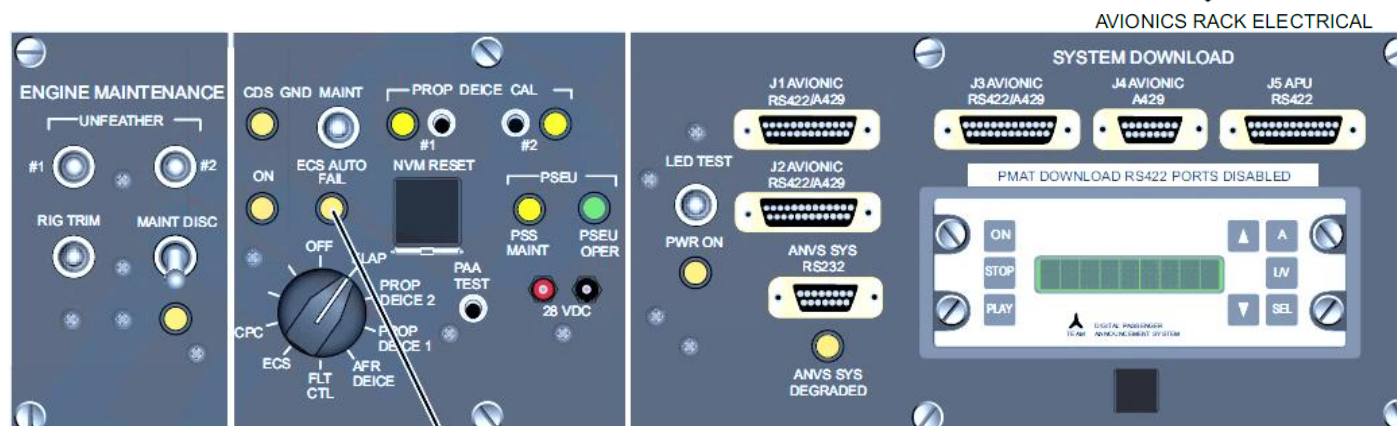
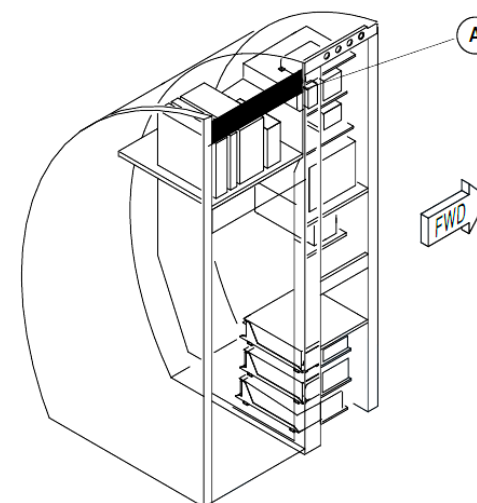
NOTE: ALL CHANNELS ARE LOCATED IN ECU CONTROLLER

Figure 21- 52 ECS ECU Block Diagram

Built-in Test

Refer Figure 21-53

The ECU performs system level BUILT IN TESTS (BIT) for both the bleed air system and the air conditioning system. The ECU stores information about system and component faults in its non-volatile memory. These faults can then be accessed by maintenance personnel through the Central Diagnostic System (CDS). If a fault is logged by the ECU, the ECS AUTO FAIL light on the maintenance panel comes on. When the faults have been corrected, the non-volatile memory (NVM) must be cleared to erase the fault information.



ECS AUTO FAIL Indicator

Figure 21- 53 Maintenance Panel

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Temperature Control Electrical Operation

[Refer Figure 21-54](#)

The ECS ECU provides solenoid power and pressure sensor excitation. The ECU controls the units as follows:

- Recirculation fan speed
- PFCSOV modulation
- Pack temperature reduction SOV
- Pack turbine SOV
- Pack Bypass SOV and
- Distribution damper valve

1. The BLEED switches are set to the 1 and 2 position (PACK switches are not in the OFF position):

- This discrete will cause the ECU to open the Nacelle Flow Control Shutoff Valve (NFCSOV), the PFCSOV and the two Turbine SOVs. ACMs will start to operate.

2. The PACK switches are set to the AUTO position:

- Pack switches send a ground discrete to connector J5 pin 37 and connector J2 pin 41 on the ECU
- Send an open circuit discrete to connector J5 pin 10 and connector J2 pin 16 on the ECU
- The signals will cause the ECU to start controlling the two ACMs to maintain the selected compartment temperatures and pack air flow rate.

During normal operation, the ECU logic will automatically:

- Monitor the two supply duct and two zone temperature sensors and perform temperature control calculation
- Monitor two pack over-temperature sensors and reduce the pack flow if the temperatures are too high

- Monitor the pack over-temperature switch to stop the pack flow if the temperature is too high
- Open the two turbine SOVs
- Modulate the two pack bypass valves to warm or cool the pack outlet air
- Open temperature reduction SOV to cool the pack inlet air when required
- Modulate the PFCSOV to control air flow into the packs. The PFCSOV open switch sends a ground signal from the PFCSOV connector P23 pin C to the ECU connector P5 pin 59
- Perform continuous system BIT

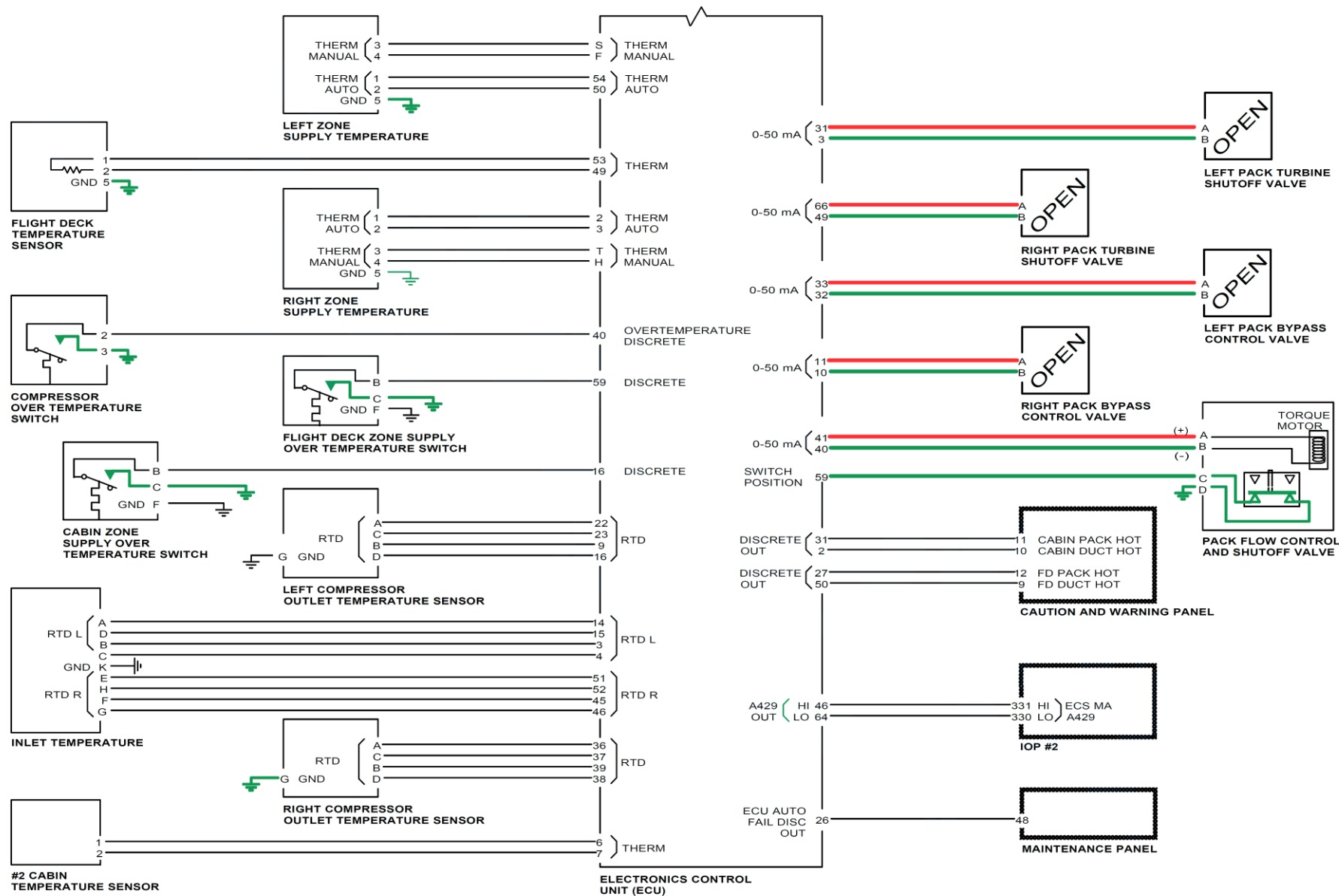


Figure 21- 54 Temperature Control Electrical Schematic

Refer Figure 21-55

The two PACK switches are set to the MAN position to give manual control of the packs:

- These actions will send open circuit discretes to the ECU (Figure 21-55. Temperature Control and Indication Electrical Schematic.):

- Connector J5 pins 10 and 37
- Connector J2 pins 16 and 41
- The ECU logic uses these discretes to send an analog (0 to 50 ma) signal to open or close the pack bypass valve
- The ECU signals are sent from the: (refer to Figure 21-54)
 - Connector P4 pins 10 and 11 to the right bypass valve and
 - Connector P3 pins 33 and 32 to the left bypass valve
- The ECU digital channel transfers the control of the control of the NFCSOV to the analog channel
- The ECU logic uses these discretes to send an analog (0 to 50 ma) signal to open or close the pack bypass valve

- The ECU signals are sent from the: (Refer to Figure 21-54)

- Connector P4 pins 10 and 11 to the right bypass valve and
- Connector P3 pins 33 and 32 to the left bypass valve

- The ECU digital channel transfers the control of the NFCSOV to the analog channel.

Refer to Figure 21-55. Temperature Control and Indication Electrical Schematic.

Rotating the CABIN control knob to the full WARM position raises the cabin temperature.

- An open circuit discrete is sent to connector J5 pin 78 on the ECU and connector J3 pin 37 on the Flight Attendant (F/A) panel

- This signal provides cabin temperature primary control and indication to the A/C panel
- The A/C panel will use the No.1 temperature sensor data from connector P101 pins 4 and 5
- The ECU sends a 50 mA signal to the torque motor to close the turbine SOV

- If AUTO is selected, a low resistance discrete is sent to connector J4 from pin 27 to pin 20 on the ECU

- If MAN is selected, a low resistance discrete is sent to connector J6 from pin U to pin K on the ECU. Component Description, Temperature Control

In the full COOL (F/A) position, the A/C panel output is as follows:

- A ground discrete is sent to connector J5 pin 78 on the ECU and connector J3 pin 37 on the flight attendant panel, this:

- This gives cabin temperature primary control and indication to the F/A panel
- The ENABLED light on the F/A panel will illuminate
- The F/A panel will use the No.2 cabin temperature sensor data on connector J3 pins 5 and 24

- If AUTO is selected, a 10 kΩ discrete is sent to connector J4 from pin 27 to pin 20 on ECU

- If MAN is selected, a 10 kΩ discrete is sent to connector J6 from pin U to pin K on ECU.

The Crew rotates the FLT COMP control knob to the full WARM position.

- If AUTO is selected, a low resistance discrete is sent to connector J3 from pin 27 to pin 35 on ECU
- If MAN is selected, a low resistance discrete is sent to connector J1 from pin R to pin D on ECU
 - The ECU uses this signal to send a 50 mA signal to the torque motor to close the turbine SOV.

In the full COOL position, the output is:

- If AUTO is selected, a 10 kΩ discrete is sent to connector J3 from pin 27 to pin 35 on ECU
- If MAN is selected, a 10 kΩ discrete is sent to connector J1 from pin R to pin D on ECU.

The GAUGE knob shows three temperature selections on the A/C panel DUCT TEMP gauge. The temperature indicator received 28 VDC from pin C on connector J102. The indicator measures the current flow through the selected sensor. The current flow is through the A/C panel pins that follow:

- The cabin duct sensor (connector J101 pins 6 and 11)
- The flight compartment duct sensor (connector J101 pins 9 and 13)
- The cabin compartment zone sensor (connector J101 pins 4 and 5)
- The indication ground pins on connector J101 are pins 4, 8, 11 and 13.

The RECIRC switch on the A/C panel will start the recirculation fan. The A/C panel sends out the discretes signal that follow:

- A ground discrete to the recirculation fan (connector J22 pin D)
 - Fan starts at low speed
 - The fan stays at slow speed during single pack operation
 - The fan operates at high speed during all other operations

- A ground discrete to the ECU (connector J2 pin 60)

- The ECU uses this ground to operate the recirculation fan.

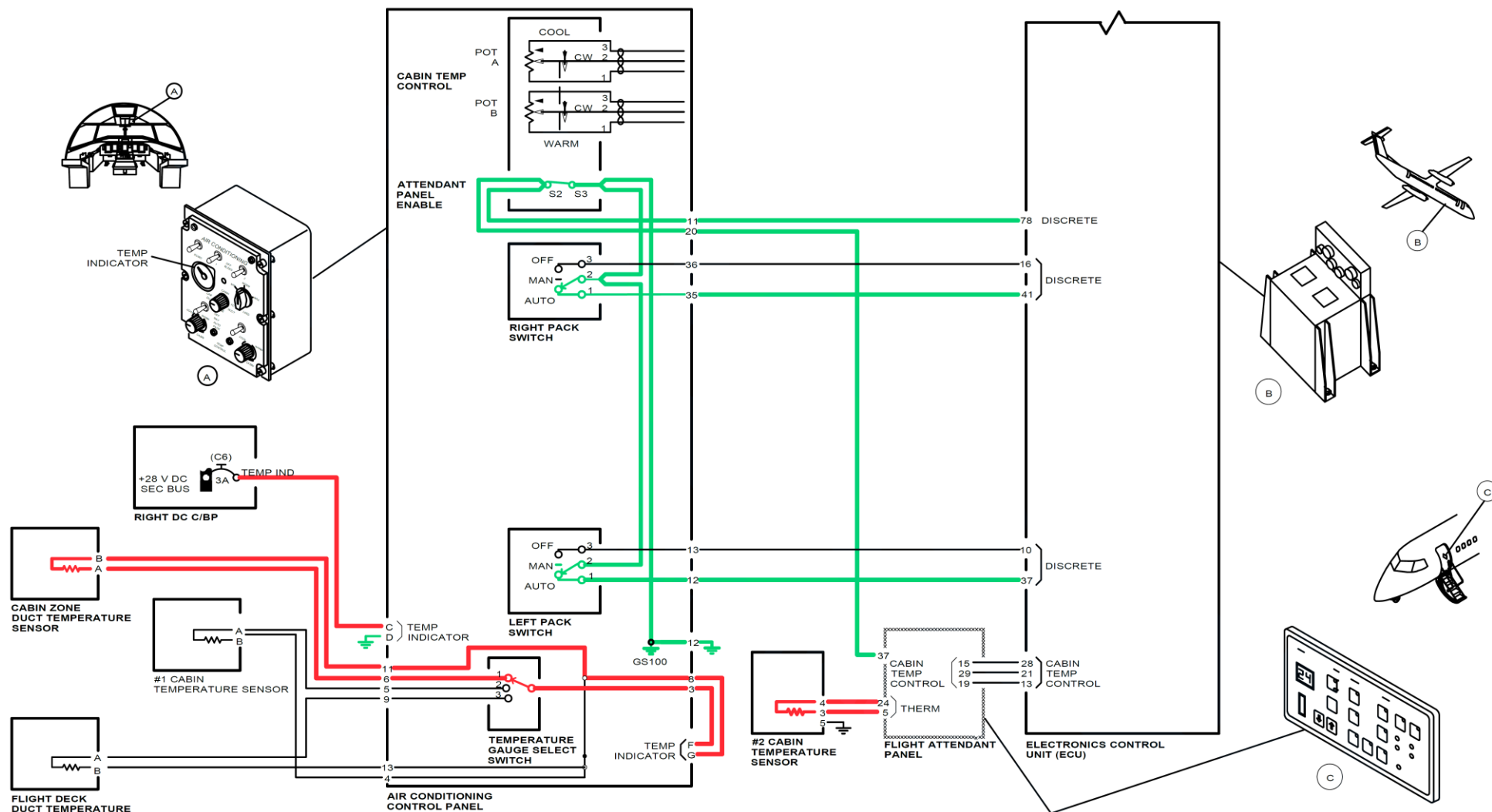


Figure 21- 55 Temperature Control and Indication Electrical Schematic

Component Description

Electronic Control Unit

[Refer Figure 21-56](#)

The ECU is installed in the avionics bay below the cabin floor. It uses the temperature signals from the two zone supply temperature sensors and two of the zone temperature sensors (one each in the flight and cabin compartments) as feedback signals. These feedback signals are used to regulate the air temperatures in the flight and cabin compartments.

The ECU has two channels (left and right). Each channel has a digital (automatic) part that has a power supply, processor, low level Input/Output (I/O) and motor driver board. Each channel also has an analog (manual) part that has an analog backup board.

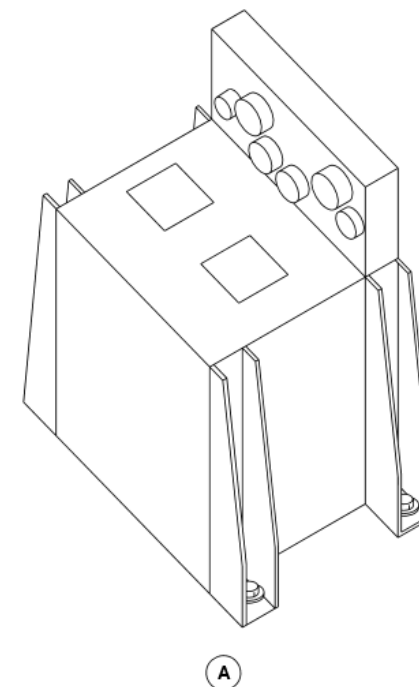
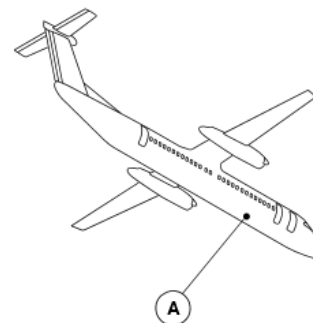


Figure 21- 56 Electronic Control Unit

Zone Temperature Sensors

There are three zone temperature sensors (one in the flight compartment and two in the cabin compartments) that sense the flight and cabin compartment temperatures. The ECU uses the feedback temperature signals from the temperature sensors to regulate the conditioned air temperature. The temperature sensors are dual thermistor sensors.

Flight Compartment Zone Temperature Sensor

[Refer Figure 21-57](#)

The flight compartment temperature sensor is mounted on the inlet housing assembly installed on the co-pilot's rudder pedal cover assembly. The flow of the air past duct inlet draws flight compartment air over the temperature sensor. This makes the temperature sensor to measure accurate ambient air temperature in the flight compartment and provides feedback signal to the ECU for AUTO temperature control.

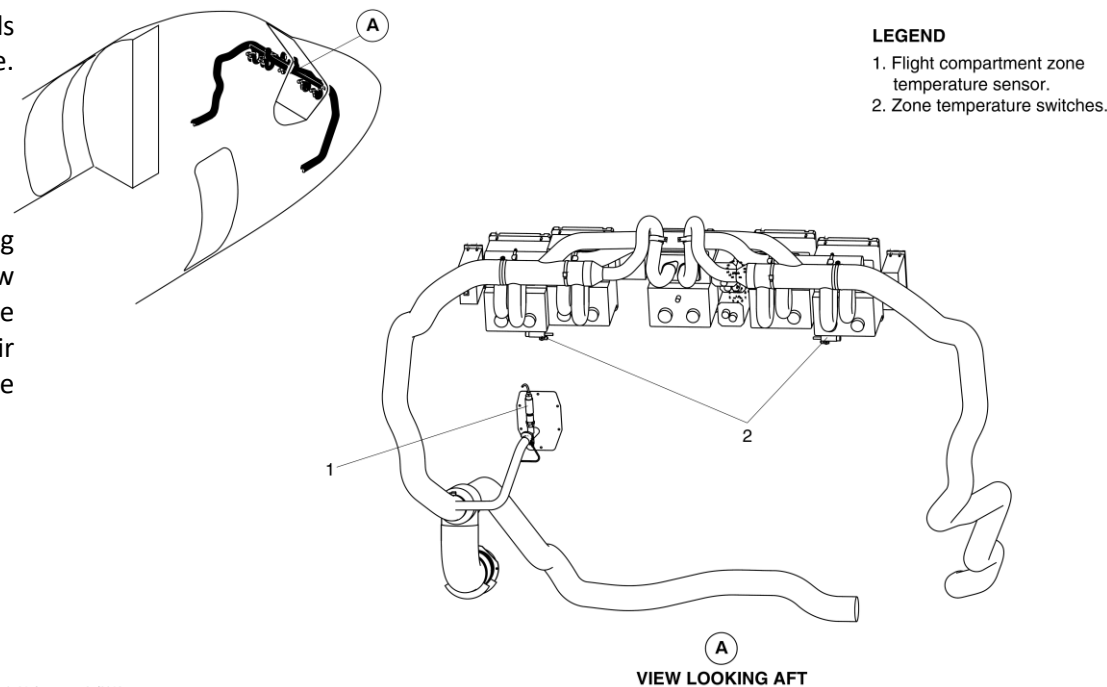


Figure 21- 57 Flight Compartment Zone Temp Sensor

Cabin Zone Temperature Sensors

[Refer Figure 21-58](#)

The cabin zone temperature sensors, located on the cabin temperature ejector by the fourth dado panel (Stn X 352) on the left side of the aircraft, measure the cabin zone temperature and provide signals to the ECU for control and indication.

The zone temperature sensors are dual thermistor sensors. Cabin supply air flows through a restriction in the dado duct. This accelerates the flow of air and causes a low pressure air which draws cabin air over the zone temperature sensor. This ensures an accurate measurement of the cabin ambient air.

The No. 1 cabin temperature sensor acts as a temperature indication sensor. It supplies a signal (cabin temperature) to the temperature gauge on the AIR CONDITIONING control panel.

The No. 2 cabin temperature sensor supplies the signal (cabin temperature) to the temperature gauge on the flight attendant's control panel.

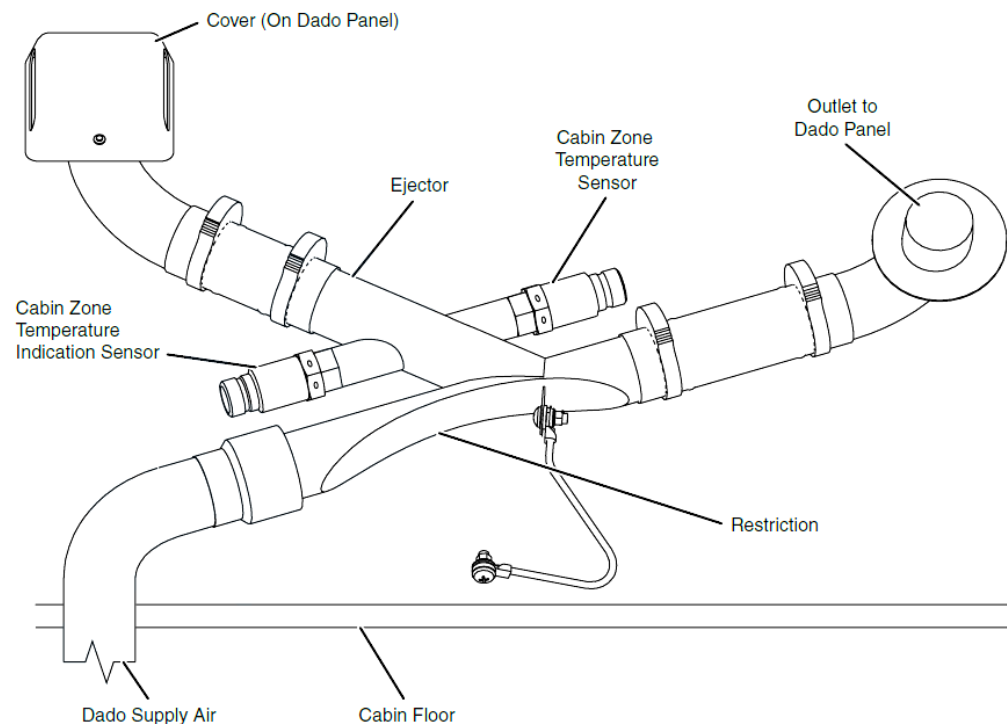


Figure 21- 58 Cabin Zone Temperature Sensors

Zone Supply Temperature Sensors

[Refer Figure 21-59](#)

The zone supply temperature sensors are dual thermistor sensors that supply independent signals to the digital and analog channels of the ECU. These sensors send supply duct temperature information to the ECU so it can control the temperature of the cabin and flight compartment.

The zone supply temperature sensors are installed in the cabin and flight compartment supply ducts under the aft baggage compartment floor (Stn X 765). These ducts supply conditioned air from the air conditioning system to the flight and cabin compartments

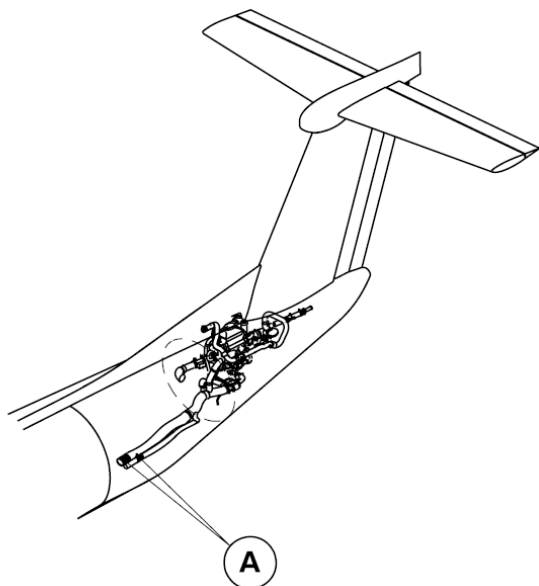


Figure 21- 60 Location of Zone Supply (Duct) Temperature Sensors

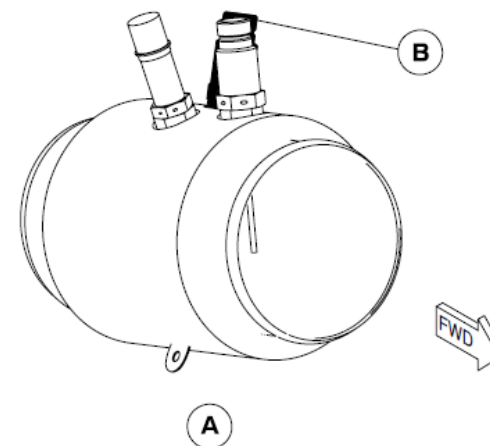
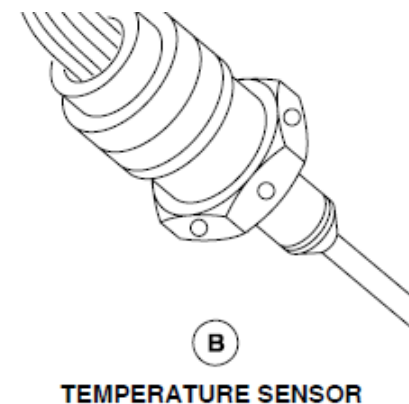


Figure 21- 59 Duct Temperature Sensor



Zone Supply Over-temperature Switches

Refer Figure 21-61

The zone supply over-temperature switches send a discrete signal to the ECU if the temperature in either supply duct is greater than 190 °F (87.8 °C). This signal causes the ECU to turn on the related CABIN DUCT HOT or FLT COMP DUCT HOT caution light on the caution and warning panel. The ECU also shuts off the related air conditioning pack (by closing the turbine and ACM bypass shutoff valves).

The zone supply over-temperature switches are installed in the cabin and flight compartment supply ducts under the aft baggage compartment floor (Stn X 765). These ducts supply conditioned air from the air conditioning system to the flight and cabin compartments.

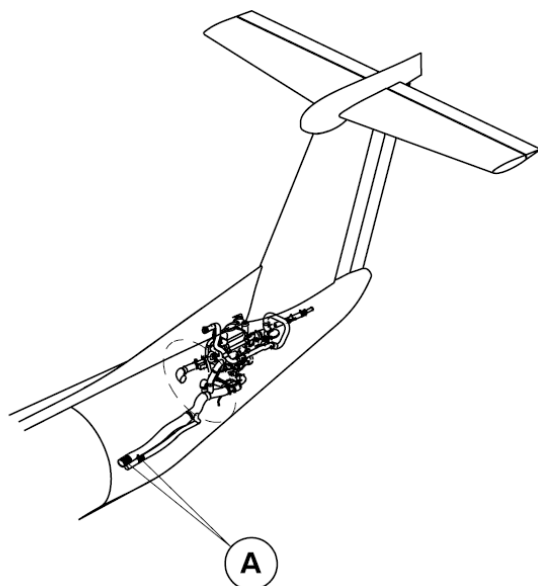


Figure 21- 62 Location of Zone Supply (Duct) Over Temperature Switch

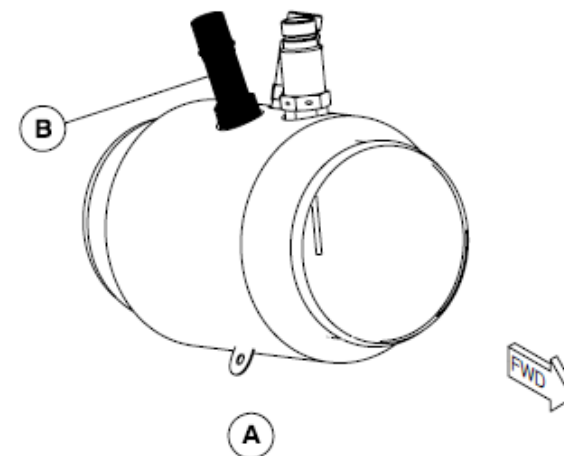
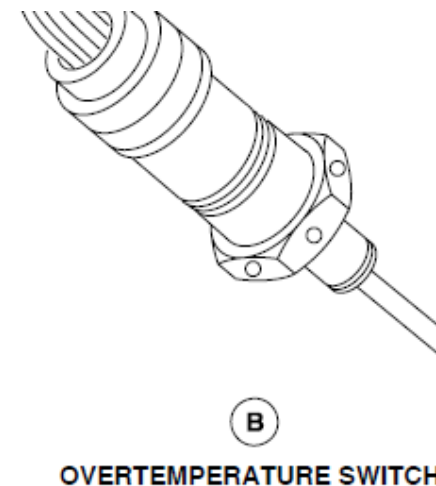


Figure 21- 61 Duct Over Temperature Switch



Distribution

Introduction

The Environmental Control System (ECS) packs supply the flight compartment and cabin air distribution systems with conditioned air. The conditioned air is used for cabin and flight compartment heating, cooling, ventilation and pressurization.

General Description

[Refer Figure 21-63](#)

The air distribution system takes conditioned air from the ECS packs and distributes it throughout the aircraft. Rigid, semi-rigid and flexible ducts are used in the distribution system. The ducts are routed under the cabin floor and have separate distribution systems for the cabin and for the flight compartment.

The conditioned air is distributed to the cabin through lower dado panel vents, overhead vents and passenger gaspers. A recirculation fan helps to boost the supply of conditioned air. A gasper system distributes air to individual outlets in the cabin, flight compartment and the lavatory. Conditioned air is also distributed to the flight compartment through lower vents near the rudder pedals, through side console gaspers, and selectable vents for side window demisting. An avionics cooling system supplies cooling for the avionics equipment bay and the EFIS/ESID liquid crystal displays. An optional ground air conditioning connection is available to supply conditioned air from a ground source.

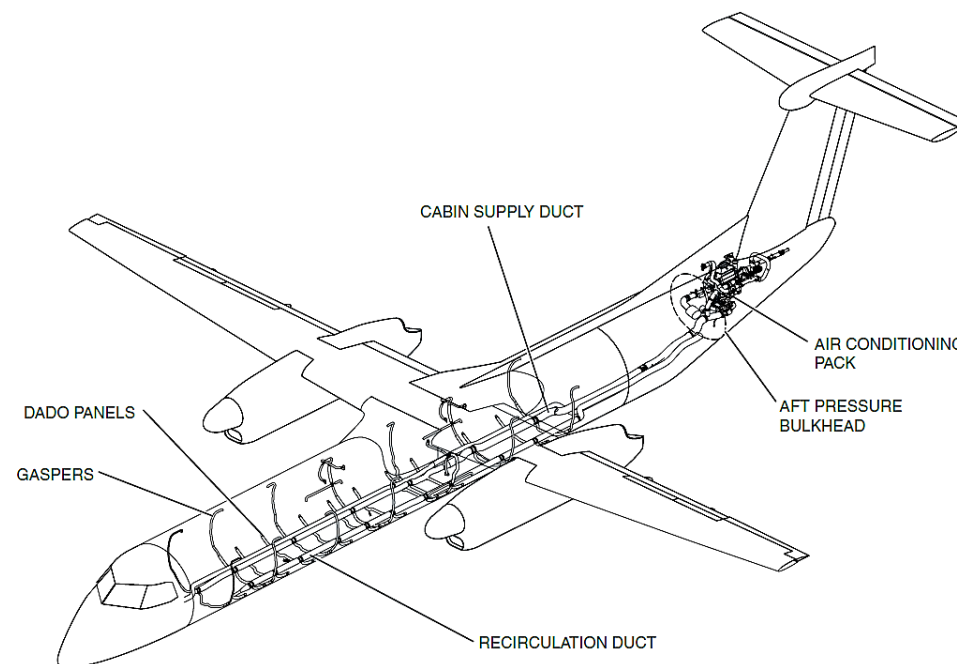


Figure 21- 63 Cabin Air Distribution

Flight Compartment Air Distribution

[Refer Figure 21-64](#)

Air supply to the flight compartment is ducted from the air conditioning pack, through the rear pressure bulkhead, to the left side of the splitter duct. From the splitter duct, the flight compartment air is ducted along the right side of the aircraft below the cabin floor. Before reaching the flight compartment, the distribution system also supplies conditioned air to the aft baggage compartment inlet, forward lavatory gasper and the forward flight attendant's gasper.

At the flight compartment bulkhead, a wye duct splits the airflow into two individual but identical distribution systems, one for the pilot side and the other for the copilot side. In the flight compartment, the distribution system has lower level and upper level outlets. The upper level outlets are demist nozzles for the pilot's and copilot's side windows. The lower level outlets include a foot warming piccolo tube (near the rudder pedals), a fixed grille near knee height and a large torso gasper. Flow control levers located near the window sill allow the pilot and copilot to control the flow of air into the flight compartment. A small manually controlled gasper at window height is also supplied.

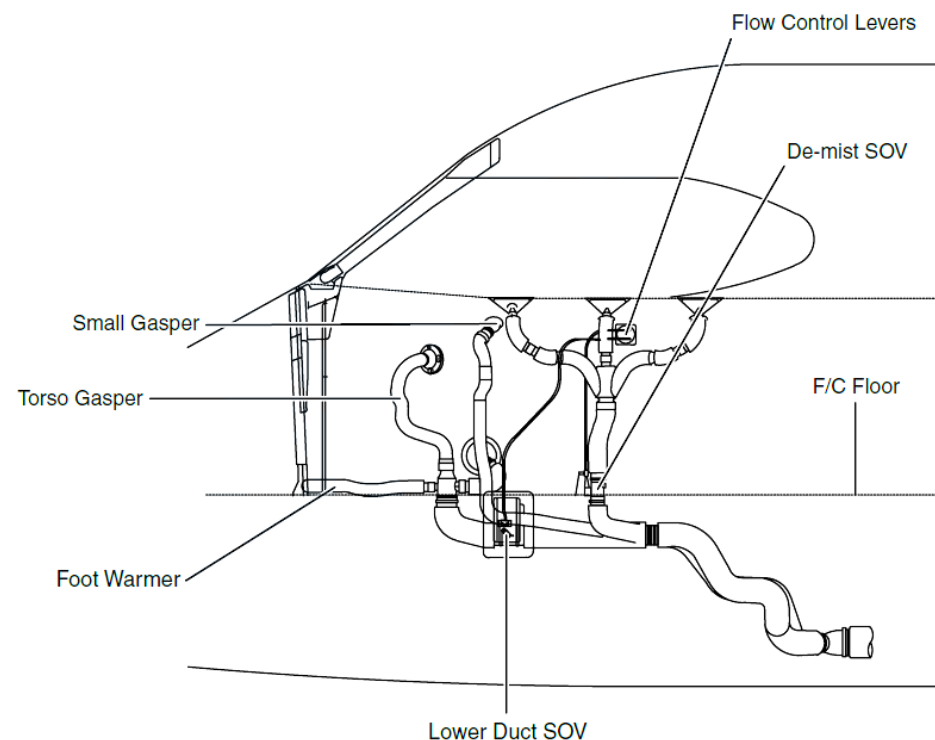


Figure 21- 64 Flight Compartment Air Distribution

Cabin Air Distribution

[Refer Figure 21-65 & 21-66](#)

Conditioned air is supplied to the cabin to maintain a comfortable environment for the passengers and crew. Conditioned air is also used for aircraft pressurization.

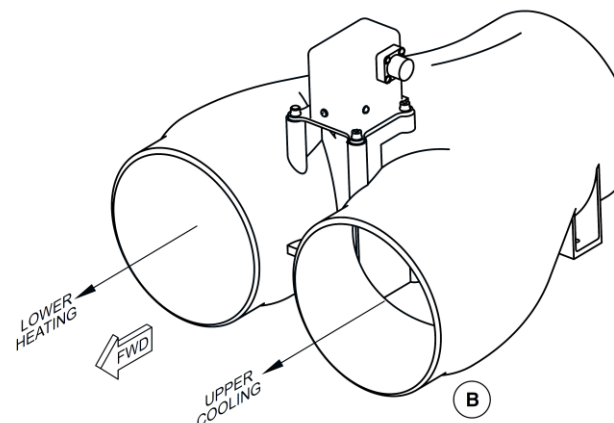
Air supply to the cabin is ducted from the air conditioning pack into the fuselage at the centre of the rear pressure bulkhead. The air is then ducted under the baggage compartment floor, where it splits into an upper and lower supply duct for each side of the fuselage. The lower duct supplies air to the dado panel vents, while the upper duct supplies air through sidewall risers to the overhead vents and gasper system. A distribution damper is set automatically depending on the cabin duct temperature.

The left side of the damper supplies the upper cabin distribution duct which is routed under the cabin floor on the left side of the aircraft. The right side of the damper supplies the lower cabin distribution duct which is routed under the cabin floor on the right side of the aircraft. The upper cabin distribution duct supplies the Passenger Service Unit (PSU) gaspers and the sidewall downwash and ceiling upwash vents. The lower cabin distribution duct supplies the dado panels.

The right digital channel of the ECU controls the distribution damper. The ECU uses the signal from the cabin zone supply temperature sensor to determine which mode to apply to the distribution damper. In the heating mode, the distribution damper sends more air to the lower cabin distribution duct (dado panels) and less air to the upper cabin distribution duct (PSU gaspers). In the cooling mode, the distribution damper reverses the proportions. During standard temperature, the distribution damper divides the air equally between the upper and lower cabin distribution ducts.

The baggage compartment has solenoid controlled inlet and outlet valves (ventilation control valves), which close when smoke is detected in the

compartment and or when electrical power is lost.



NOTE

Depending on Cabin Zone Supply Sensor (Right Channel in ECU):

- Upper for Cooling, Lower for Heating.
- Standard - 50% Lower, 50% Upper.
- Heating - 70% Lower, 30% Upper.
- Cooling - 30% Lower, 70% Upper.

Figure 21- 65 Distribution Damper

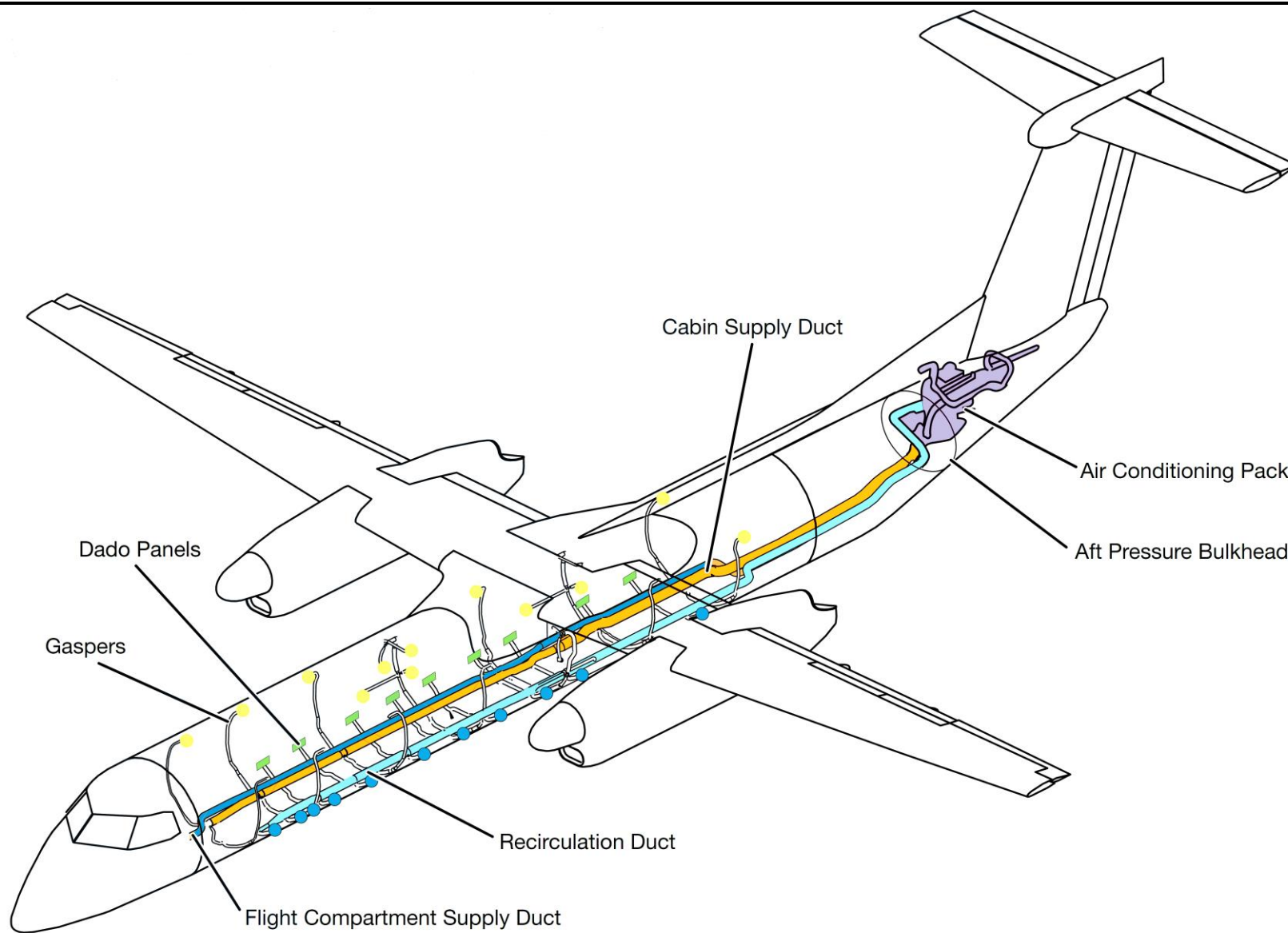


Figure 21- 66 Cabin Air Distribution

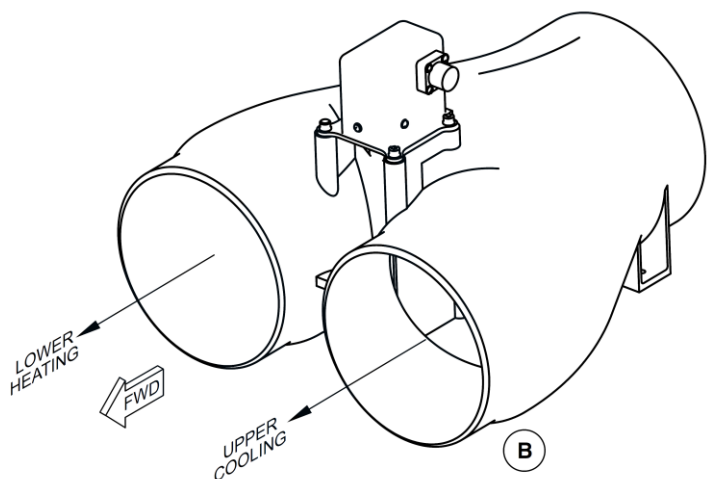
Component Description

Distribution Damper

[Refer Figure 21-67](#)

The distribution damper is an electronically controlled and electrically actuated valve with a 7 in. (177.8 mm) diameter inlet and two 6.5 in. (165.1 mm) diameter outlets. It is installed just aft of station X604.5.

The right digital channel of the ECU controls the electric motor of the distribution damper valve. If the right digital channel or the electric motor fails, the damper valve will remain in its last position. Two position switches in the damper valve send discrete signals to the ECU indicating whether the valve is in the full warm of the full cool position.



NOTE

Depending on Cabin Zone Supply Sensor (Right Channel in ECU):

- Upper for Cooling, Lower for Heating.
- Standard - 50% Lower, 50% Upper.
- Heating - 70% Lower, 30% Upper.
- Cooling - 30% Lower, 70% Upper.

Figure 21- 67 Distribution Damper

Cabin Air Distribution Electrical Schematic

[Refer Figure 21-68](#)

Operation of the switches that follow on the AIR CONDITIONING control panel operate the ECS:

- The BLEED switches are set to the 1 and/or 2 position.
- A PACKS switch is set to the AUTO position.
- The distribution damper is controlled by the ECU (connector J5 pins 29, 52 and 68). The damper controls the airflow in the positions that follow:
 - Warm operation; 70% of the warm air goes to the lower vents and 30% goes to the overhead vents.
 - Cooling operation; 30% of the airflow goes to the lower vents and the overhead receives 70%.
 - At standard temperatures air is directed evenly (50-50%) to the overhead and lower vents.
 - If the right digital channel in the ECU fails, the damper valve can not change position.

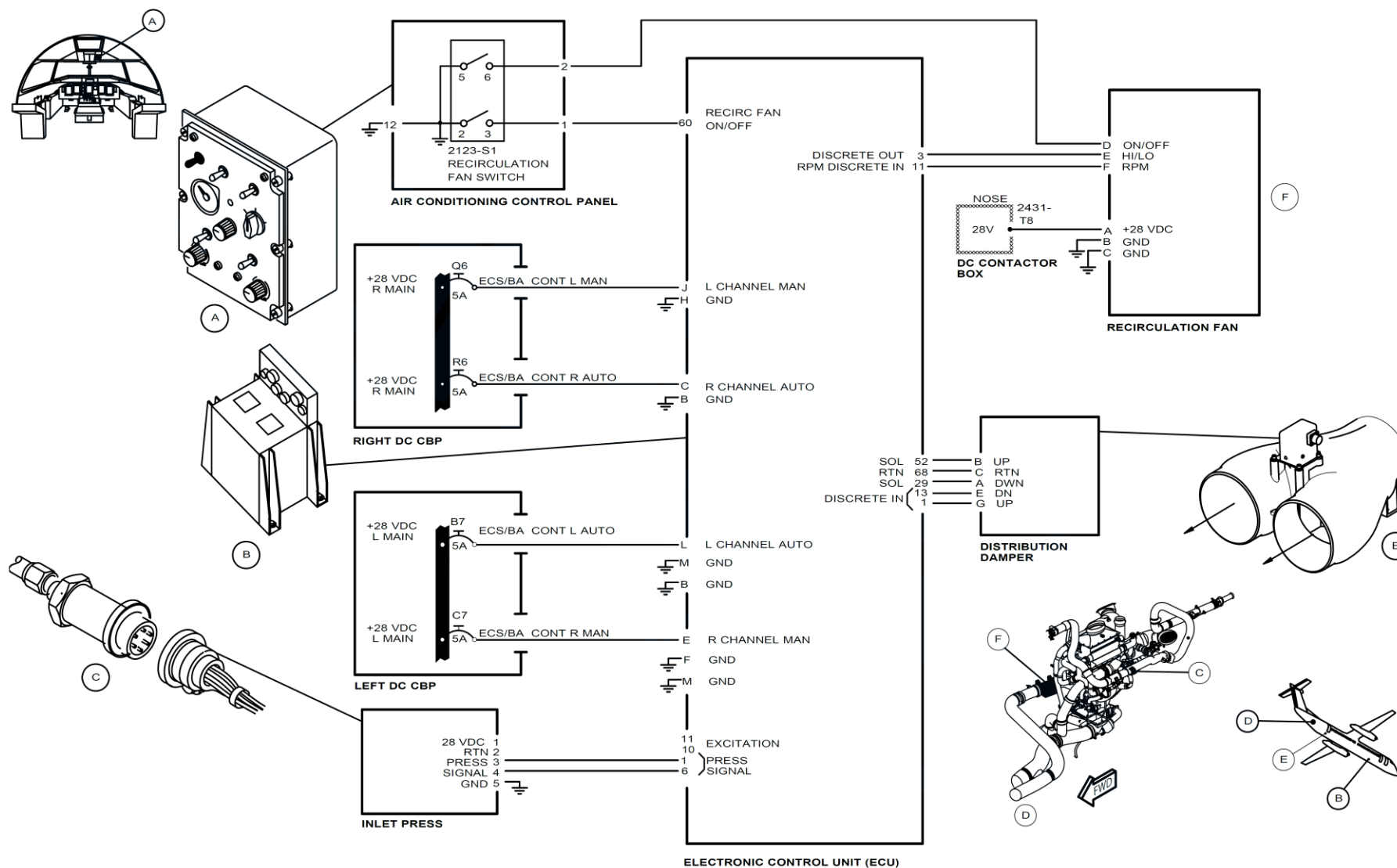


Figure 21- 68 Cabin Air Distribution Electrical Schematic

Aft Baggage Compartment Vent Valves

[Refer Figure 21-69](#)

The aft baggage compartment vent valves are installed in the aft baggage compartment.

One valve is installed at the inlet supply below the floor on the right hand fuselage. The other vent is installed at the outlet on the left hand fuselage.

These valves allow the supply of conditioned air to the baggage compartment when the ECS packs are in operation.

When smoke is detected in the aft baggage compartment, the control amplifier will remove electrical power from the aft baggage vent inlet and vent outlet shutoff valves. This will minimize the air flow within the baggage compartment. Indication of vent valve status will be shown as VENT INLT and VENT OTLT lights on the Fire Protection Panel.

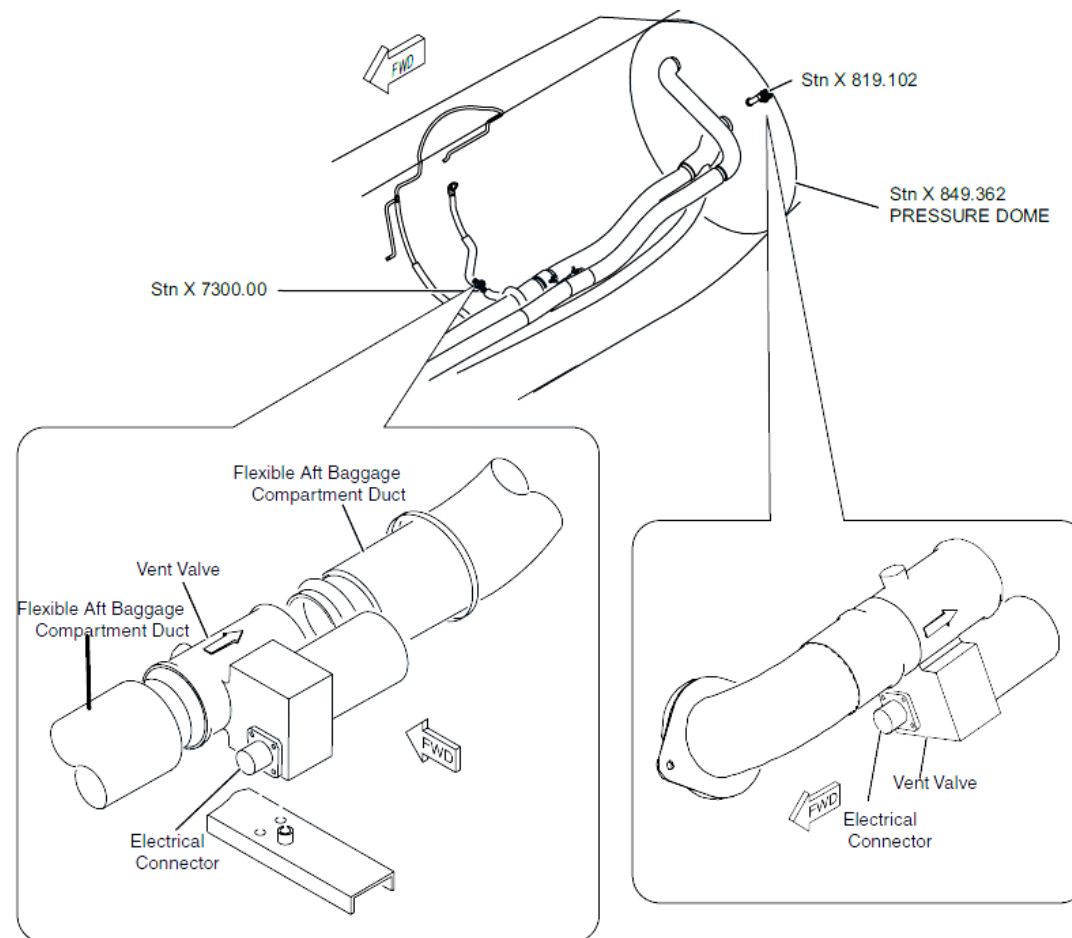


Figure 21- 69 Aft Baggage Vent Valves

Ground Air Conditioning Connection

[Refer Figure 21-70](#)

The ground connection is a duct, accessed through a circular shaped, hinged, and sealed door, located at the rear fuselage right hand side.

The connection point has a cover to prevent contamination from entering the system. The 5" diameter check valve prevents pressurized air from the air conditioning packs or the emergency ram air system from venting overboard. When there is no ground conditioned air pressure available, the check valve is held closed by a spring to prevent dirt from entering the system.

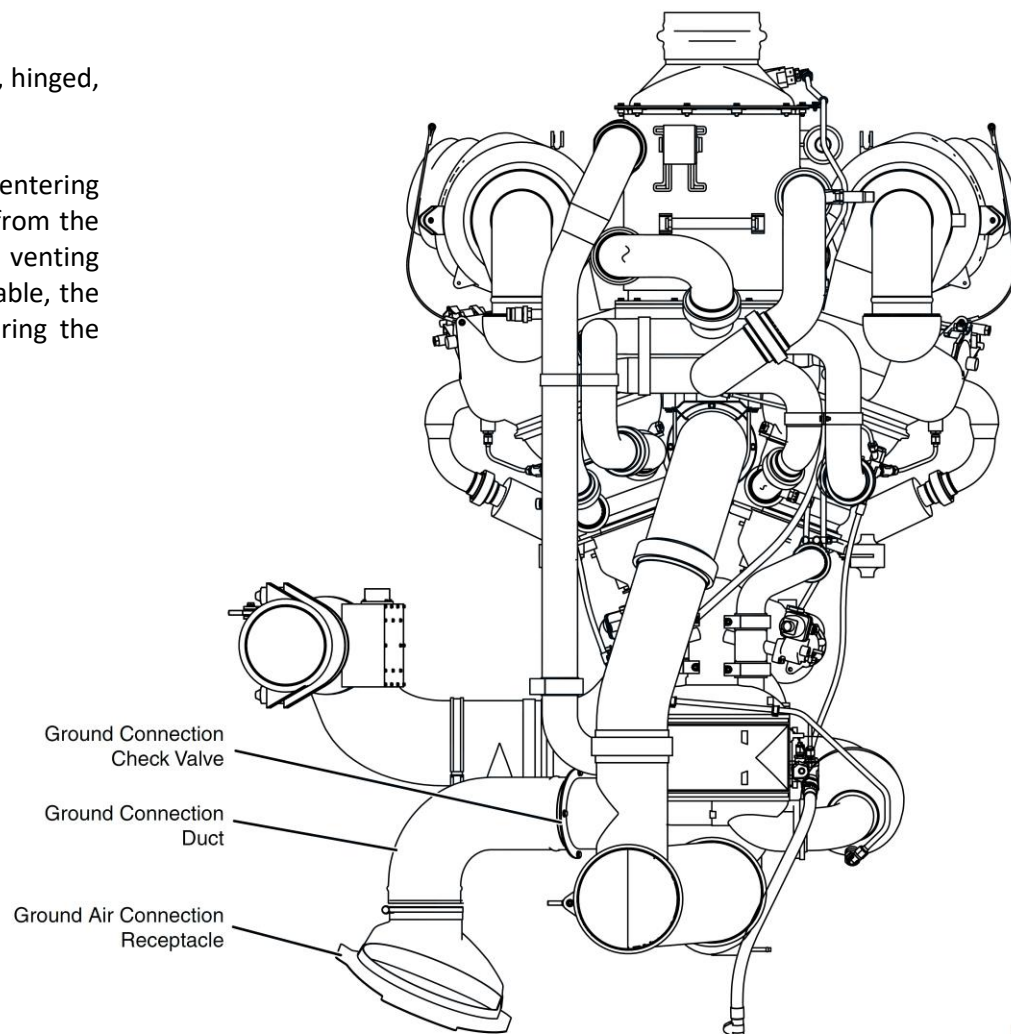


Figure 21- 70 Ground Air Conditioning Connection

NOTE

View looking aft.

Splitter-Distribution Duct

The splitter duct is installed in the aft baggage compartment below the floor panels and behind the canted frame bulkhead panel. The flight compartment and cabin air flows are separated by the splitter duct. The splitter duct is a semi-rigid bifurcated duct made from silicone impregnated glass cloth.

Distribution-Noise Treatment Duct

The distribution noise treatment duct reduces the overall cabin noise coming from the Air Cycle Machines (ACMs). The duct is a perforated aluminum tube surrounded by an insulating material and silicone impregnated glass cloth. The duct is installed in the cabin supply ducting between station X664.5 and X683.5.

Avionics Cooling

Introduction

The avionics cooling system has two independent duct networks and three fans. The system removes heat from the avionics rack, the wardrobe rack and the five liquid crystal displays. This improves the reliability and availability of the avionics systems.

General Description

[Refer Figure 21-71](#)

The aircraft has an extraction type cooling system for the avionics and liquid crystal displays (LCDs). Control of the system is automatic and requires no pilot action for both normal and abnormal operation. The cooling system has three identical fans, each of which can supply half the required cooling flow. Only two of the three fans are required to be operational for dispatch. The electrical and ducting systems have been designed so that single failures do not result in the loss of all the displays.

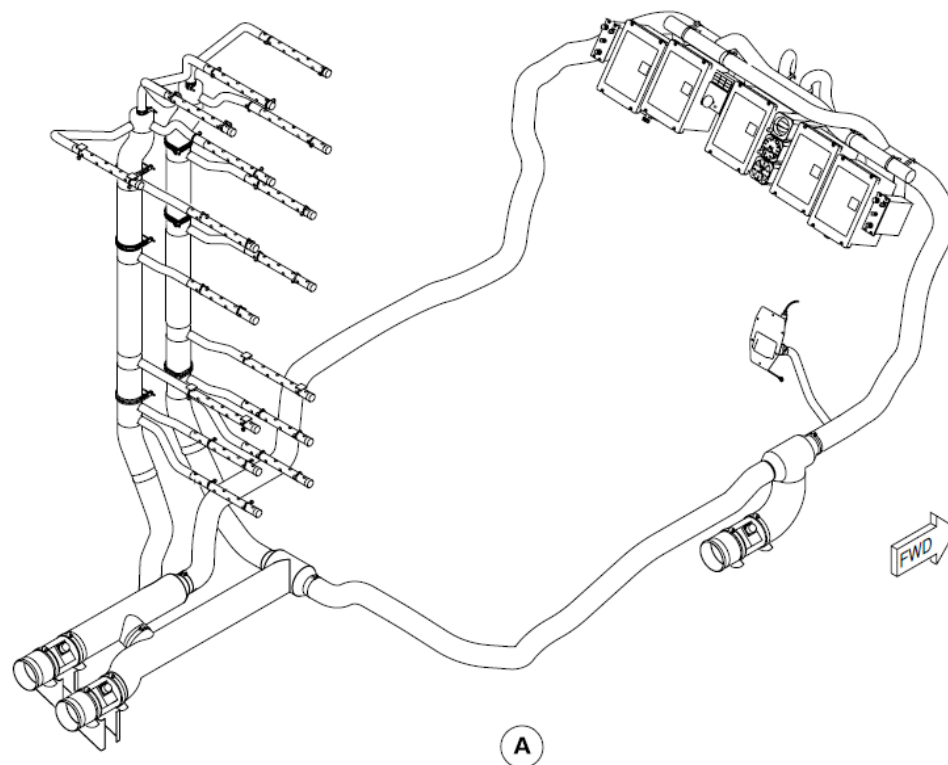
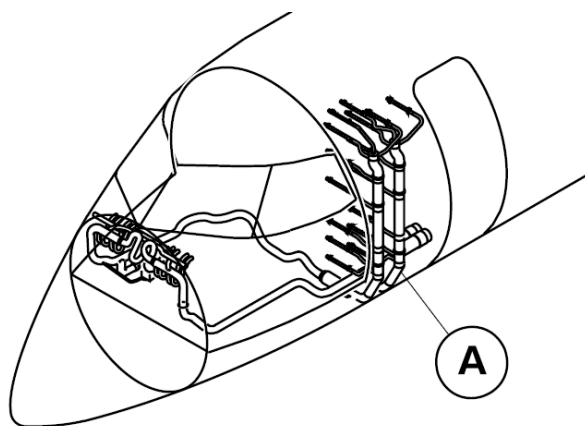


Figure 21- 71 Avionics Cooling Ducts

Detailed Description

Refer Figure 21-72

The avionics cooling system has dual redundant duct assemblies. Each duct assembly has a cooling fan and branches that are connected to:

- 5 Liquid Crystal Displays (LCDs)
- Avionics Rack
- Wardrobe Rack.

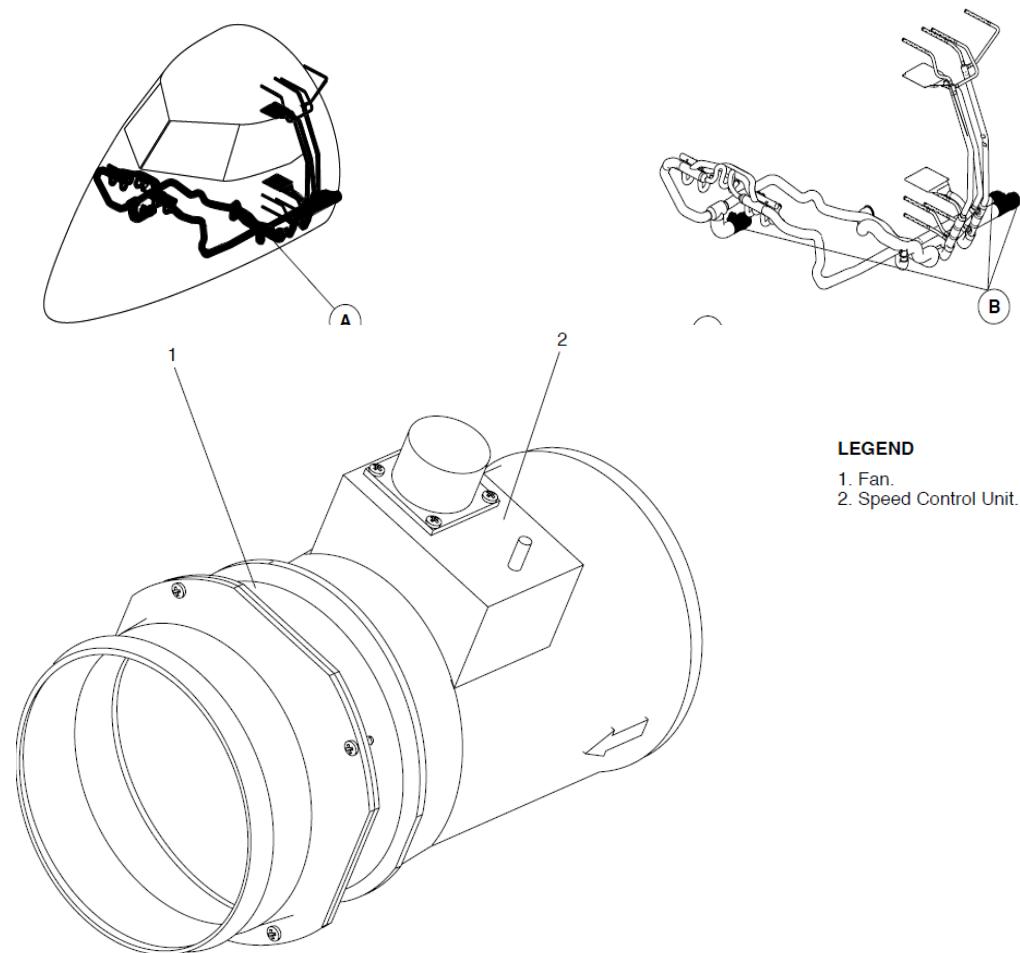
One of the duct assemblies is routed to the instrument panel from the pilot side, and the other assembly is routed from the copilot side. The duct assembly on the copilot side also has the standby fan #3 attached. The hot air is exhausted under the floor behind the flight compartment.

The cooling system removes the hot air produced by the five LCDs in the instrument panel, the avionics rack and the wardrobe rack. Fan #1 and fan #2 are running whenever the electrical power is applied to the aircraft dc main bus. If either of these fans fails, the standby fan #3 automatically starts operating. Each fan operating at high speed supplies one half of the required air flow. When the standby fan takes the place of the failed fan, the cooling system becomes fully operational again.

Because of the location and the high density design of the LCD displays, the operational time of the equipment without forced air cooling is limited to 60 minutes. When the internal temperature increases, the displays will reduce internal power dissipation by reducing the screen brightness to 60%. This reduction in screen brightness increases display availability. If there is the possibility of a misleading or total loss of display due to internal temperature increases, the display unit will shut down. In case of electrical power failure, when only the battery power is available, fan #1 operates at low speed and fan #2 turns off. If fan #1 is not available, fan #2 operates at low speed. The standby unit fan #3 is not available when the main buses are not powered. This operational mode is capable of supplying the minimum air flow

required for the LCDs to operate at reduced brightness. With minimum air

flow and reduced brightness, the LCDs will operate with higher internal temperatures, but below the automatic shutoff threshold.



LEGEND

1. Fan.
2. Speed Control Unit.

Figure 21- 72 Avionics Cooling Fan

Refer Figure 21-73

The avionics cooling fan has an electronically controlled, two speed, brushless DC motor. Each fan has a built-in electronic low speed warning detector (LSWD) which sends a signal to the input/output module (IOM) when the fan is rotating slower than 8000 rpm. The IOM uses the low speed warning signal to activate the standby fan if there is a failure of fan #1 or fan #2. The IOM continuously monitors the fans for signals from the LSWD. Both IOM 1 and IOM 2 monitor all three fan fail outputs. Failure of any fan is recorded in the central diagnostic system (CDS) for further maintenance action. There is no indication of a single fan failure to the flight crew. If the failure of two fans occurs while on the ground, an advisory message FANS FAIL is sent to the engine display (ED) to inform the pilot that dispatch is not permitted. At the same time, the IOM turns on the AVIONICS caution light on the caution and warning panel. If the aircraft is in the air when a double fan failure occurs, the AVIONICS caution light does not come on and an advisory message FANS FAIL is sent to the ED.

A fan speed control unit is installed on the side wall of each of the three avionics cooling fans. At start-up, electrical power is supplied to all three fans. If there is no fan fail signal from the LSWD, fan #3 will shut down and fan #1 and fan #2 will operate in high speed mode (HSM). This is to ensure that a hidden failure does not exist in any of the three fans.

When dc generated power is available and the aircraft is in flight (i.e., no weight on wheels signal), the fan speed control unit operates as follows:

- Three fans available, two fans operating at HSM
- Two fans available, two fans operating at HSM
- One fan available, one fan operating at HSM.

If only battery power is available, the system automatically selects an operational fan (fan #1 or fan #2) based on signals from the LSWD. The selected fan operates from battery power at low speed mode (LSM). A fan operating at LSM can supply enough airflow to meet the avionics and LCD cooling requirements while still meeting the battery loading requirements. The standby fan #3 is not available when operating on battery power.

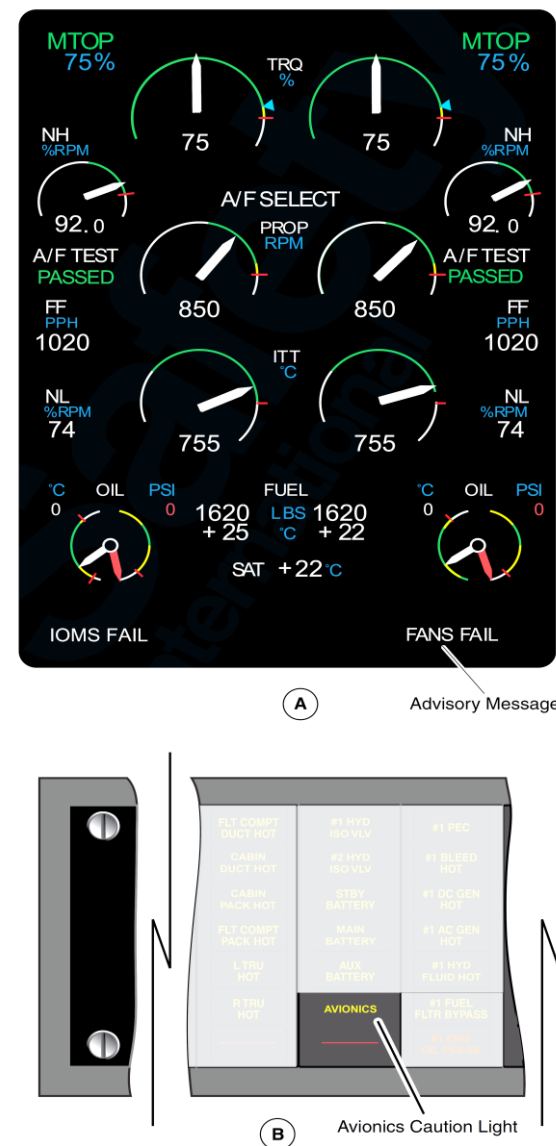


Figure 21- 73 FANS FAIL message and AVIONICS caution

[Refer Figure 21-74](#)

Zone Temperature Sensors

The flight compartment zone temperature sensor is mounted on the inlet housing assembly installed on the co-pilot's rudder-pedal cover assembly. The flow of the air past duct inlet draws flight compartment air over the temperature sensor. This makes the zone temperature sensor to measure accurate ambient air temperature in the flight compartment. The electronic control unit (ECU) uses the feedback temperature signals from the zone temperature sensor to regulate conditioned air temperature within the flight compartment. The zone temperature sensor is a dual thermistor sensor.

Zone Temperature Switches

The ducting for each avionics cooling fan has a connection to one of two zone temperature switches located under the LCDs. These switches inhibit the fan operation on the ground, when the flight compartment temperature is below 41 °F (5 °C). This allows the LCD internal heater to operate during cold day starts without interference from the fans. The temperature switches are disabled by a relay if the WOW signal indicates that the aircraft is airborne. Each zone temperature switch has a second section which sends a signal to the IOM. The signal causes the IOM to inhibit the recording of a fan failure by the CDS when the temperature is below 50 °F (10 °C).

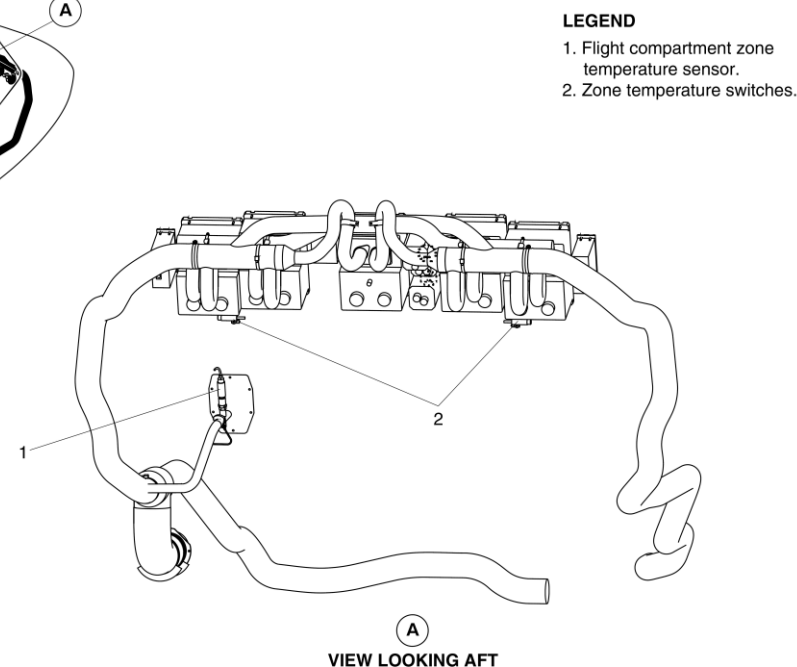
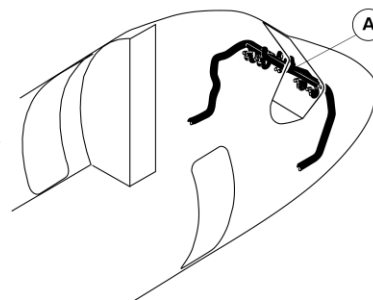
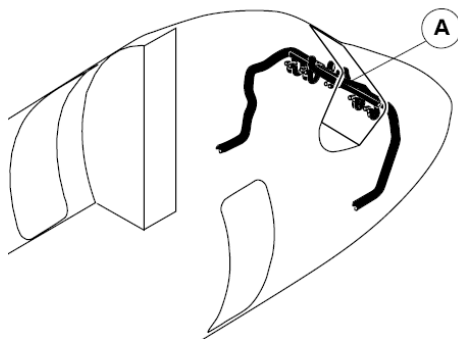


Figure 21- 74 Zone Temp Sensors and Switches

Avionics Cooling Electrical Schematic

[Refer Figure 21-75](#)

All three fans have dual redundant power supply connections and power selection is controlled by relays. Relay 2126-K1 controls power to fan #1 and relay 2126-K2 controls power to fans #2 and #3. When 2126-K1 is energized by the left main bus, fan #1 is then powered by the left main bus.

If power from the left main bus is lost, 2126-K1 is de-energized and fan #1 is powered from the left essential bus. Power from the right main bus energizes relay 2126-K2, which connects power from the bus to fan #2. If power from the right main bus is lost, 2126-K2 is de-energized and fan #2 connects to the right essential bus. Fan #3 receives power from the right main bus through relay 2126-K3, but if the right main bus is lost, de-energizing 2126-K3, fan #3 receives power from the left main bus. Each fan has an ON/OFF control input and a HI/LO control input. If the ON/OFF input is a grounded circuit, the fan stops running and if the input is an open circuit, the fan will run at high speed. An open HI/LO control input circuit will cause the fan to operate at a high speed and a grounded input causes the fan to operate at a low speed. Each fan has one fan fail output for monitoring. A ground output signal indicates normal operation and an open output signal indicates a fan fail.

An ON/OFF line connects each fan to one of the two zone temperature switches. These switches inhibit fan operation on the ground when the cockpit temperature is below 41 °F (5 °C) so the fans do not interfere with the operation of the LCD display internal heater during cold day starts. Both switches have an output set at 41 °F (5 °C) and one at 50 °F (10 °C). When the temperature rises above each of these temperatures, the switch becomes an open circuit. The difference in output signals allows for sensor tolerances. The landing gear system supplies a WOW signal to the cooling system through relay 2126-K5. This signal inhibits the temperature sensors

in flight. When the WOW input changes to flight mode, relay 2126-K5 de-energizes and both switch outputs are disconnected.

IOM modules of the avionics receive the fan fail outputs of each fan and record the failures in the central diagnostic system. These cooling system connections are for monitoring purposes only and do not affect the cooling system logic.

During normal operation, power is supplied by the left and right main buses to relays 2126-K1, 2126-K2 and 2126-K4. Relay 2126-K1 connects fan #1 to the left main bus and 2126-K2 connects fan #2 to the right main bus. With power applied to relays 2126-K1 and 2126-K4, contacts through these relays become open circuits which set fan #1 and fan #2 to high. When a fan runs above the speed threshold, the fan fail output becomes shorted to ground. Similarly, grounding of the fan #2 fail output will energize relay 2126-K3. Fan #3 ON/OFF control is turned off when a grounded circuit input is received through relay 2126-K3 and the fan #1 fail output.

If fan #1 fails, the fan #1 output circuit becomes open and fan #3 is activated. If fan #2 fails, relay 2126-K3 is de-energized, which activates fan #3.

Loss of both main buses de-energizes relays 2126-K1, 2126-K2 and 2126-K4. Fan #1 then receives power from the left essential bus but with the HI/LO control input grounded, which puts fan #1 in the LSM. The LSM operation is used to minimize electrical power consumption. Fan #1 fail output is grounded when fan #1 is running in LSM. Through CR1 and relay 2126-K4, the fan #2 ON/OFF control is grounded and turns fan #2 off. With both main buses unavailable, fan #3 is not powered. If fan #1 fails, the fan #1 fail switch becomes open circuit which activates fan #2 through CR1 and relay 2126-K4. Fan #2 becomes powered by the right essential bus through relay 2126-K3. Because relay 2126-K4 is de-energized, fan #2 runs at low speed.

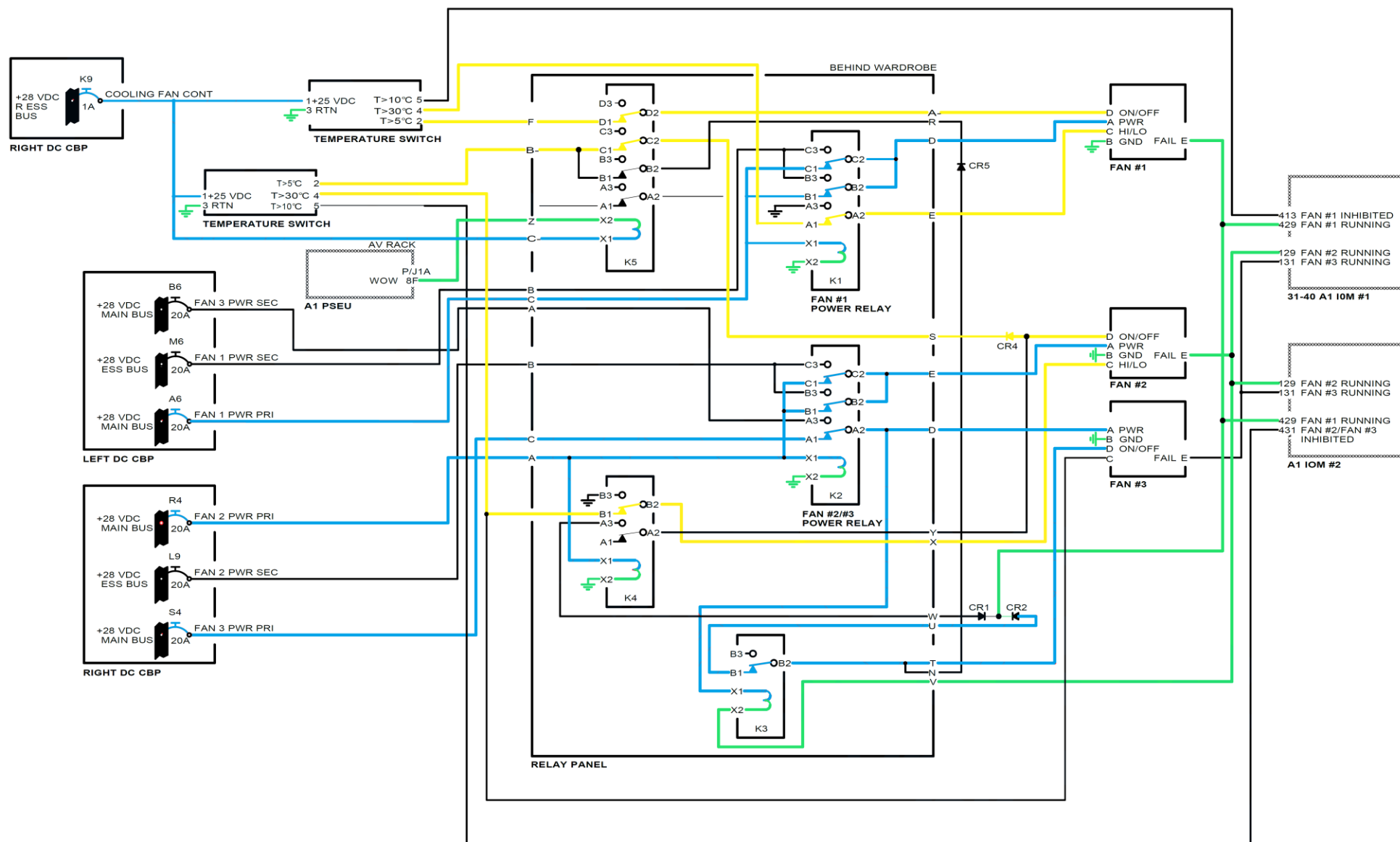


Figure 21- 75 Avionics Cooling Electrical Schematic

Pressurization Control

Introduction

The cabin pressure control system keeps a constant cabin pressure during the ground and flight modes. The system controls the cabin altitude and the cabin rate of change.

General Description

[Refer Figure 21-76](#)

The pressurized area of the fuselage extends from the front pressure bulkhead immediately forward of the windshield at aircraft station X -111 to the rear dome at Sta X 856. This area includes the flight compartment, passenger compartment, under floor space and baggage compartment.

The Cabin Pressure Control Panel controls the system in automatic and manual modes. The Cabin Indication Module shows the data to monitor the system. When cabin altitude is too high, the cabin pressure and the cabin indication module send the warning signal to the aircraft systems.

The Cabin Pressure Control Panel controls the system in automatic and manual modes. The Cabin Indication Module shows the data to monitor the system. When cabin altitude is too high, the cabin pressure and the cabin indication module send the warning signal to the aircraft systems.

Electrical power supply for pressurization control is supplied by the 28 VDC left main bus through a CABIN PRESS AUTO CONT circuit breaker.

The aft outflow valve is used for automatic and manual control of the pressurization system.

The forward safety valve is used for emergency operation and for smoke removal from the flight compartment. It also protects the aircraft against excessive positive and negative air pressures.

The aft safety valve protects the aircraft against excessive positive and negative pressures and assists in cabin pressure equalization after aircraft lands.

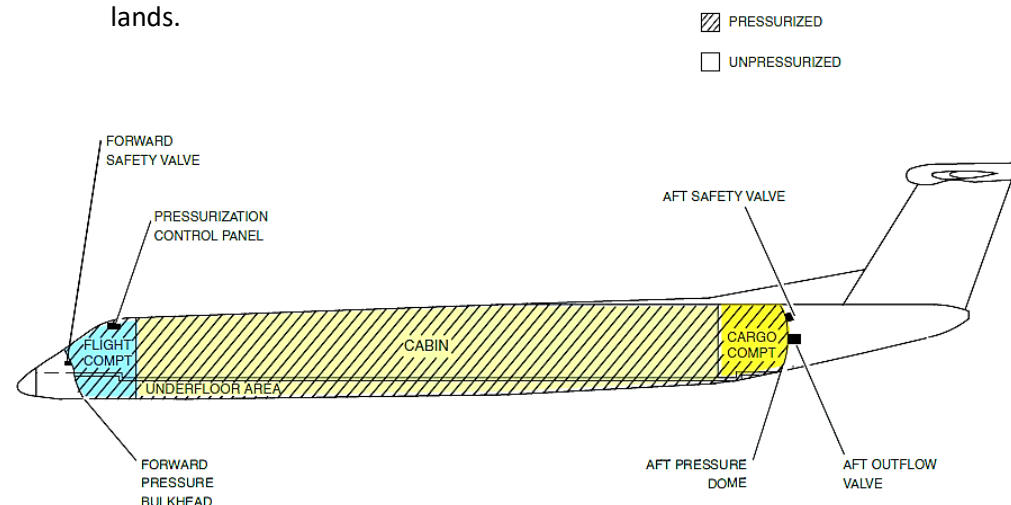


Figure 21- 76 Fuselage Pressurized Area

The pressurization control system has the components that follow:

- Cabin Pressure Control Panel
- Cabin Pressure Controller
- Cabin Indication Module
- Aft Outflow Valve
- Forward Safety Valve
- Forward Safety Valve Selector
- Aft Safety Valve
- Air Jet Pump

Detailed Description

[Refer Figure 21-77](#)

The system shows in the flight compartment:

- Cabin altitude
- Cabin altitude rate of change
- Differential pressure
- Cabin altitude warning

The aft outflow valve is for automatic and manual control of the pressurization. The aft outflow valve can also be used to dump the pressurization. The forward safety valve is for emergency operation and for smoke removal from the flight compartment. The aft safety valve and the forward safety valve keep the positive and negative pressure relief constant.

The system functions in the modes that follow:

- Normal/Automatic
- On ground
- Take-off
- Landing

- Emergency/Smoke Removal
- Pressure dump

Normal/Automatic: The pressurization control system is electrically operated. The pressurized area of the fuselage is supplied with a constant flow of conditioned air from the engine bleed air systems through the Air Conditioning Unit (ACU). The pressure in the fuselage is controlled by system modulation of the aft outflow valve, to control the amount of air let out of the fuselage. If the external ambient pressure is more than the fuselage pressure, the safety valves will open for negative pressure relief.

When electrical power is first supplied to the system, a full electrical test of the system is done. The FAULT alert light, on the Cabin Pressure Control Panel comes on during the power up test mode. If there is a failure in the system, the light will stay on. The system operation is fully automatic with the data programmed into the controller.

On ground: When the aircraft is on the ground with the weight on the wheels and the No. 2 engine power lever angle is set less than 60 degrees, electrical power is supplied through the open contacts of the energized landing gear relay. The relay is energized through the Proximity Switch Electronic Unit (PSEU). During the ground mode the aft outflow valve is at the fully open position to prevent aircraft pressurization.

Take-off: When the No. 2 engine power lever angle is set to greater than 60 degrees the controller sends a signal to the aft outflow valve to open or close, as necessary, to pressurize the aircraft to 400 ft (121.9 m) less than ambient. The aft outflow valve moves from the fully open position and starts to modulate to control the pressure changes that occur after take-off. When the landing gear relay is de-energized after take-off (through the PSEU), the aft outflow valve modulates to keep the set aircraft pressure.

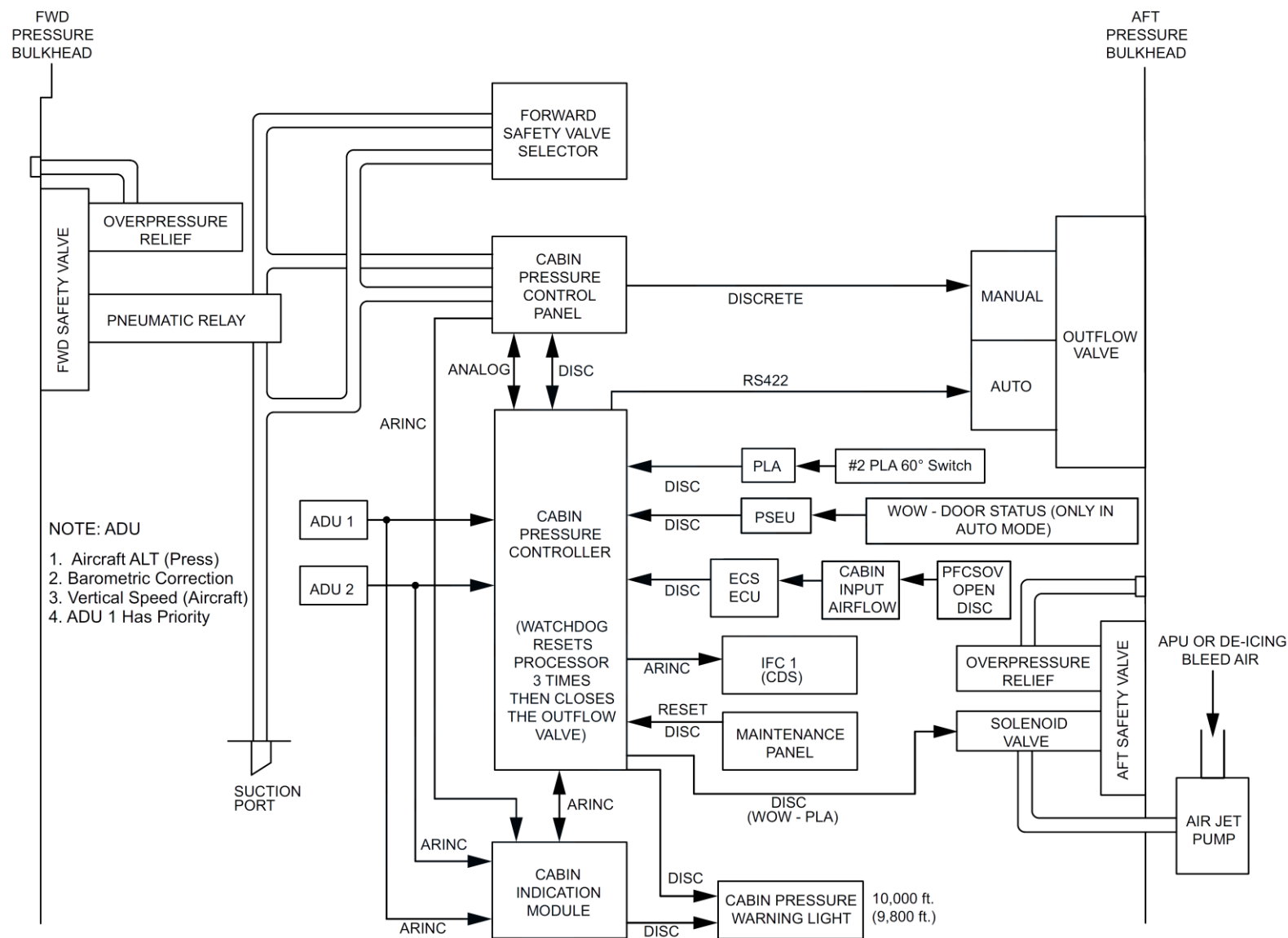


Figure 21- 77 Pressurization Control

Landing: The aircraft depressurization is controlled automatically. If the set field altitude is higher than actual field altitude, the aircraft will land unpressurized. If the field altitude is set less than actual field altitude, the aircraft will land pressurized. On landing, cabin altitude will go back to field altitude at the rate programmed into the controller, for one minute before cabin pressure is bled to ambient.

Manual: The manual mode is used if the automatic pressurization mode does not operate. Pressurization can be controlled through the aft outflow valve, when the AUTO-MAN-DUMP switch is set to MAN. The cabin pressure is set with the toggle switch moved and held to the DECR position, to open the aft outflow valve and increase the cabin altitude. When the toggle switch is moved and held to INCR, the aft outflow valve closes and the cabin altitude decreases.

Emergency/Smoke Removal: Pressurization can be controlled through the forward safety valve when the AUTO-MAN-DUMP switch is set to MAN. Cabin pressure can be regulated by turning the FWD OUTFLOW knob, as necessary, to adjust the amount of pressure bled to get the required pressurization selection. When the control knob is turned clockwise the forward safety valve opens and the cabin pressure decreases. Pressurization can also be reduced by operating the forward safety valve selector on the copilot's side console.

If the AUTO/MAN/DUMP switch is set in automatic mode, and the FWD OUTFLOW knob is turned to open the forward safety valve, the cabin pressure will decrease and the aft outflow valve will start to close. When the FWD OUTFLOW knob is set to a higher cabin altitude (lower cabin pressure) than that set in automatic mode it will override the automatic selection.

During ground operation with the AUTO/MAN/DUMP switch selected to MAN, the FWD OUTFLOW knob turned fully clockwise (forward safety valve set to open) and all doors and hatches closed, the aircraft will begin to

pressurize. The forward safety valve will be slow to start to modulate and

cause an increase of cabin pressure.

Pressure dump: The fast depressurization function may be done in the automatic and the manual modes. The AUTO/MAN/DUMP switch set to DUMP, fully opens the aft outflow valve. In the manual mode, the aft outflow valve opens when the toggle switch is moved and held in the DECR position. On the ground the AUTO/MAN/DUMP switch is set to DUMP, to make sure that all pressure is bled off, before any doors or hatches are opened.

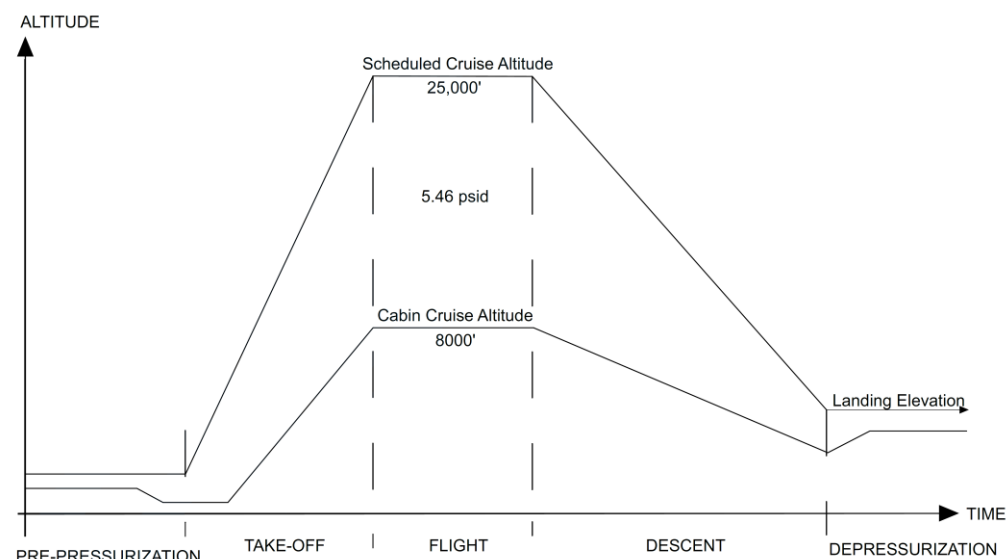


Figure 21- 78 Pressurization Graph

Component Description

Cabin Pressure Control Panel

[Refer Figure 21-79](#)

The cabin pressure control panel is located in the flight compartment on the overhead panel. The panel has mechanical and hardware subassemblies. The mechanical subassembly has:

- FWD OUTFLOW rotary select knob
- AUTO/MAN/DUMP select switch
- Fault alert light
- Landing altitude indicator
- LDG ALT rotary select knob
- Needle mechanically driven by the landing altitude potentiometer
- MAN DIFF INC/DECR toggle switch.

The FWD OUTFLOW knob is connected to the cabin pressure and the reference pressure. The knob controls the pneumatic opening of the forward safety valve. When the knob is turned clockwise, the valve opens and it increases the cabin rate of change. The AUTO/MAN/DUMP switch is used to set the mode of operation for pressurization control. In the automatic mode, the pressurization control is fully automatic. The manual mode is used if the automatic pressurization mode does not operate. In the DUMP mode, the aft outflow valve is in the open position and the aircraft can be operated unpressurized. The FAULT alert light gives a visual indication of failures in the cabin pressure control system. The LDG ALT select knob is set before take-off. A needle is mechanically driven by the LDG ALT knob and gives the set landing altitude information to the flight crew. The LDG ALT knob sends a signal to the Cabin Pressure Controller. If the landing altitude is not set, the controller input becomes the landing altitude. The potentiometer selections are from -2000 to 14,000 ft (-609.6 to 4267 m).

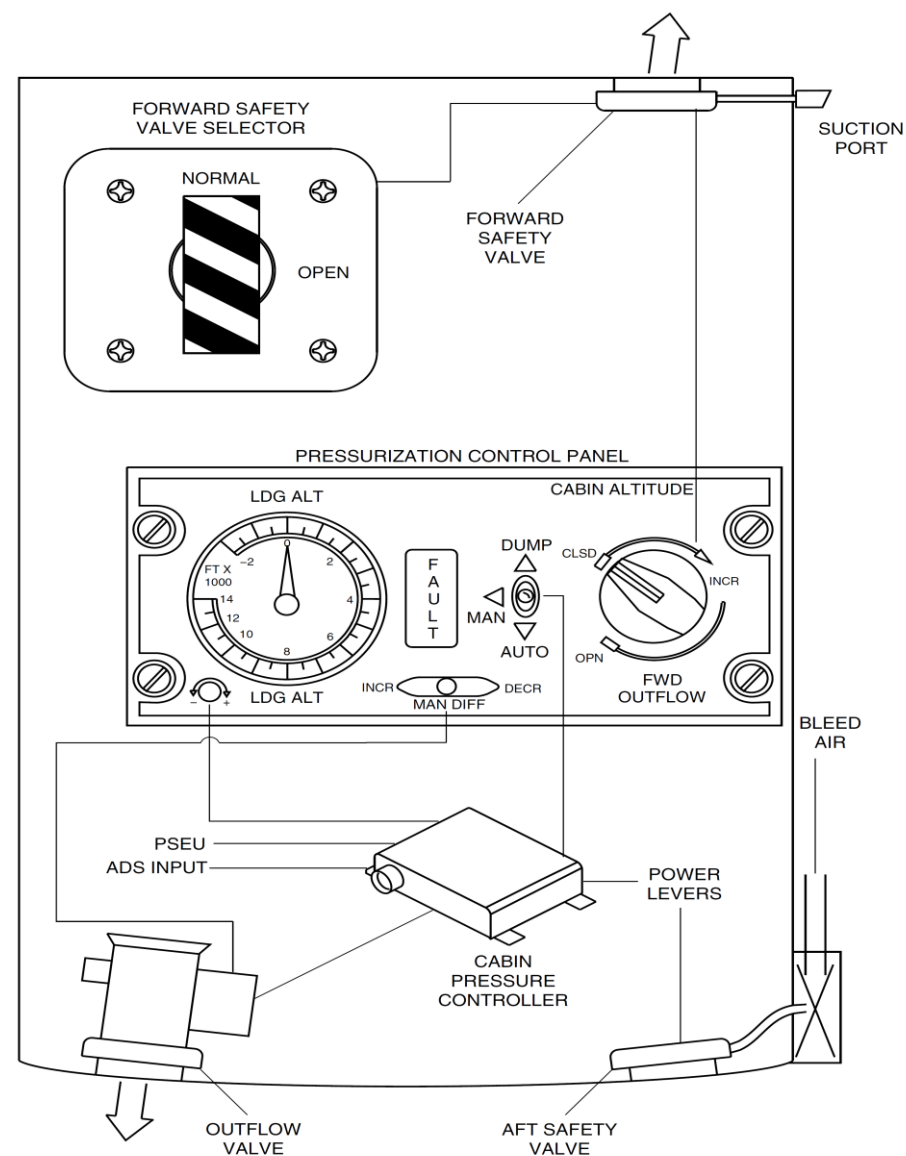


Figure 21- 79 Pressurization Schematic

Cabin Pressure Controller

[Refer Figure 21-80](#)

The cabin pressure controller is located on the avionics rack, and operates the auto channel of the electrical actuator of the aft outflow valve to control the cabin pressure. A pressure sensor measures the cabin pressure. The sensor gives a warning, of cabin altitude too high, to the Cabin Indication Module.

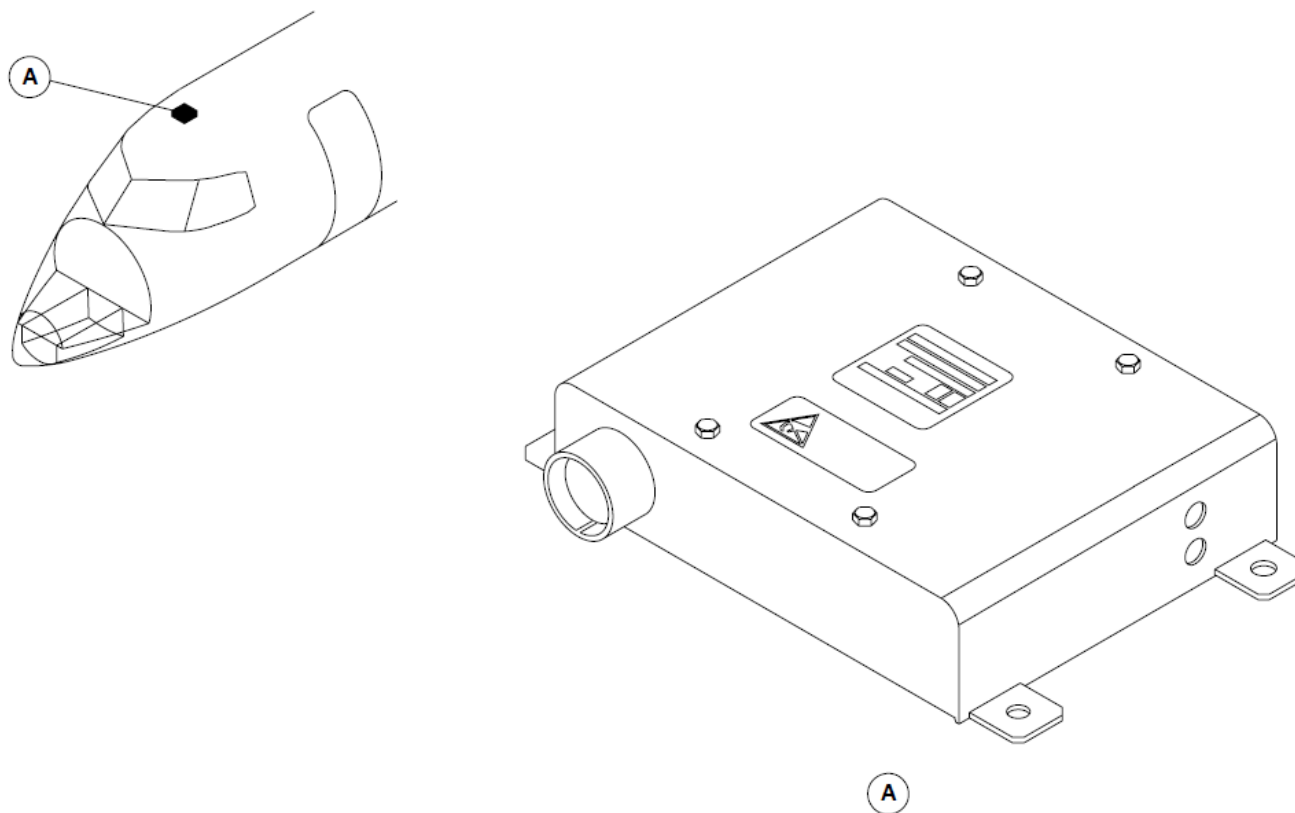
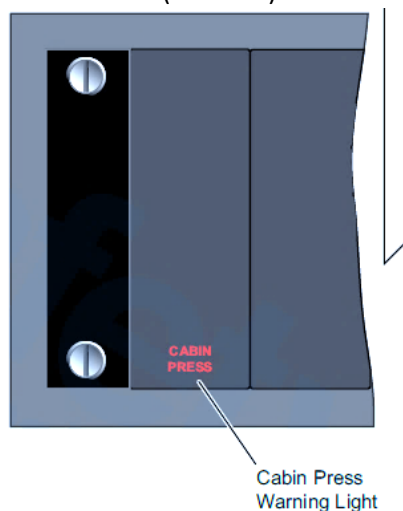


Figure 21- 80 Cabin Pressure Controller

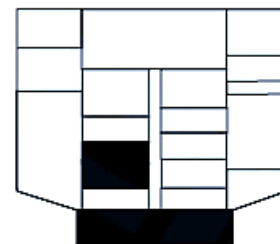
Cabin Indication Module

The cabin indication module is located in the flight compartment on the overhead panel and has a differential pressure indicator, a cabin altitude indicator, a cabin rate of change indicator, and a pressure sensor. The differential pressure indicator gives a differential pressure from -0.5 to 5.8 ± 0.15 psi (-3.45 to 40 ± 1.03 kPa) between the cabin pressure and the outside static pressure. The pressure sensor gives the excessive altitude warning received from the Cabin Pressure Controller to the CABIN PRESS warning light. The CABIN PRESS warning light on the Caution and Warning panel comes on when the cabin altitude reaches 10,000 ft (3048 m), except when take-off or landing altitude is above 8000 ft (2438.4 m). For take-off or landing altitudes above 8000 ft (2438.4 m) the warning light will come on in the conditions that follow:

- The aircraft take-off altitude + 1000 ft (304.8 m)
- Landing altitude + 1000 ft (304.8 m)



CAUTION WARNING PANEL



OVERHEAD CONSOLE

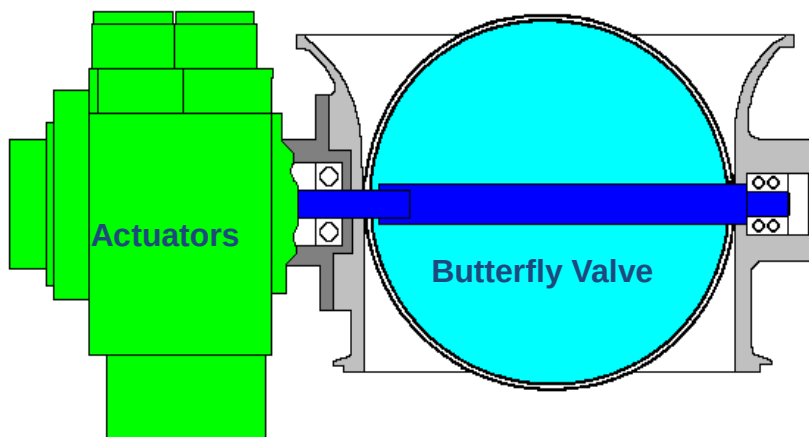


Figure 21- 81 Cabin Indication Module

Aft Outflow Valve

[Refer Figure 21-82](#)

The aft outflow valve, located on the rear pressure bulkhead, has a subassembly attached to an electrical actuator. The subassembly has a body, a butterfly with a sealing gasket, and a shaft that goes through the butterfly and is supported by two bearings. The electrical actuator has a dc motor, a gearbox and a control box. When the cabin pressure controller is set to pressurize the cabin, the aft outflow valve is closed. Air from the cooling packs is supplied to the cabin. Since no air can be released, the cabin pressure increases. Once the cabin has reached the set pressure, the cabin sensor automatically sends a signal to the pressure controller to modulate the outflow valve through the electrical actuator. The dc motor of the electrical actuator moves the shaft to open the butterfly chamber of the aft outflow valve. The air is released from the cabin to regulate the pressure. The chamber pressure is kept below the cabin pressure.



Valve Construction

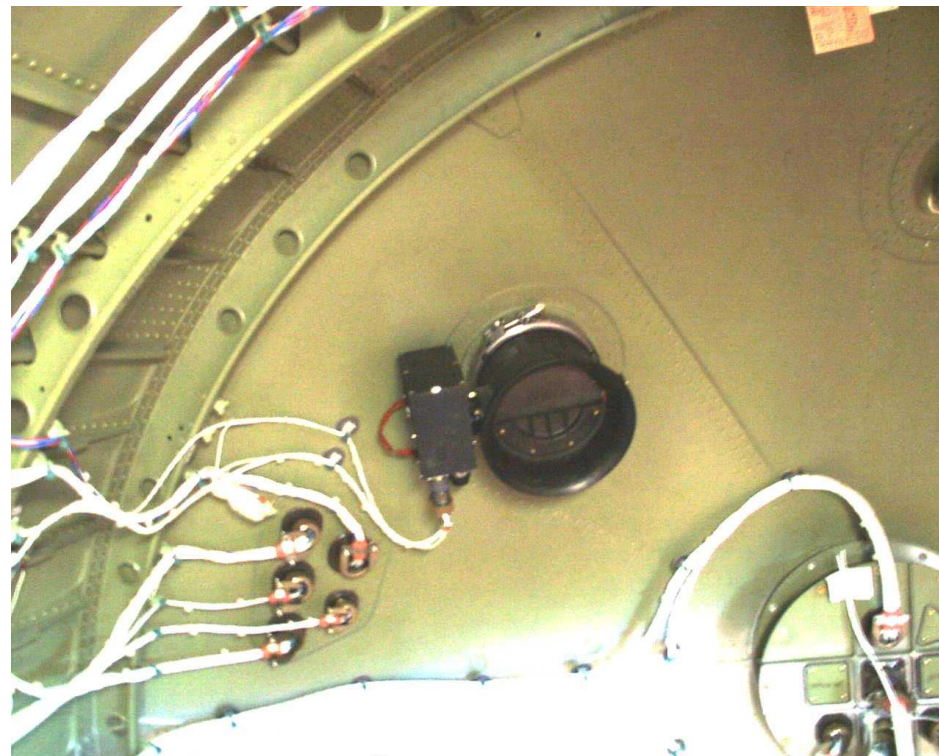


Figure 21- 81 Aft Outflow Valve

Installation on Rear Pressure Bulkhead

Forward Safety Valve

Refer Figure 21-83

The forward safety valve body is located in the nose compartment and is installed on the bulkhead flange. The valve has a cover that includes an overpressure relief box and a filter box, a poppet/diaphragm assembly, a deflector, a spring, and a pneumatic relay that includes a manometric capsule and a check valve. The forward safety valve automatically releases the cabin pressure when preset pressure limits are approached. The cabin pressure is applied to the inside of the overpressure relief box and the pneumatic relay. The poppet/diaphragm assembly releases the cabin pressure through the filter box. When the cabin pressure is more than the combined force of the outside static pressure and the spring, the check valve of the manometric capsule is opened. When the check valve opens, the cabin pressure of the poppet/diaphragm assembly starts to flow out to outside static pressure through the hollow center of the forward safety valve. The reduced pressure in the chamber of the poppet/diaphragm assembly lets the spring open the poppet/diaphragm assembly. The cabin air flows out.

To prevent high cabin pressure, the safety valves have an automatic positive pressure-relief control system operation. The force of the cabin pressure applied on one side of the poppet/diaphragm assembly is more than the force of the outside static pressure and the spring on the other side of the poppet/diaphragm assembly. The cabin pressure in the chamber is released and the cabin pressure decreases.

The safety valves also have a negative pressure relief function. During an emergency descent or without cabin airflow, the outside static pressure becomes more than the cabin pressure. The force of the cabin pressure applied on one side of the poppet/diaphragm assembly is less than the force of the outside static pressure and the spring on the other side of the poppet/diaphragm assembly. The differential pressure between the outside and the cabin moves the deflector and the poppet/diaphragm assembly

opens. The air in the chamber is released into the cabin through the filter box. The outside air flows into the cabin and the cabin pressure increases.

The forward safety valve has a pneumatic emergency RAM air and smoke evacuation function. The forward safety valve can be opened with the forward safety valve selector or with the FWD OUTFLOW knob of the cabin pressure control panel. The pneumatic relay opens and the outside static pressure creates a negative pressure in the chamber and the poppet/diaphragm assembly opens. The air in the chamber is released.

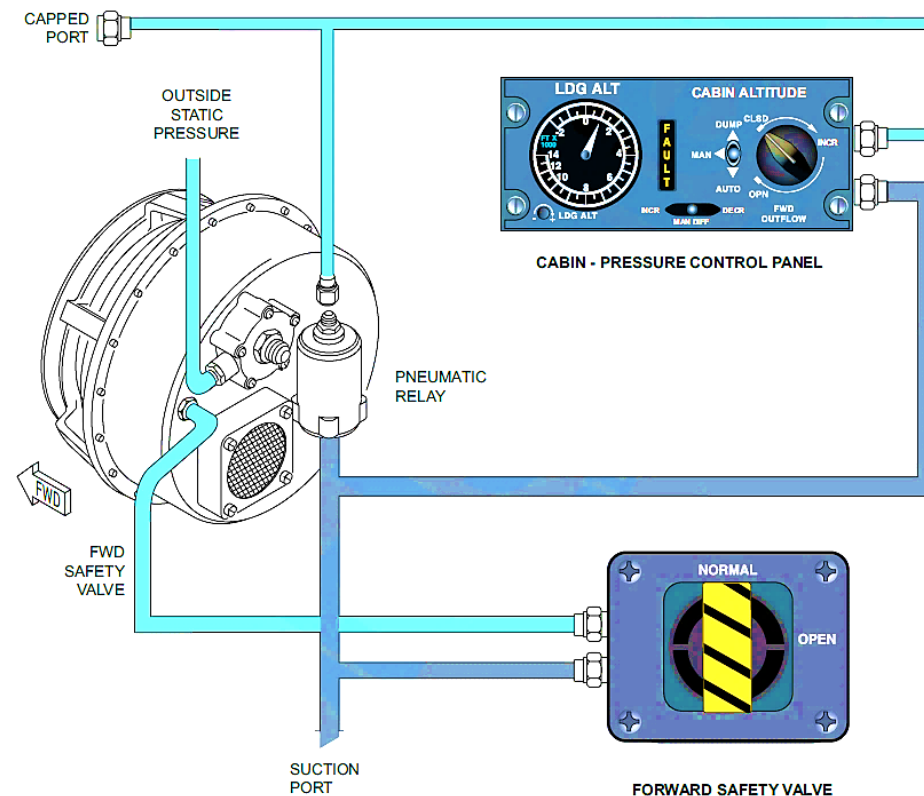


Figure 21- 82 Forward Safety Valve Operation

Forward Safety Valve Selector

[Refer Figure 21-85](#)

The forward safety valve selector is located on the copilot side console and completely opens the forward safety valve to dump the cabin pressure. The selector is a shut-off valve. NORMAL or OPEN positions can be selected with the rotary switch.

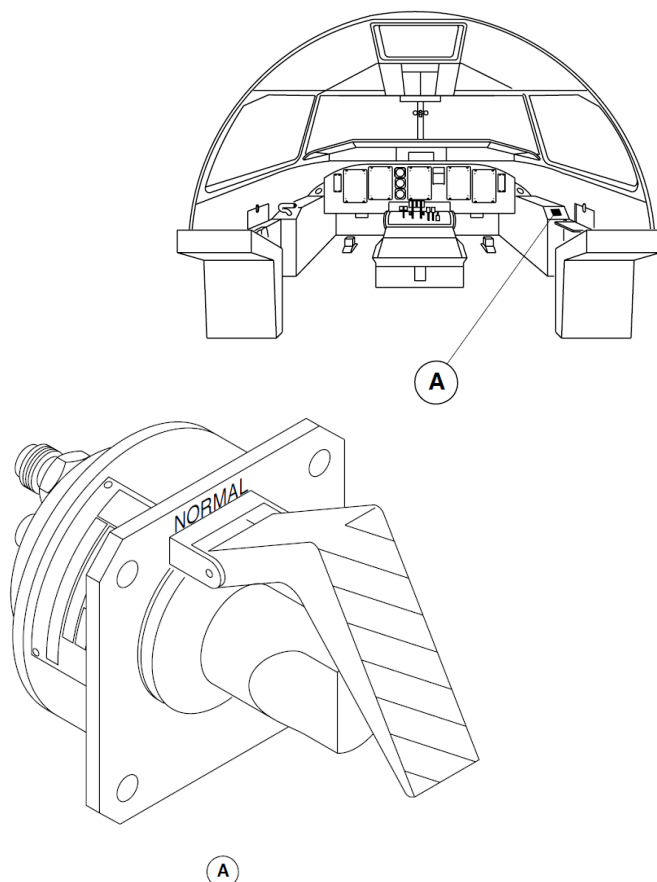


Figure 21- 84 Safety Valve Selector

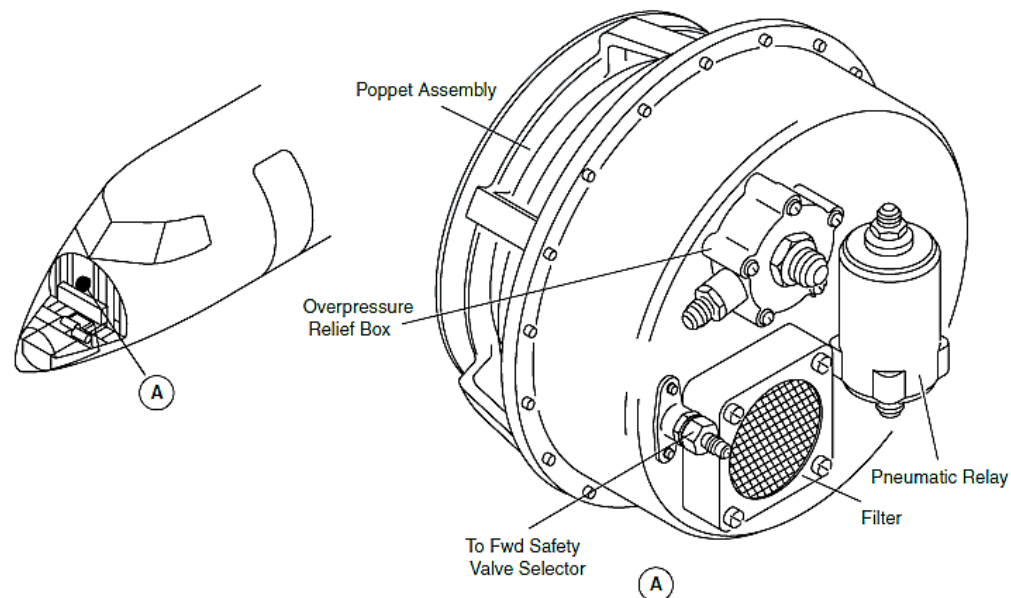
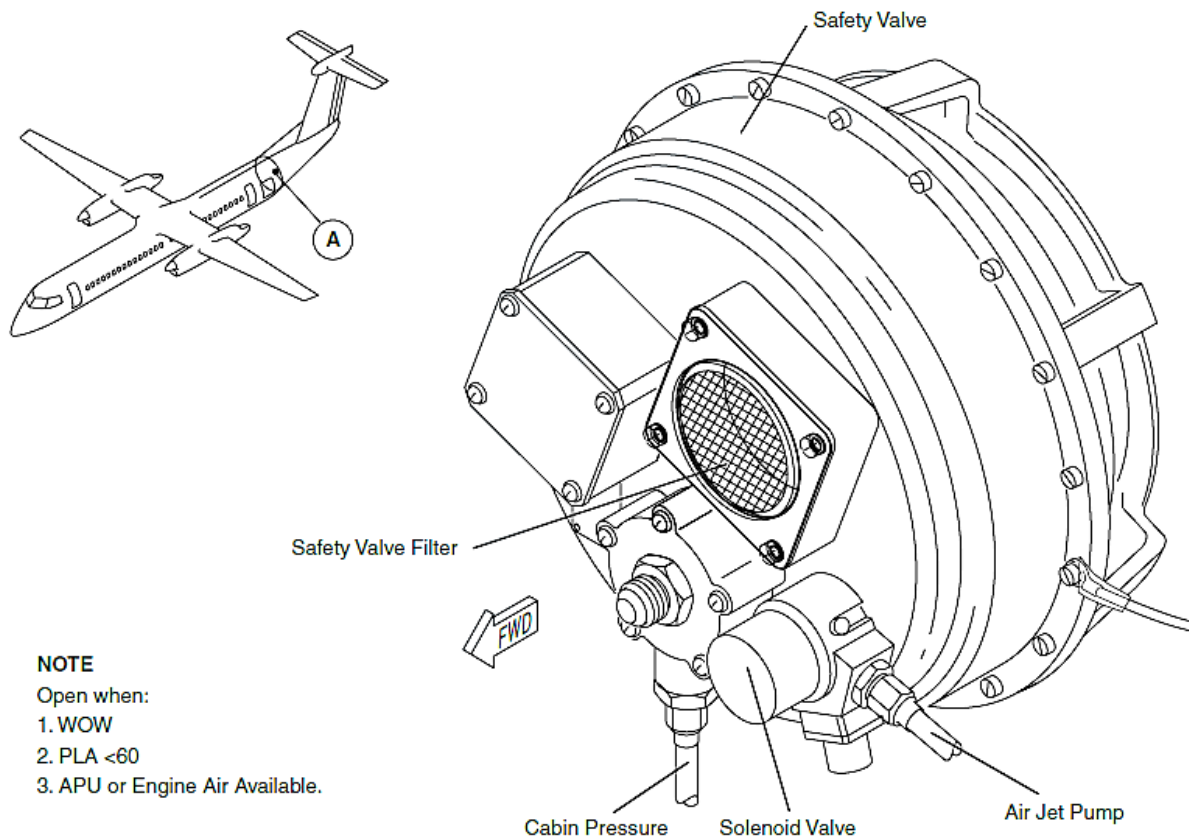


Figure 21- 83 Forward Safety Valve

Aft Safety Valve

[Refer Figure 21-86](#)

The aft safety valve is located on the aft pressure bulkhead and opens on the ground when the engines are in the idle condition. The aft safety valve is the same as the forward safety valve, but has a solenoid valve instead of a pneumatic relay.



NOTE

Open when:

1. WOW
2. PLA <60
3. APU or Engine Air Available.

Figure 21- 85 Aft Safety Valve

Air Jet Pump

The air jet pump located on the rear pressure bulkhead, has a body and three orifices. The pump uses the bleed air to cause a negative pressure on the solenoid valve of the aft safety valve. The pump fully opens the aft safety valve on the ground. The pump is supplied with air from the de-icing line or the APU.